

ADDIS ABABA UNIVERSITY

SCHOOL OF MEDICINE

DEPARTMENT OF PHYSIOLOGY

**MOTOR FUNCTIONS
OF THE
NERVOUS SYSTEM**

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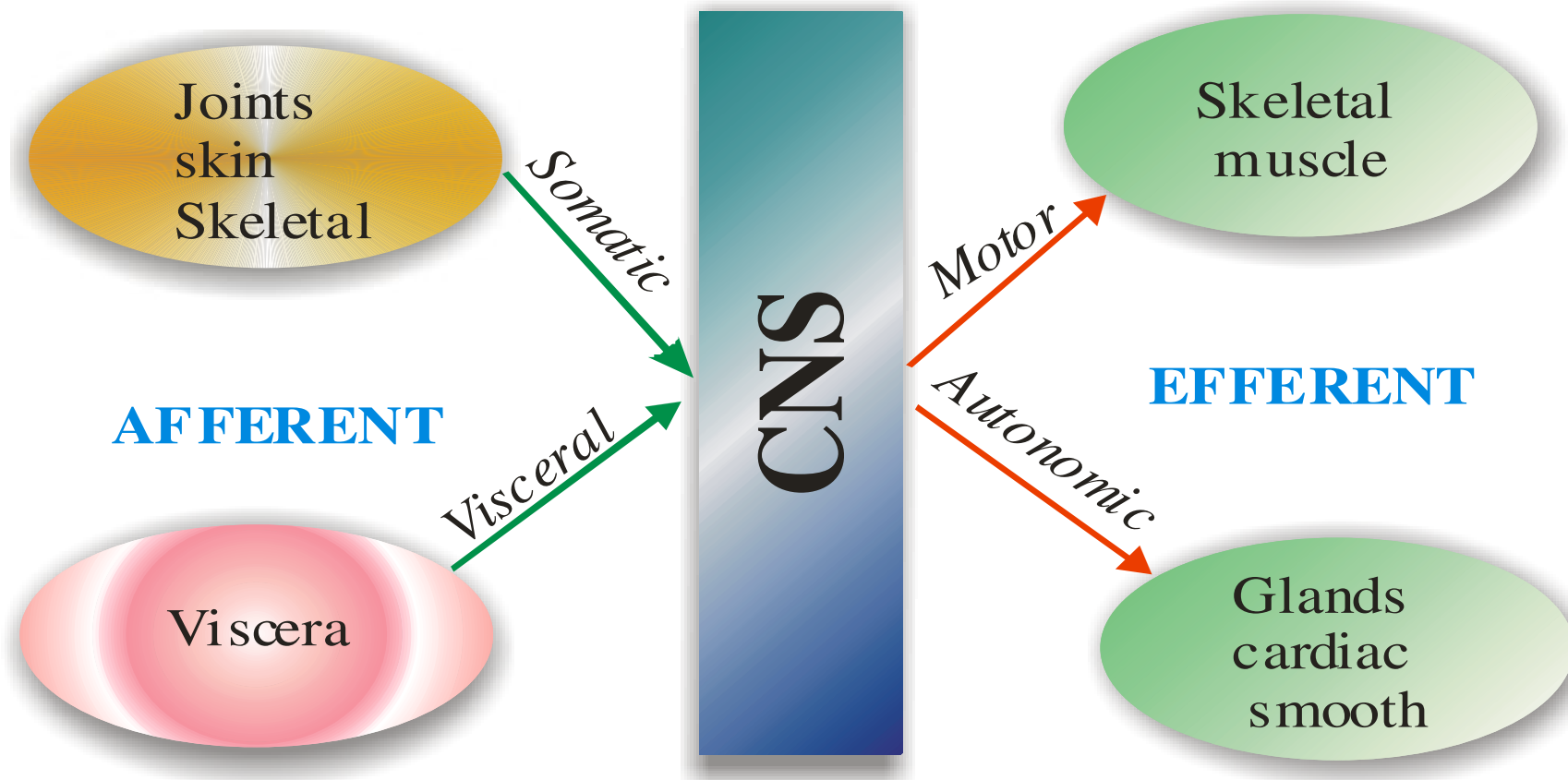
2024

Outlines of Motor Function of the CNS

- Organization of the nervous system
- Cerebral blood flow, CSF and metabolism of brain
- Overview of the arterial blood supply to the CNS
- Muscle receptors (spindles and golgi tendon organ),
- Extrafusal and intrafusal muscle fibers and their innervations
- Motor functions of spinal cord: the cord reflexes and clinical implications
- Cortical Control of Motor Function and clinical implications
- Brain Stem Control of Motor Function
- Cerebellum and its motor function and clinical implications
- The basal ganglia and their motor functions and clinical implications

Chapter ONE: Organization of Nervous System

● Functional organization of Nervous System



- ❖ Each minute CNS receives millions of bits of info from the different sensory nerves and sensory organs
- ❖ The CNS integrates all these to determine responses to be made by the body.
 - ✓ The CNS is estimated to contain *80 to 100 billion neurons for integration*

Organization of CNS

➤ Brain

- Fore brain

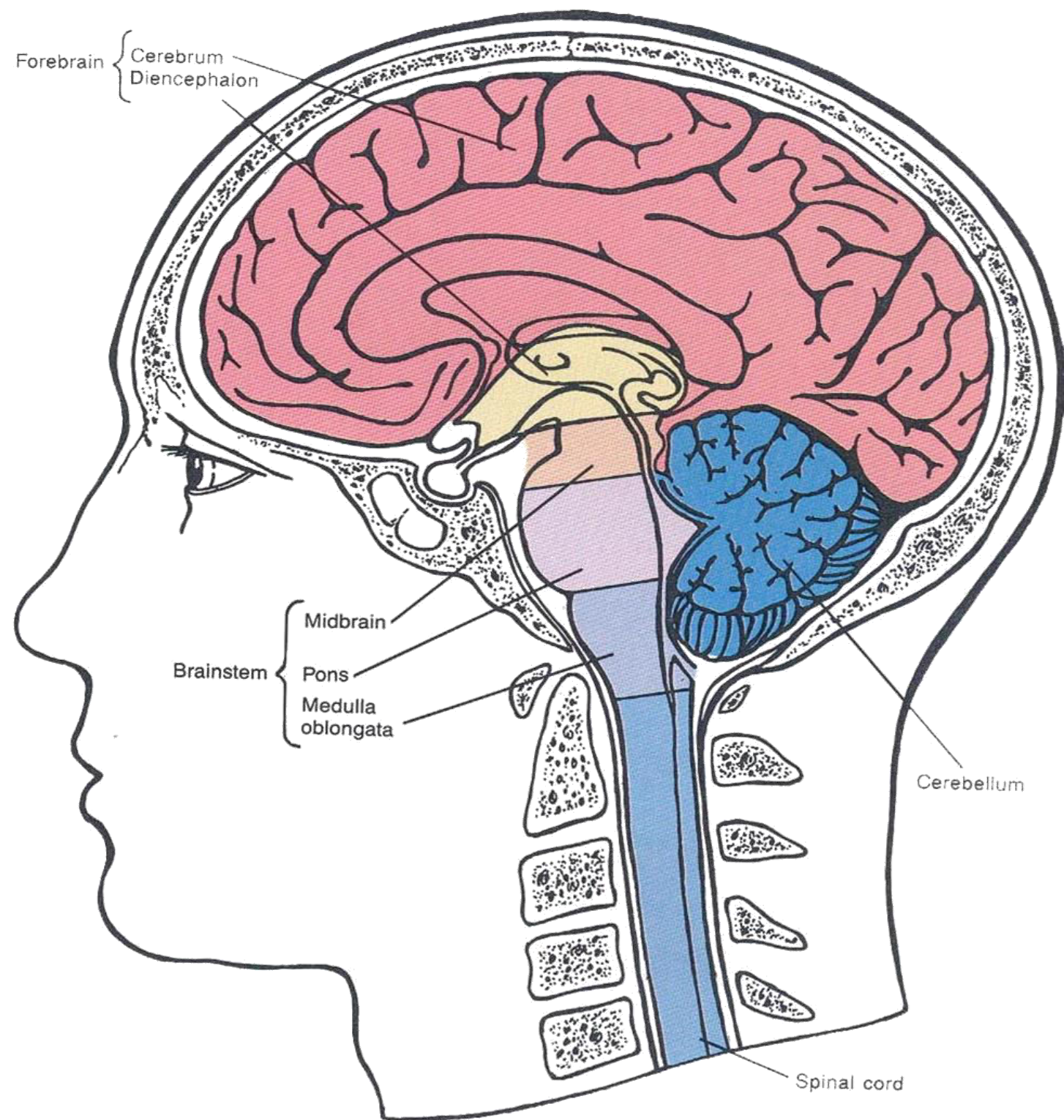
- ✓ Cerebral hemispheres
- ✓ Diencephalon

- Brain stem area

- ✓ Midbrain
- ✓ Pons
- ✓ Medulla

- Cerebellum

➤ Spinal cord



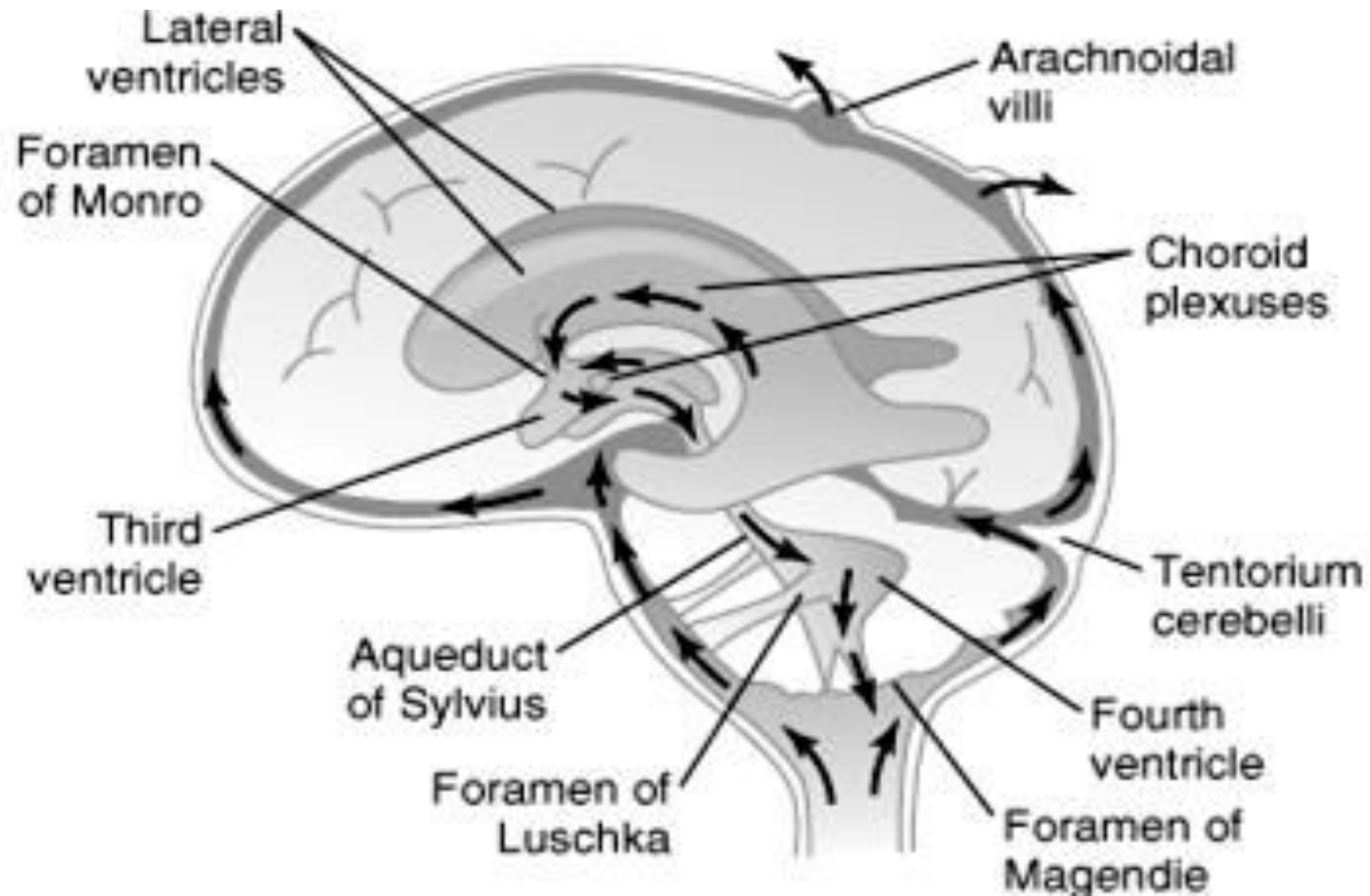
INTRODUCTION TO PHYSIOLOGY OF THE CNS

- Cerebral Blood Flow, CSF, and Brain Metabolism
- Normal Rate of Cerebral Blood Flow
 - 750-900 ml/min
 - 50 - 65 ml/100 grams of brain tissue/minute.
- Cerebral BF is highly related to metabolism of the tissue.
 - 3 metabolic factors have potent effects in controlling cerebral BF:
1. $[CO_2]$ 2. $[H^+]$, and 3. $[O_2]$
- Excess $[CO_2]$ or Excess $[H^+]$ or $\downarrow [O_2] \Rightarrow$ Increase of Cerebral BF
- Cerebral BF is autoregulated within MABP limits of 60 - 140 mm Hg

Cerebrospinal Fluid System

- CSF- is a clear fluid that bathes the brain and the spinal cord.
- About 250 - 300 ml in volume
 - About 500 ml is formed and reabsorbed daily.
 - > 2/3rd of fluid originates as *secretion from choroid plexuses* in the ventricles
- Major function → to cushion the brain within its solid vault.
- Intracranial pressure (ICP) is the pressure exerted by CSF
 - The normal ICP is 7–15 mmHg (average **13** mm H₂O (10 mm Hg)) in adult
 - **Cerebral perfusion pressure(CPP)=MABP-ICP**
- ↑ CSF pressure in Pathological Conditions of the Brain (>20mmHg)
 - Causes: Inflammation, mass, bleeding, obstruction of CSF flow in hydrocephalus
 - symptoms and signs of rise in ICP :
 - headache, vomiting , blurred vision/papilledema
 - altered level of consciousness, behavioral change
 - Change in V/S: ↓PR, Irregular respirations, ↑BP [Cushing triad]

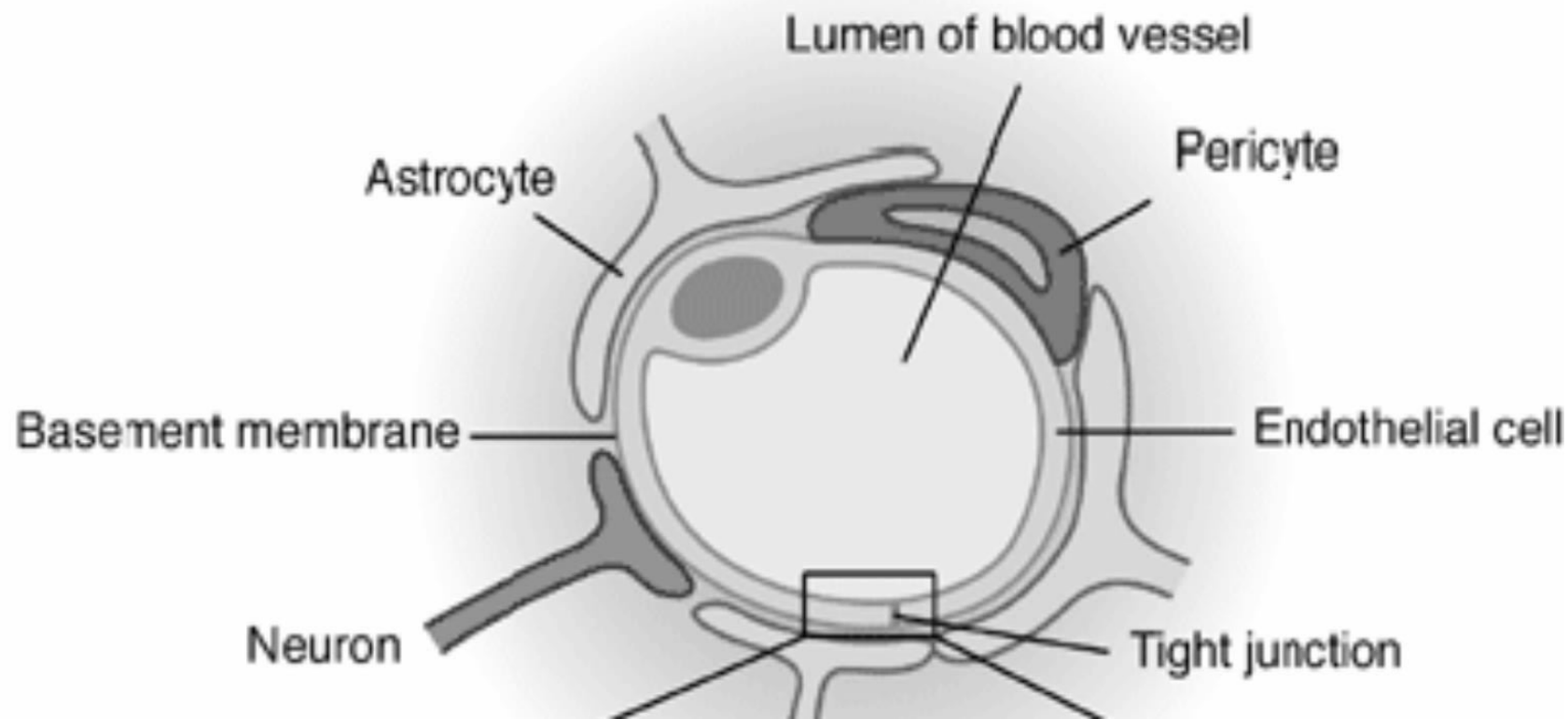
Secretion	Choroid plexus (lateral, third and fourth ventricles), interstitial space and dura of the nerve root sleeves
Circulation	Lateral ventricle(via Monro) → third ventricle → fourth ventricle, then: → subarachnoid space (via foramen of Luschka) → central canal of spinal cord (via foramen of Magendie)
Absorption	Arachnoid granulations , choroid plexuses and lymphatics



Blood-Cerebrospinal Fluid and Blood-Brain Barriers

- large molecular substances hardly pass:
 - from the blood into the CSF or
 - From blood into the interstitial fluids of the brain
- Barriers, called the *blood-CSF barrier* and the BBB
 - exist between the blood and the CSF and brain fluid, respectively.
 - In both at choroid plexus and at brain tissue capillary membranes
 - in all parts of the parenchyma *except in few other areas*
- the blood-CSF and blood-brain barriers (BBB) are:
 - highly permeable to H₂O, CO₂, O₂ and most lipophilics
 - slightly permeable to electrolytes: Na⁺, Cl⁻, K⁺
 - almost impermeable to plasma proteins and most non-lipid-soluble large organic molecules

- non lipid soluble drugs, protein antibodies $\Rightarrow$ fail to cross the barriers
- The BBB is formed by specialized capillaries in the brain
 - Such capillaries are less permeable than those elsewhere in the body
 - Contain tight junctions between capillary cells
 - Covered by astrocytes
 - Have specific transporters and carriers

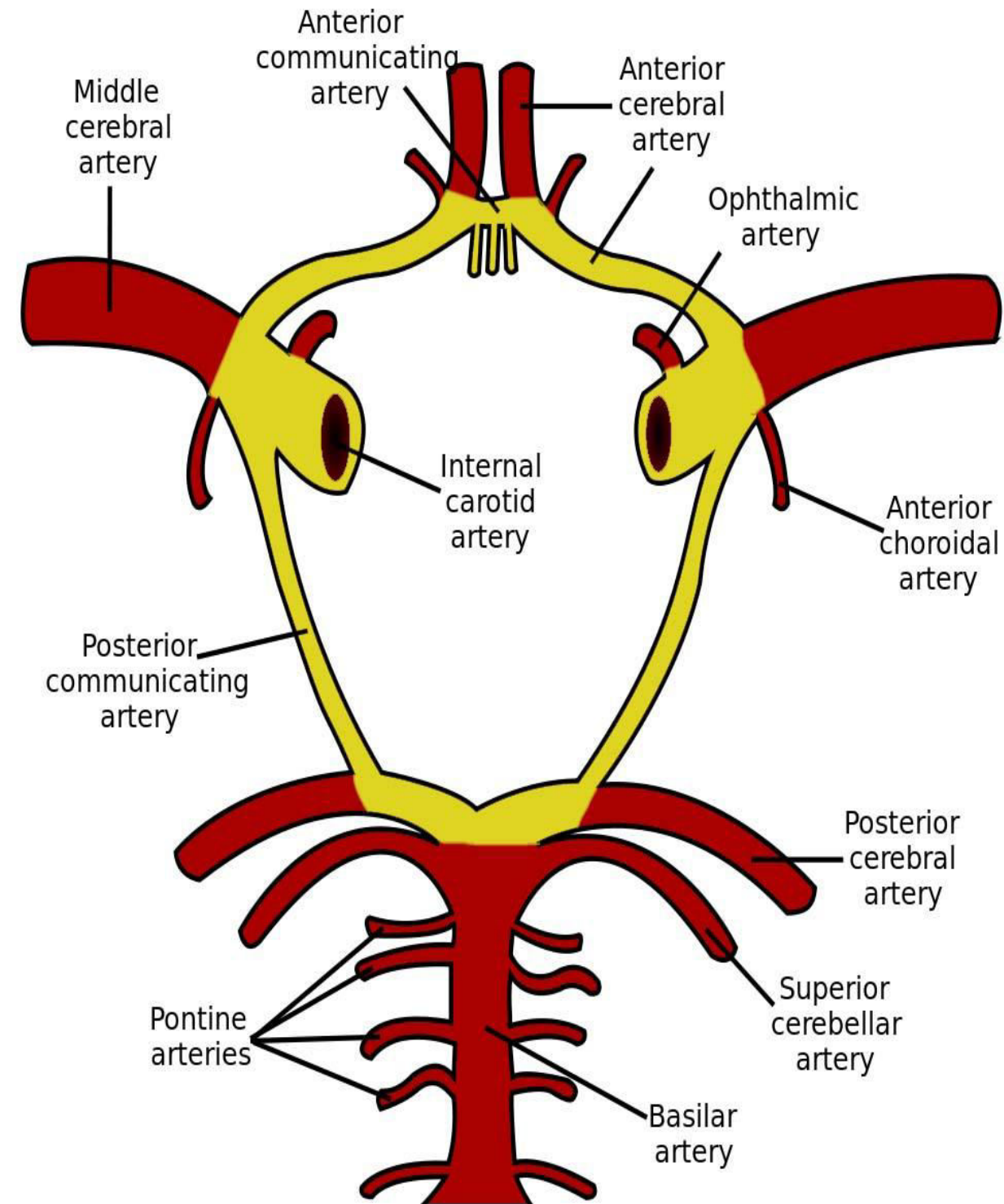


Brain Metabolism

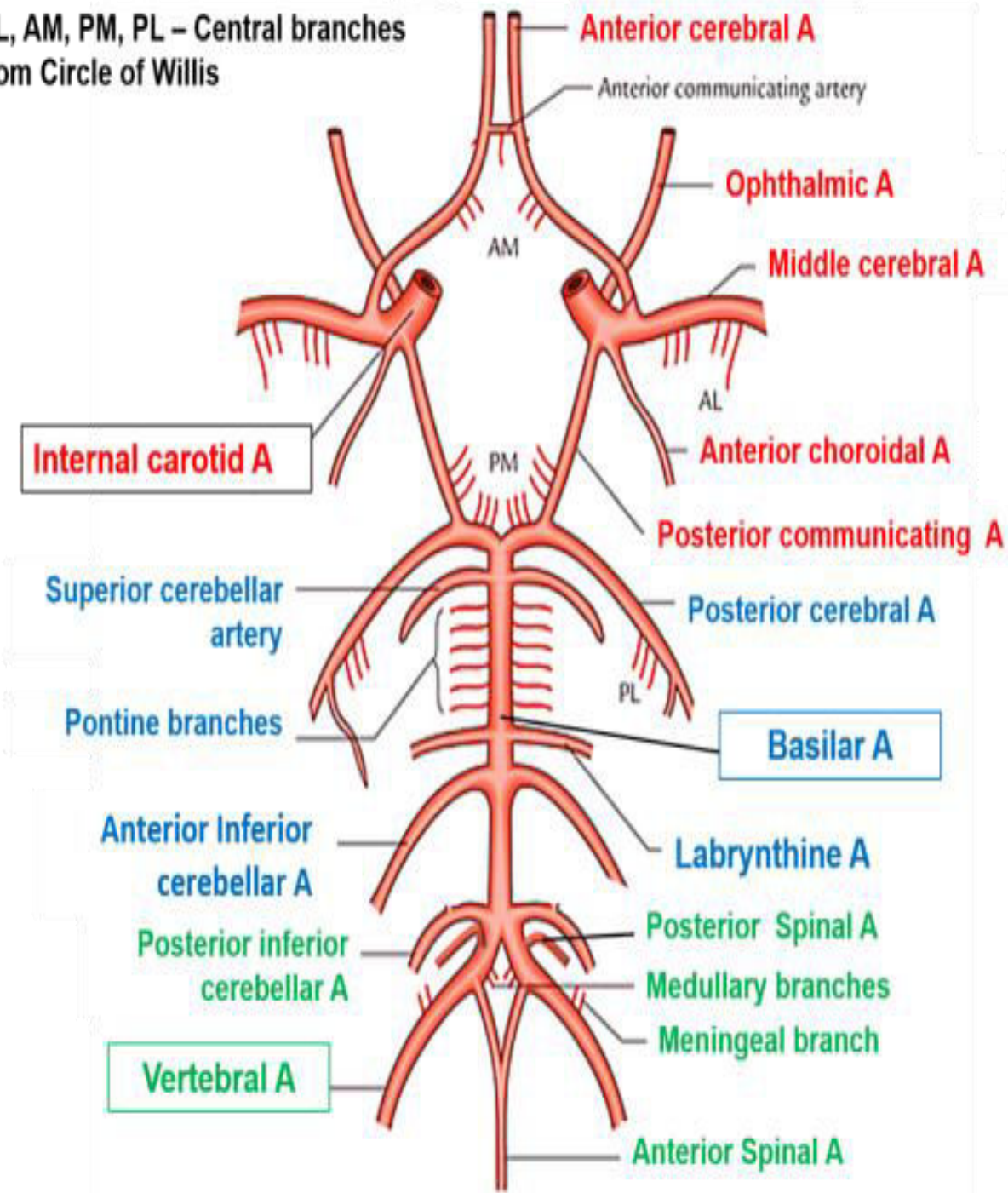
- the mass of brain is only 2% of the total body mass
- BMR of brain is about 15% of the total metabolism in the body
 - can ↑ by ~ 100 to 150% during excessive brain activity
- The high neuronal metabolic rate ⇒ require sec-by-sec delivery of O₂
 - The brain is not capable of much anaerobic metabolism.
- Sudden cessation of BF to the brain or sudden total lack of O₂ in the blood can cause unconsciousness within 5 to 10 seconds

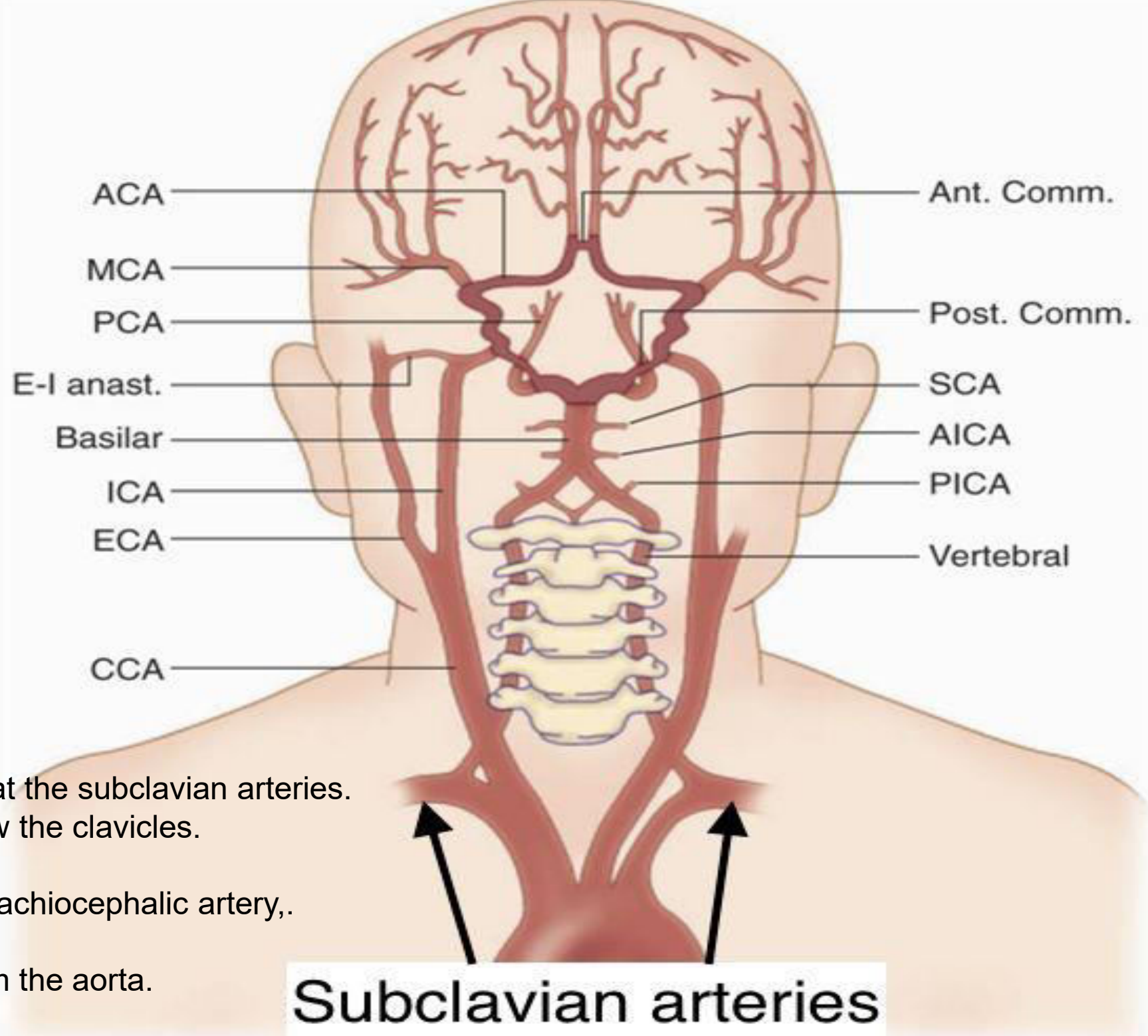
Overview of Arterial blood supply of brain

- There are two paired arteries responsible for the blood supply to the brain;
 - the **vertebral arteries**, and the **internal carotid arteries**.
 - The vertebral artery provides 20% of BF to the brain Vs carotid artery supplies the other 80%
 - These arteries arise in the neck, and ascend to the cranium.
 - Their terminal branches form anastomotic circle, called **Circle of Willis**.
- There are 3 main (paired) constituents of the Circle of Willis:
 - **Anterior cerebral arteries** – terminal branches of the internal carotid arteries.
 - **Internal carotid arteries** – located immediately proximal to the origin of the middle cerebral arteries.
 - **Posterior Communicating Artery**, and **Posterior cerebral arteries** – terminal branches of the basilar artery and



AL, AM, PM, PL – Central branches from Circle of Willis

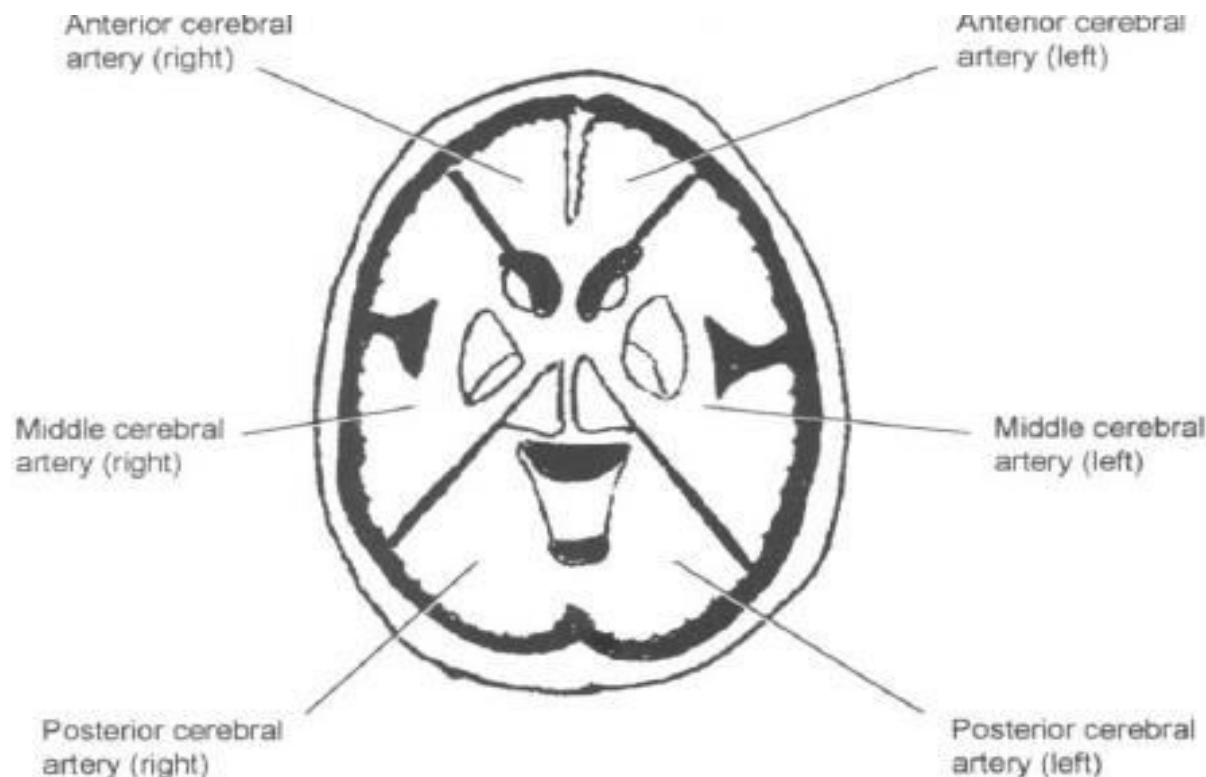




- ❖ The two vertebral arteries start at the subclavian arteries.
- ❖ The subclavian arteries sit below the clavicles.
- ❖ Rt subclavian arises from the brachiocephalic artery,.
- ❖ Lt subclavian arises directly from the aorta.

Three arterial territories(regional blood supply) of the brain

- **A. Anterior Cerebral Artery** : Anterior circulation
 - Supplies most of medial surface, superior surface & frontal pole.
- **B. Middle Cerebral Artery** : Middle circulation
 - Supplies most of lateral surface & temporal pole.
- **C. Posterior Cerebral Artery**: posterior circulation
 - Supplies most of inferior surface & occipital Pole.



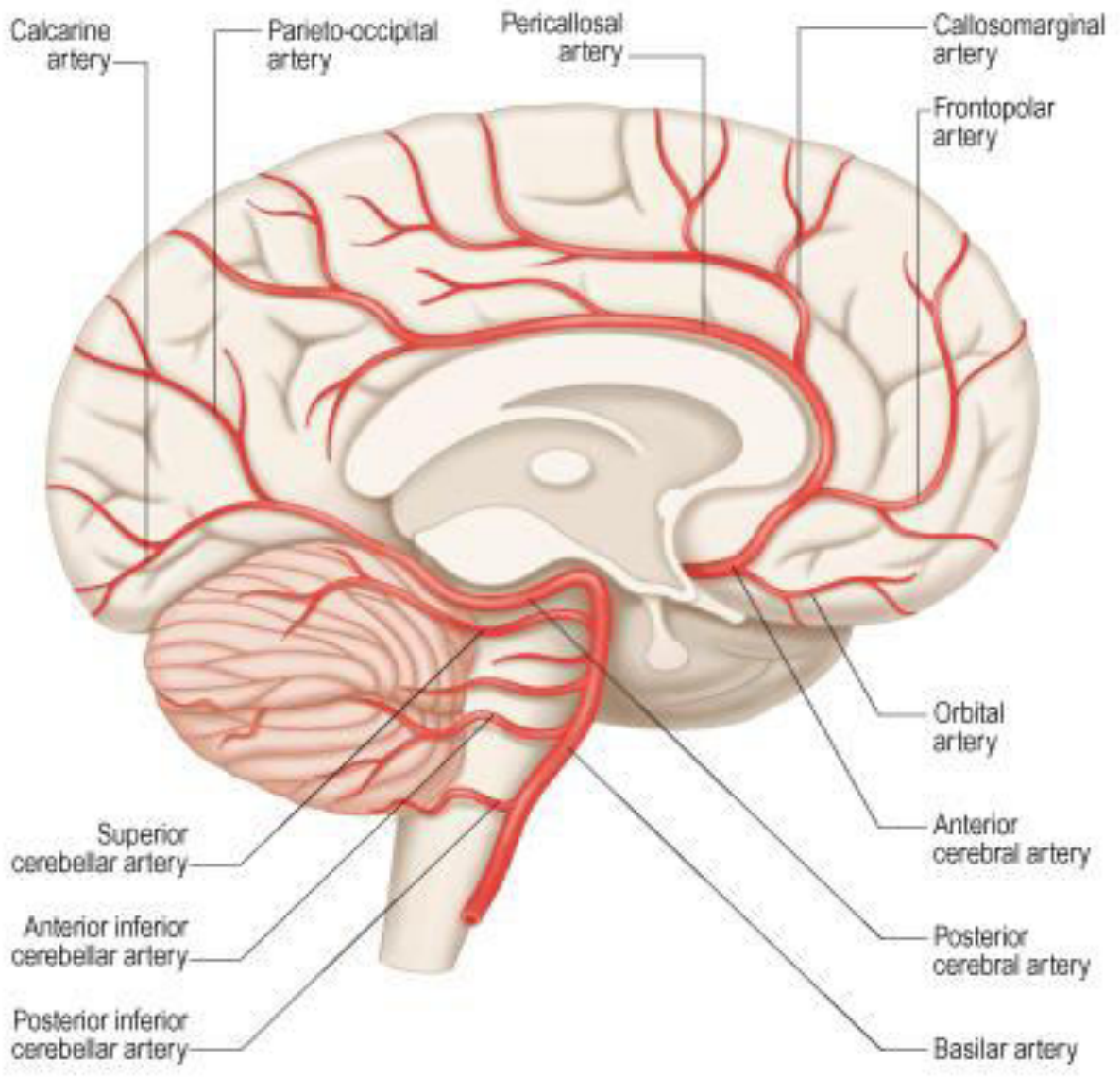
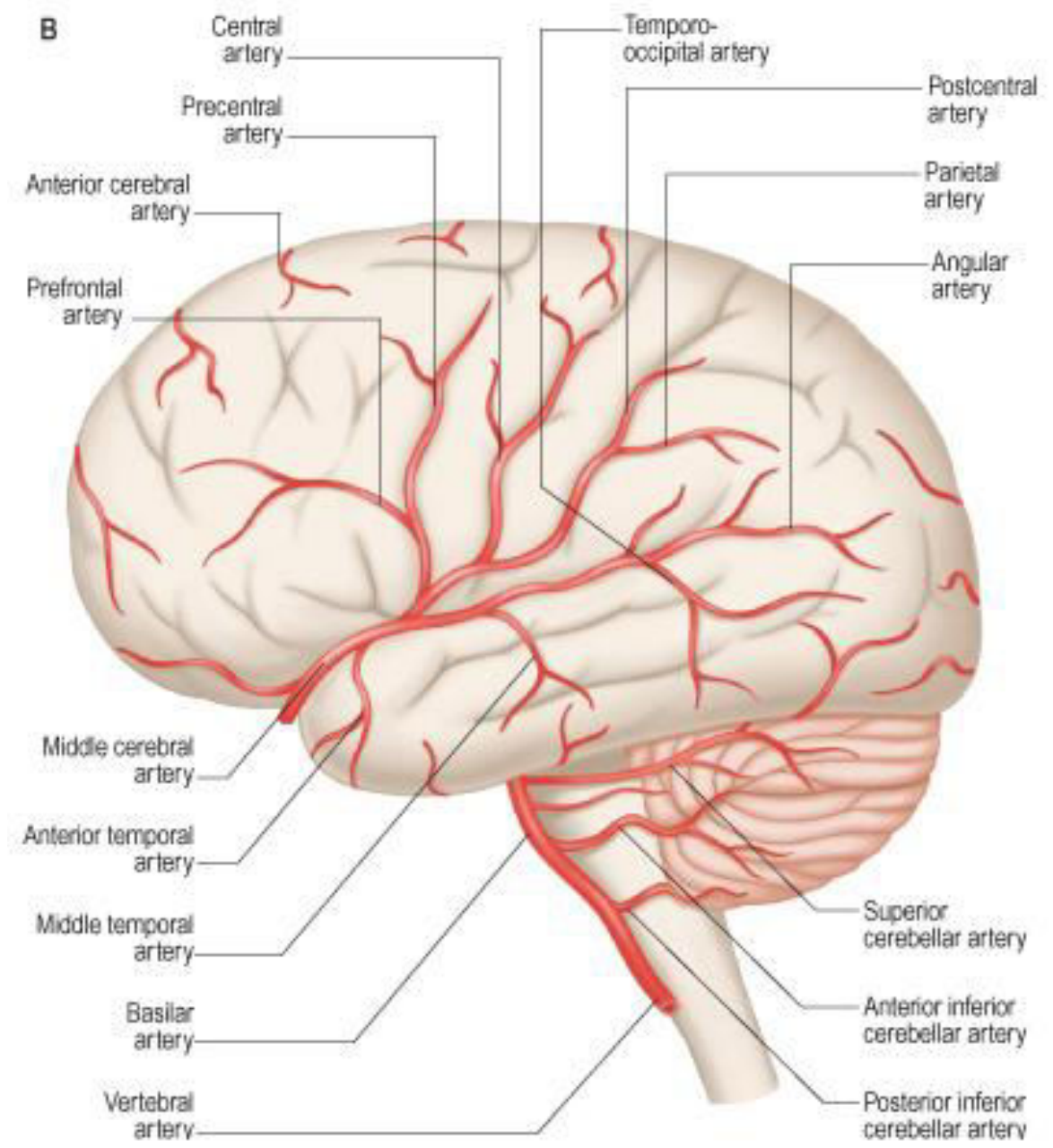
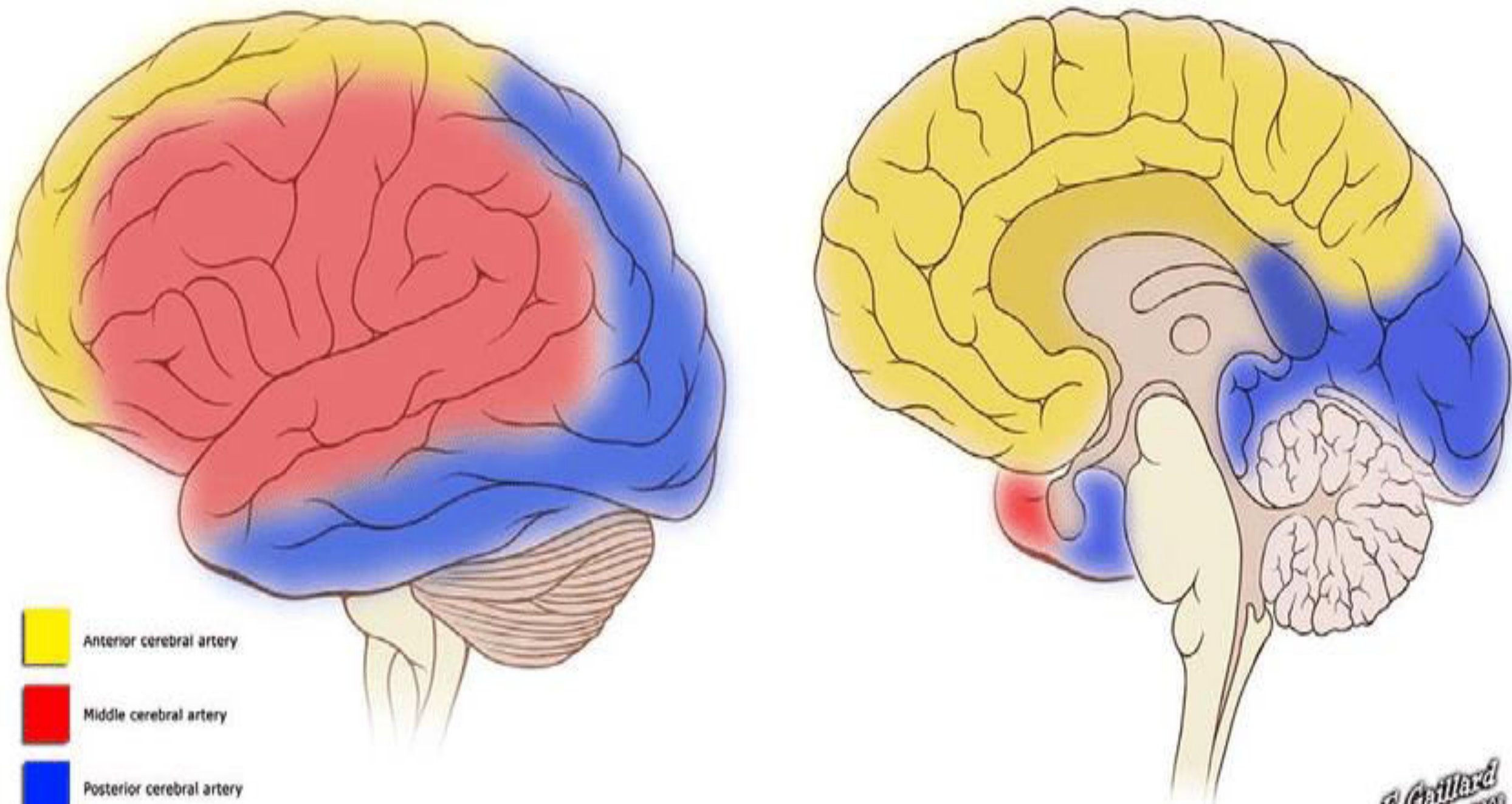
A**B**

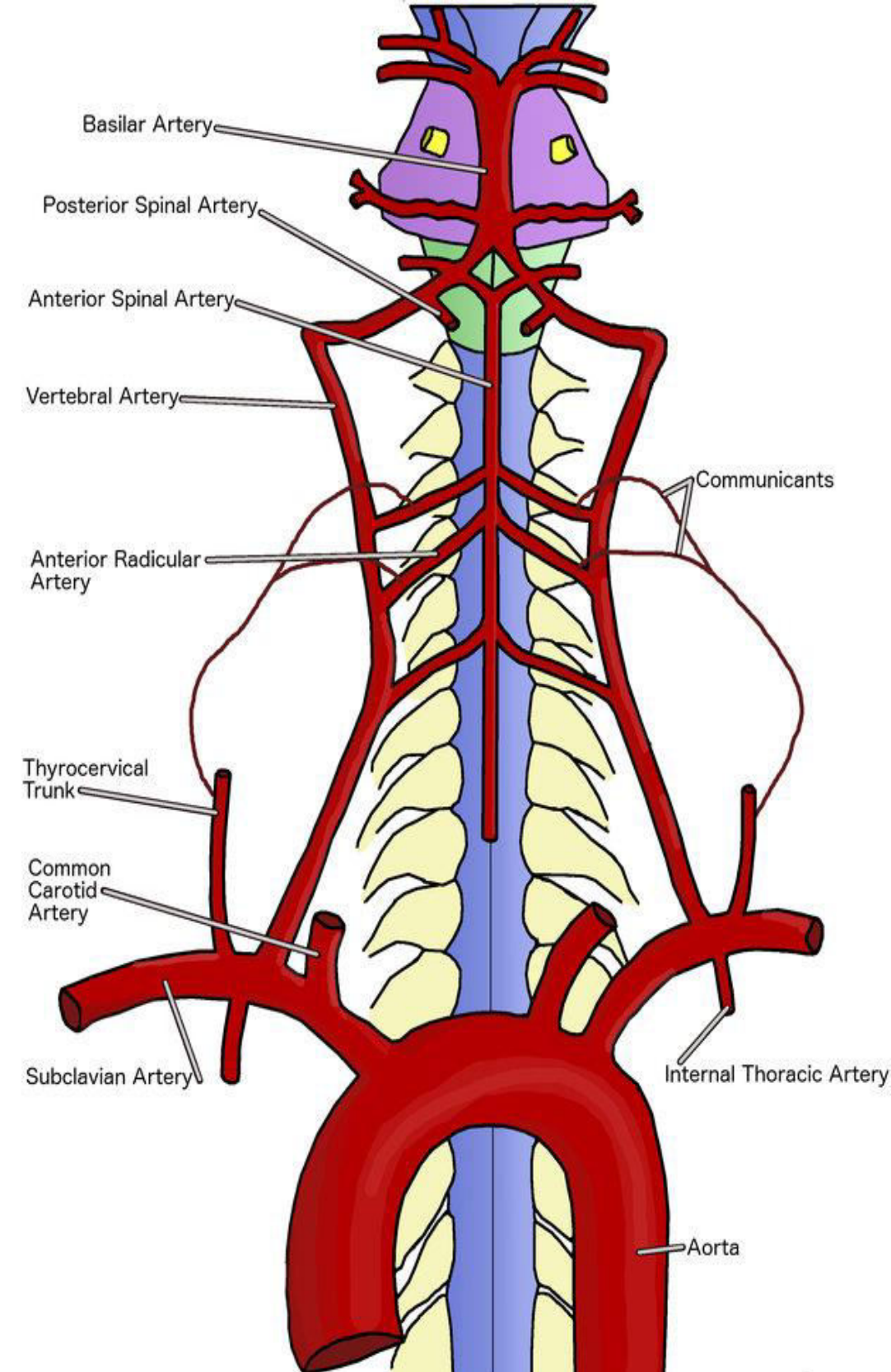
Fig. Major arteries of the brain. A, medial aspect; B, lateral aspect.



Three arterial territories of the brain

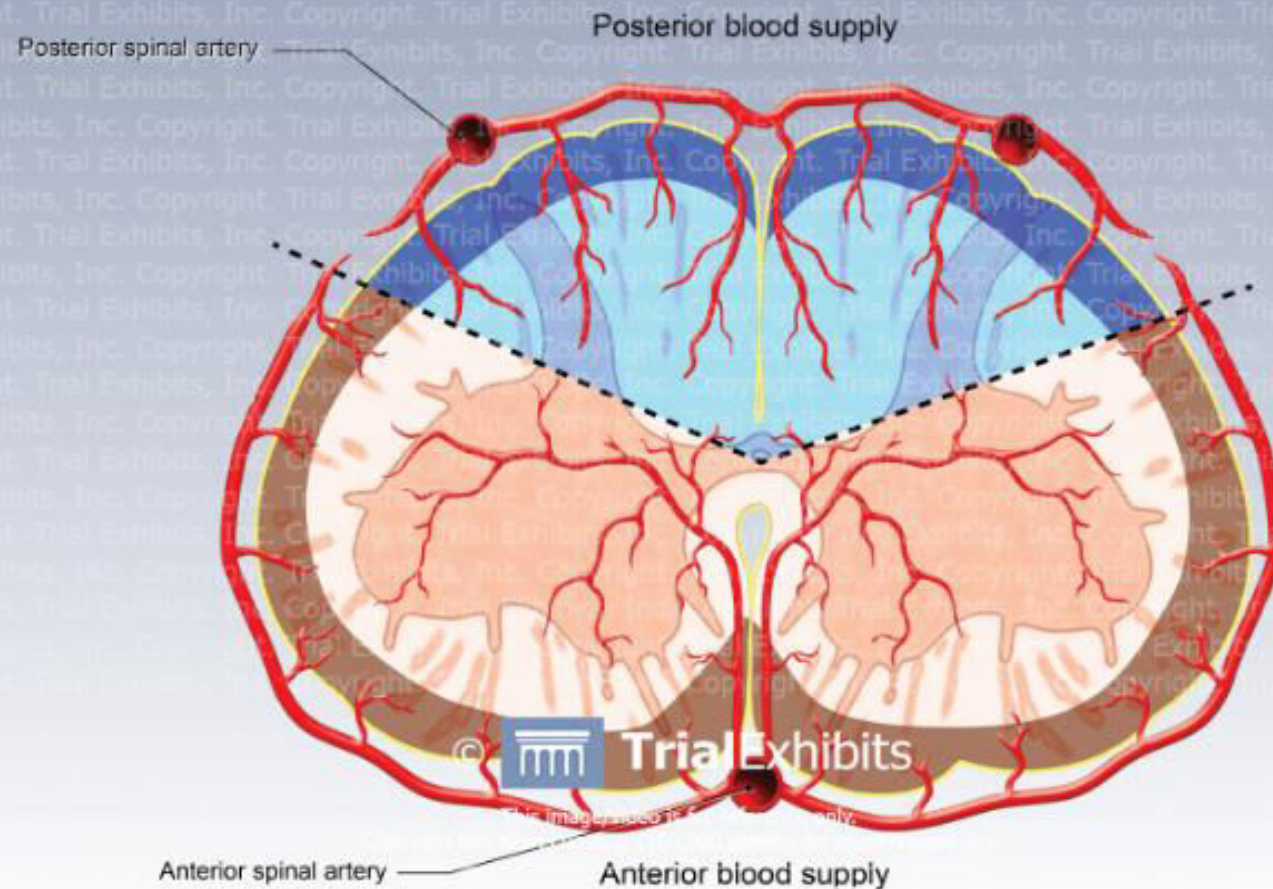
Blood supply to spinal cord

- The cord is dependent upon longitudinal arterial trunks
 - One anterior and two posterior spinal arteries
 - spinal arteries receive supply via the vertebral arteries only up to the cervical segments of the spinal cord.
 - Inferiorly the spinal arteries receive blood through the lumbar and radicular arteries



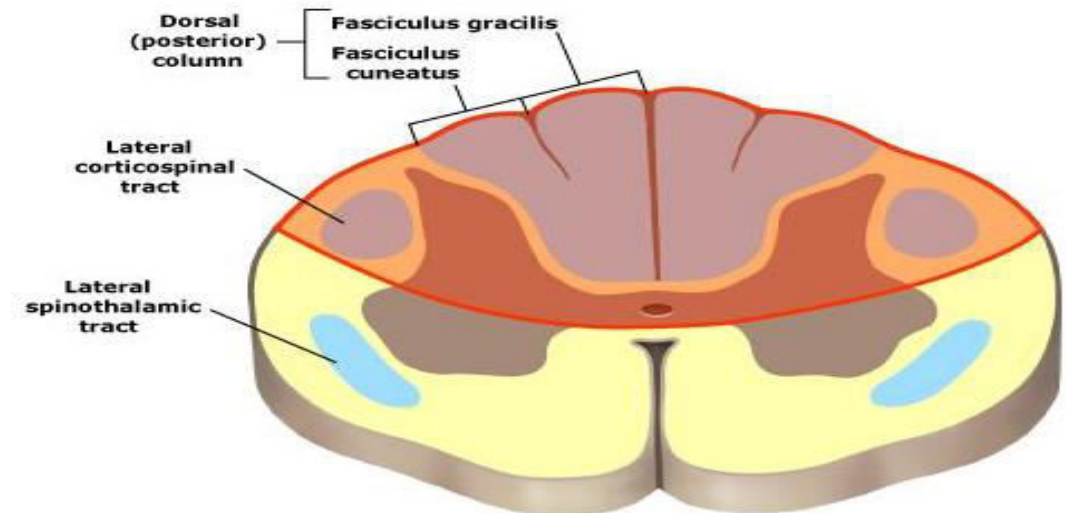
- **Anterior spinal artery (ASA):**
 - Runs the length of the spinal cord longitudinally along the anterior median fissure
 - **Primary blood supply of anterior two-thirds of the spinal cord**
- **Posterior spinal artery (right and left):**
 - Run longitudinally on each side of the midline of the posterolateral sulcus consistently through the length of the spinal cord
 - The two posterior spinal arteries originate directly from the vertebral arteries
 - are the primary blood supply to the posterior columns, dorsal grey matter, dorsal sensory columns

Blood Supply to the Spinal Cord

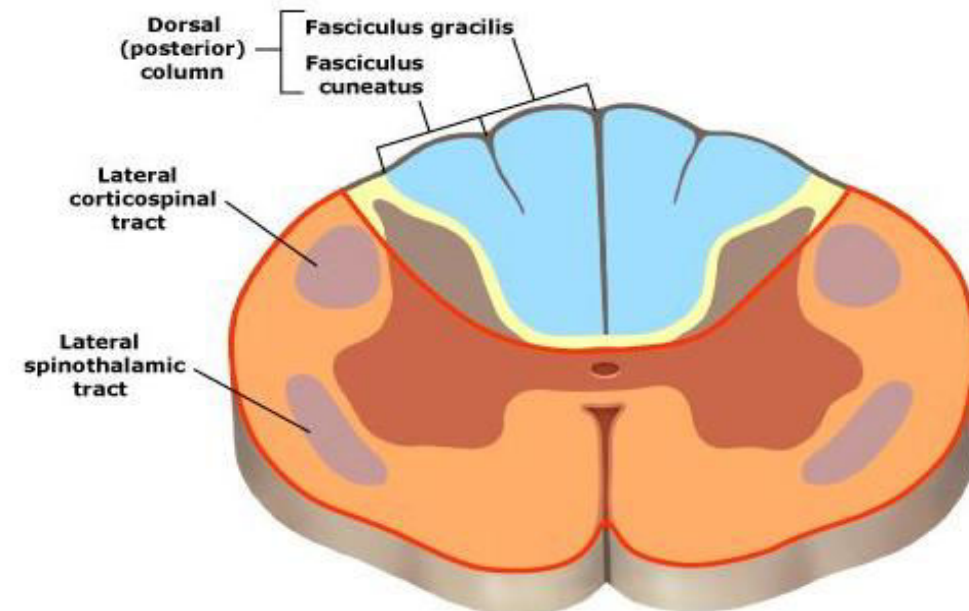


Which functions are lost if damage either to ASA or PSA?

Location of lesion in dorsal cord syndrome



Location of lesion in ventral cord syndrome



Spinal cord syndromes

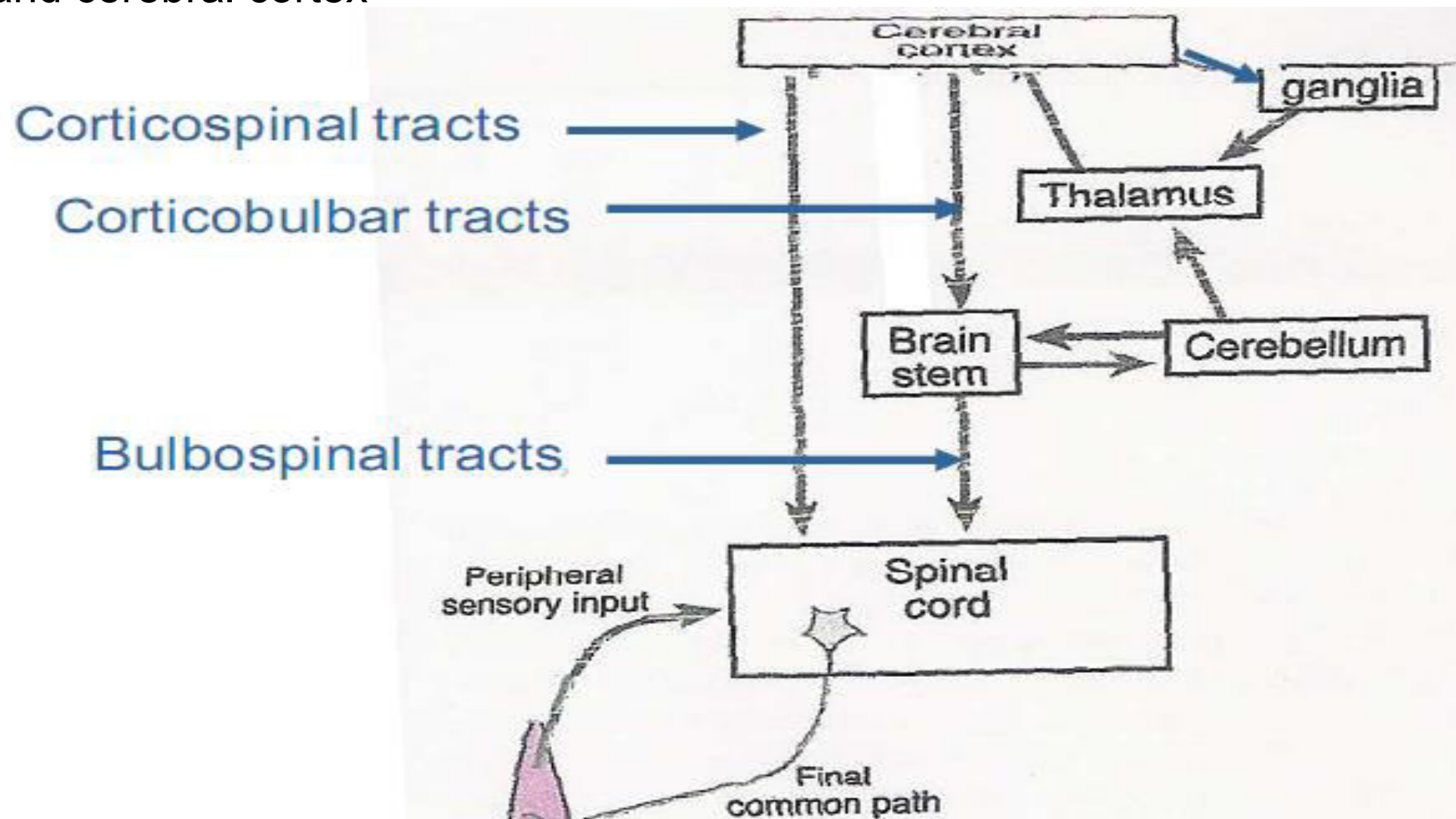
Syndrome	Clinical manifestations
Segmental (transection) syndrome	Loss of all sensory modalities, weakness below affected level; bladder dysfunction
Dorsal cord syndrome	Loss of proprioception, vibratory sensation; variable weakness and bladder dysfunction
Ventral cord syndrome (anterior spinal artery syndrome)	Loss of pain and temperature sensation, weakness, bladder dysfunction
Brown Sequard syndrome	Ipsilateral weakness and loss of proprioception; contralateral loss of pain and temperature sensation
Central cord syndrome	Segmental loss of pain and temperature, weakness often greater in the arms than legs

MOTOR FUNCTIONS OF THE CNS

- **General Design of the Nervous System**
- NS contains more than 100 billion neurons
- functional division of the nervous system:
 - sensory,
 - motor, and
 - integrative

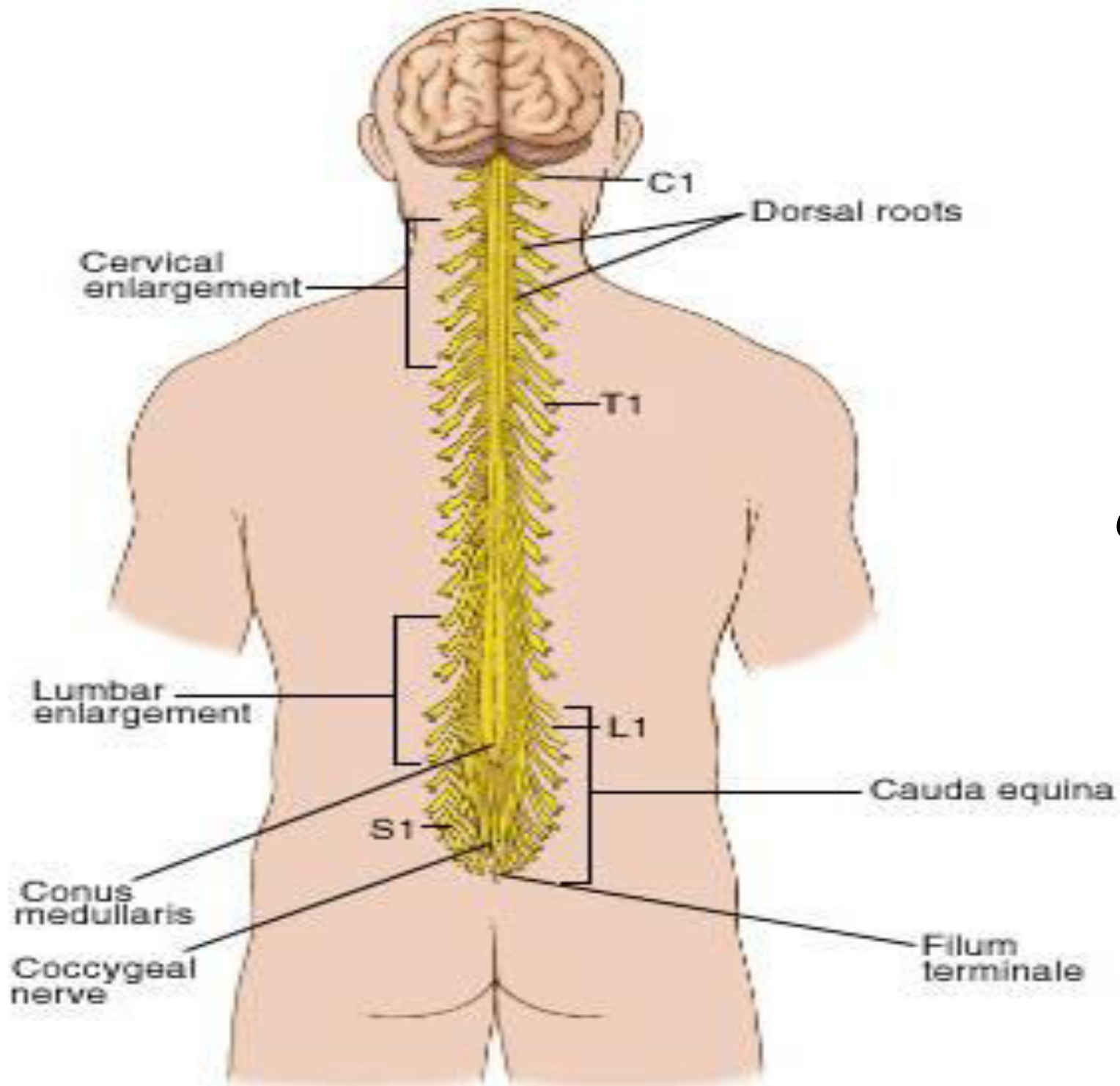
WHAT DO WE MEAN BY MOTOR?

- The word motor means movement
- Types of movement: Voluntary vs. involuntary
- **Motor functions of the CNS:**
 - ✓ is the integrated activity of spinal cord, brain stem, Cerebellum, Basal ganglia and cerebral cortex



Structure of The Spinal Cord

- The spinal cord is extremely important to the overall function of the NS
- is communication link b/n brain & peripheral NS inferior to the head,
 - integrate incoming information & producing responses through reflex.
- The spinal cord extends from **foramen magnum** to the level of **L2 (or end of L1)** where it tapers to a point called conus medullaris.
 - **Lumbar puncture: usually between L3 and L4**
- Arising from the conus medullaris is a bundle of nerve roots called **cauda equina**.
 - cauda equina is a continuation of nerve roots in the lumbar and sacral region
 - Cauda equina syndrome(CES) **occurs when there is dysfunction of multiple lumbar and sacral nerve roots**
 - occurs when the bundle of nerves below the end of the spinal cord are damaged eg. Disc herniation
- Spinal cord is considerably shorter than the vertebral column
 - because it does not grow as rapidly as vertebral column during development.



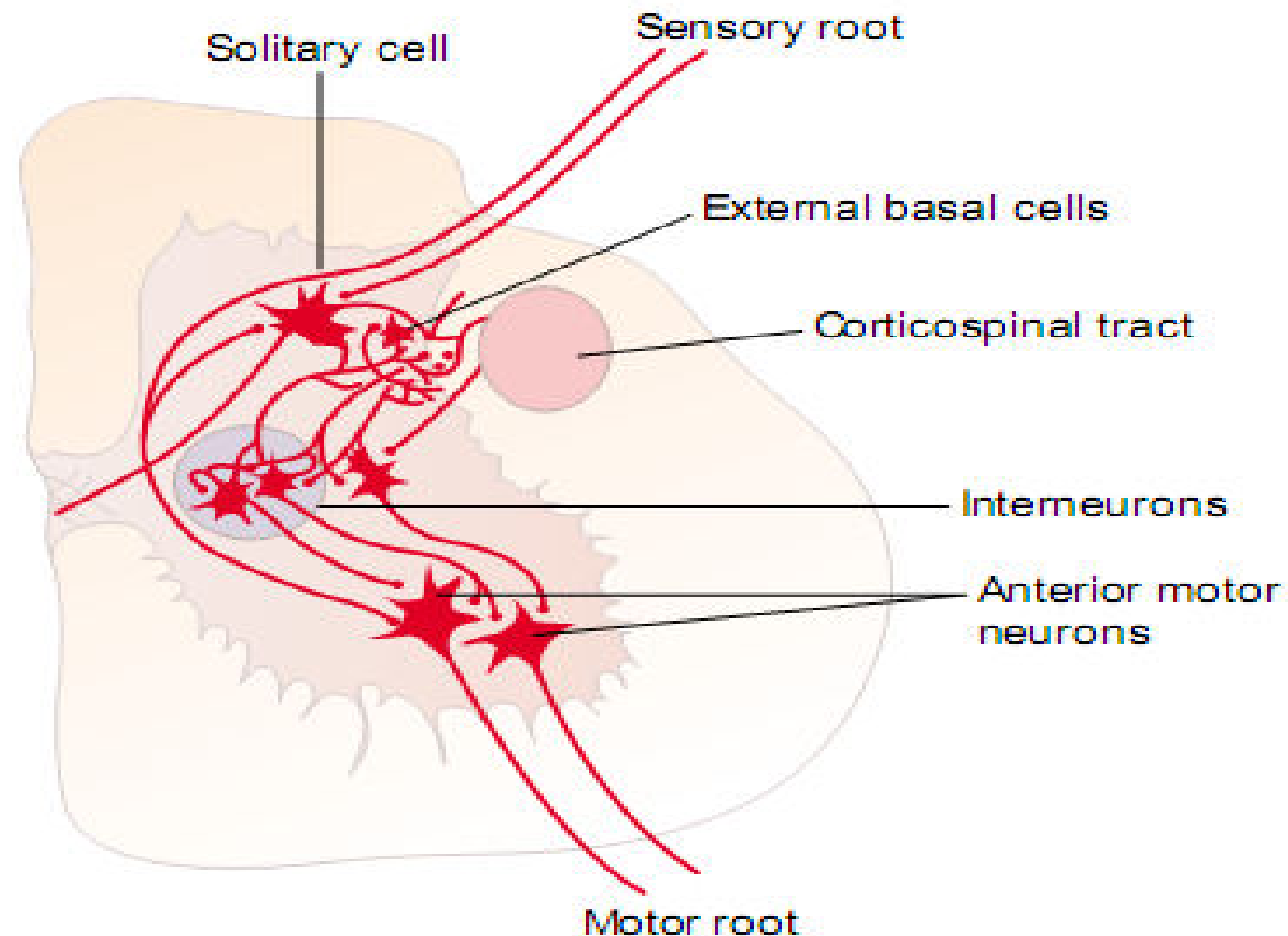
Contains 31 pairs of spinal nerves

- ✓ ***8-pairs of cervical***
- ✓ ***12-pairs of thoracic***
- ✓ ***5-pairs of lumbar***
- ✓ ***5-pairs of sacral***
- ✓ ***1-pairs of coccygeal***

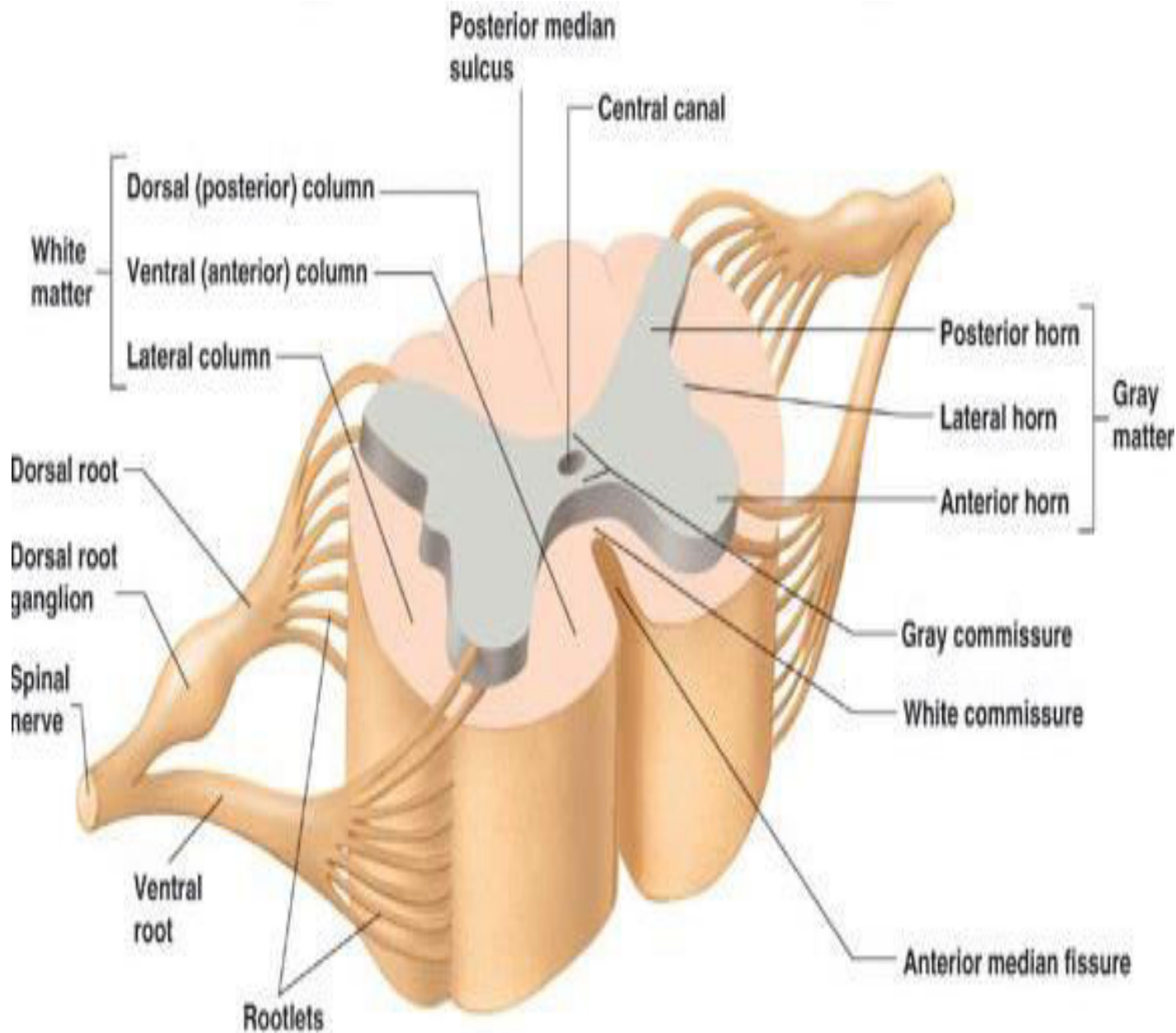
Fig. Anatomy of the Spinal Cord

- Like brain, the spinal cord consists of gray matter and white matter.
- The **gray matter** (**cell bodies & synapses**) of the cord is located centrally & is surrounded by white matter (**myelinated axons**).
- The cord gray matter is the **integrative area for the cord reflexes**

Fig. typical organization of the cord gray matter in a single cord segment: Connections of peripheral sensory fibers and corticospinal fibers with the interneurons and anterior motor neurons of the spinal cord.



Cross-sectional Anatomy of the Spinal Cord



(a) Anterolateral view

Fig. Cross-Section of the Spinal Cord

➤ **Gray matter**

- ✓ Contains neuron cell bodies, dendrites, synapse
- ✓ Divided into horns

➤ **White matter**

- ✓ contains myelinated axons
- ✓ Divided into columns (funiculi)
- ✓ Consists of ascending & descending sensory or motor tracts

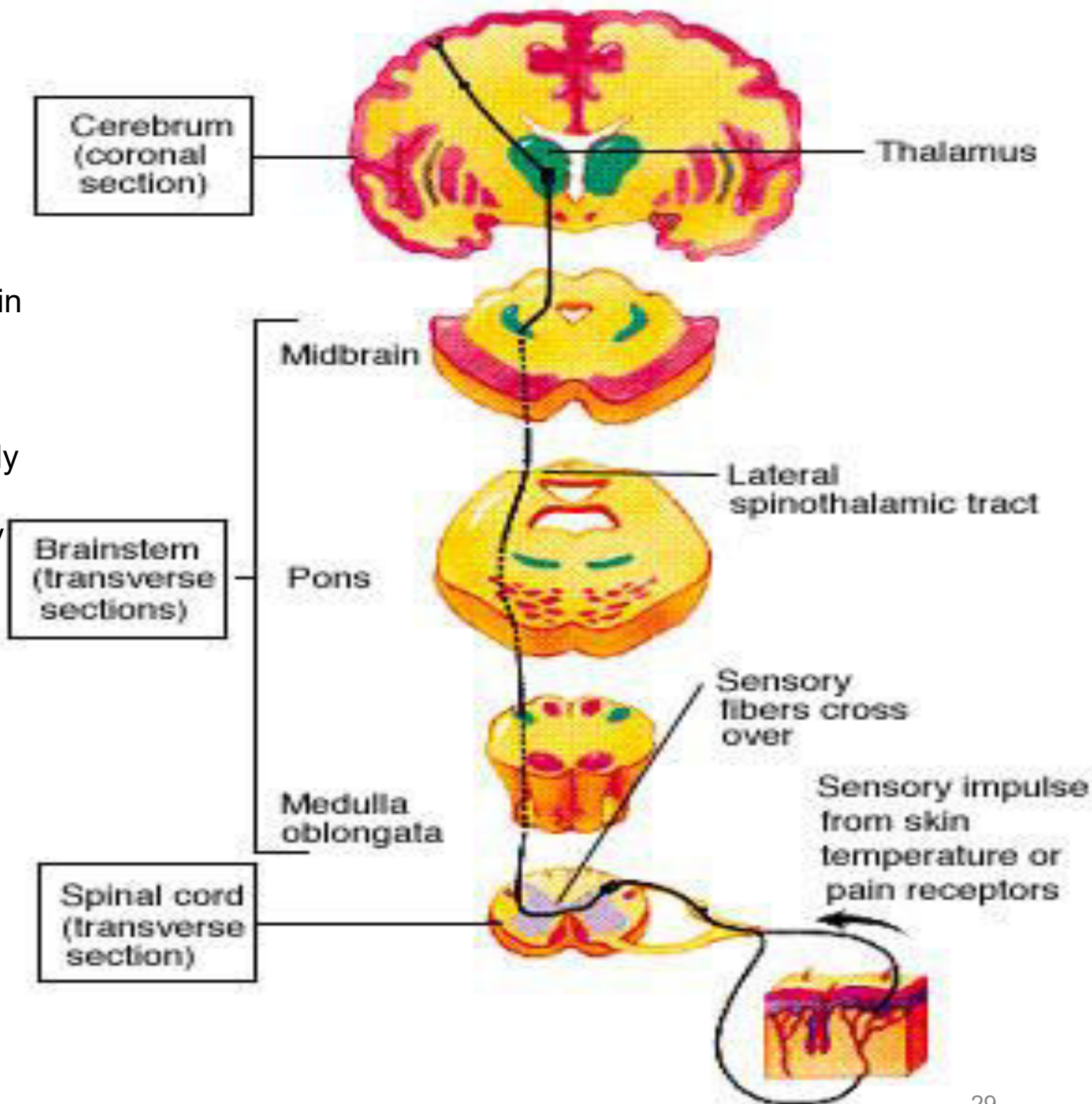
- **Functions of the spinal cord:** three major functions:
 - conducting information to and from the brain.
 - serving as a center for spinal reflexes; and
 - maintaining repetitive, coordinated skeletal muscle contractions;
- The nerve tracts of the cord provide a 2-way communication system b/n the brain & body parts outside the NS
 - **Ascending tracts-** carry sensory information to the brain
 - **Descending tracts-** conduct motor impulses to the muscles & glands
- Many of the fibers exhibit **decussation**- cross from one side of the body to the other **within the spinal cord or the medulla oblongata**.
 - Anterolateral pathways decussate @spinal cord, dorsal pathways @medulla
 - Those that do not cross(dorsal pathways) remain on the ipsilateral side of the body

Most sensory impulses (anterolateral) originating in skin receptors cross over in the spinal cord & ascend to the brain



As a result, each side of the brain mostly receives sensory information from the contralateral (opposite) side of the body

Eg. Pain, temp, crude touch



Motor Functions of the Spinal Cord: THE CORD REFLEXES

- ❖ Sensory signals enter the cord almost entirely through the sensory roots, aka the *posterior* or *dorsal roots*.

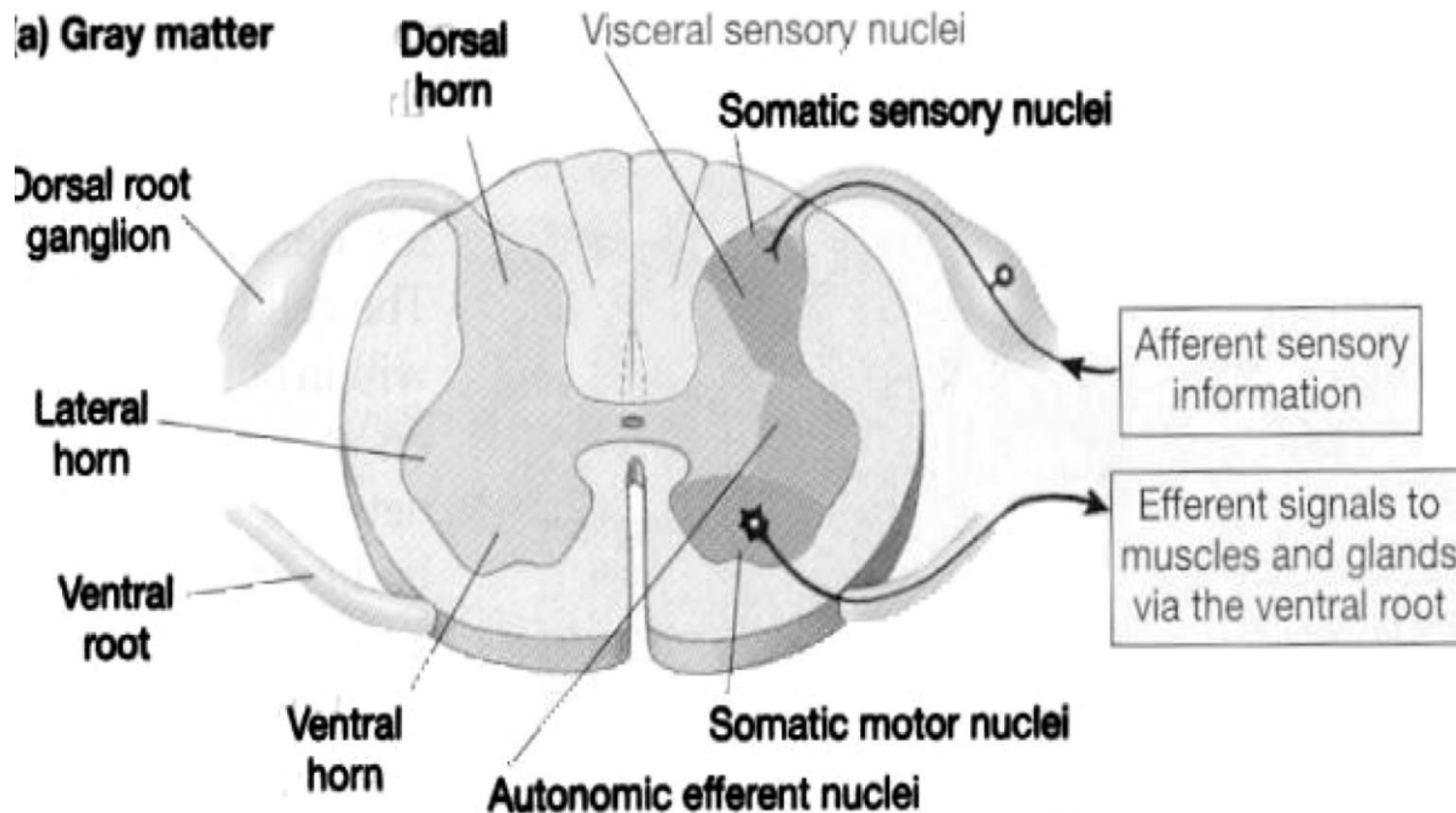


Fig. Organization of the Spinal Cord for Motor Functions

- After entering the cord, every sensory signal travels to two destinations :
 - 1. immediately in the gray matter of the cord and elicits local reflexes
 - The cord gray matter is the integrative area for the cord reflexes
 - ↓
 - **elicits local segmental cord reflexes**
 - 2. to higher levels of the nervous system - the cord itself, to the brain stem, cortex

Anterior Motor Neurons

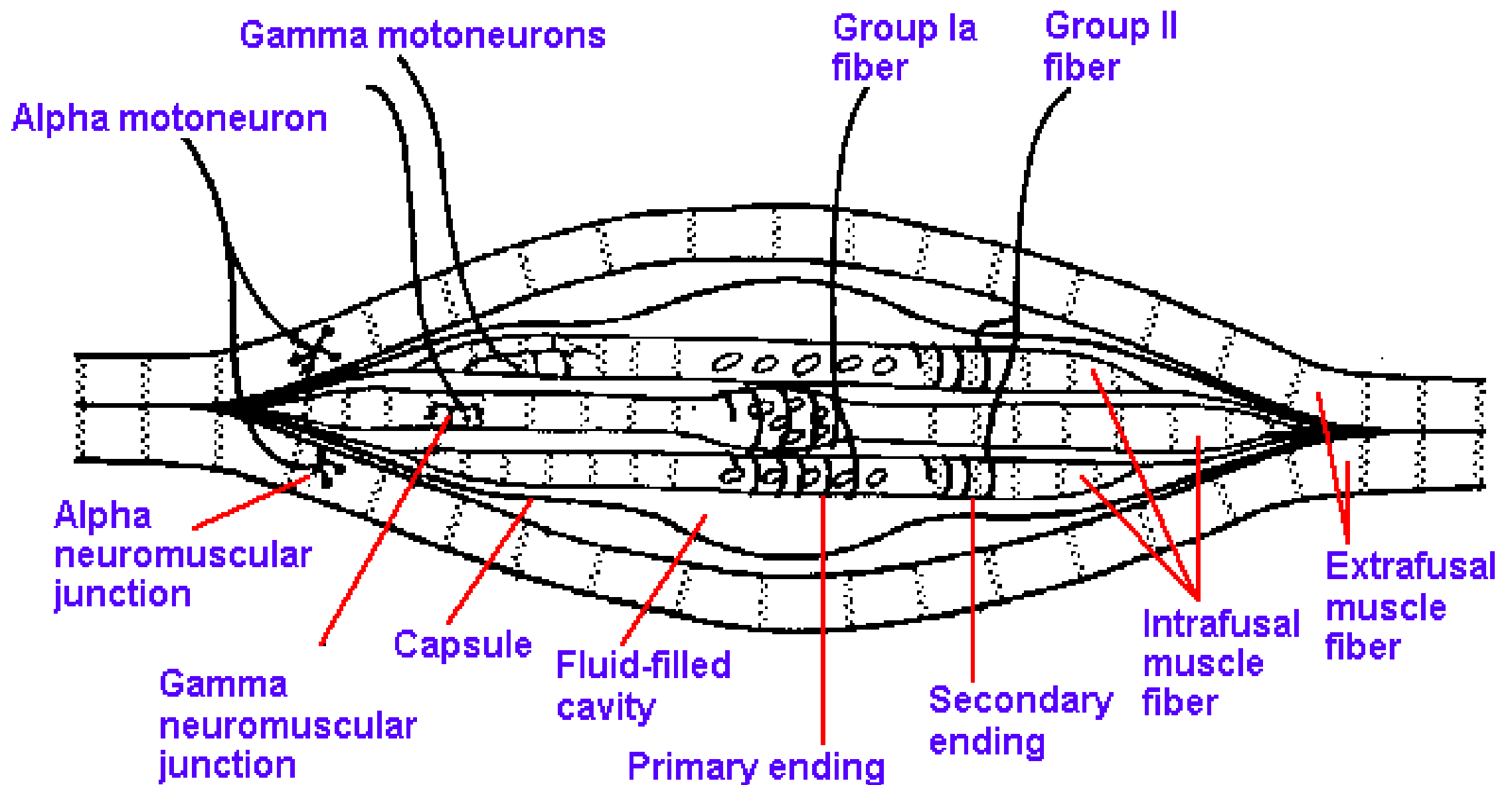
- Located in each segment of the anterior horns of the cord gray matter
- They give rise to the nerve fibers that leave the cord by way of the anterior roots and directly innervate the skeletal muscle fibers
- The neurons are of two types:
 - **alpha (α) motor neurons** and
 - **gamma (γ) motor neurons.**

a. α motoneurones:

- Give rise to large type A alpha ($A\alpha$) motor nerve fibers (14 μm in diameter)
- innervate the **extrafusal skeletal muscle, responsible for** generating skeletal movement
 - ✓ **alpha motor neurons control the contractile function of the skeletal muscles**
- Stimulation of a single alpha fiber excites anywhere from 3-100s skeletal muscle fibers,
 - ✓ These muscle fibers together with the motor neuron are collectively called **the motor unit**

b. γ motoneurones:

- Transmit impulses through much smaller $A\gamma$ motor nerve fibers (5 μm in diameter)
- Innervate the **intrafusal skeletal Muscle** (small skeletal muscle fiber)
- intrafusal fibers constitute the poles of the muscle spindle,
 - ✓ help to control basic muscle “tone”



Muscle spindle, showing its relation to the large extrafusal skeletal muscle fibers. Note also both motor and sensory innervation of the muscle spindle.

motor unit

- **The term motor unit** describes motor neuron + muscles supplied
 - a motor neuron, its neuromuscular junction at each axon branch, and
 - all of the extrafusal muscle fibers innervated by that motor neuron
- When a motor neuron is activated, all of its muscle fibers are also activated.

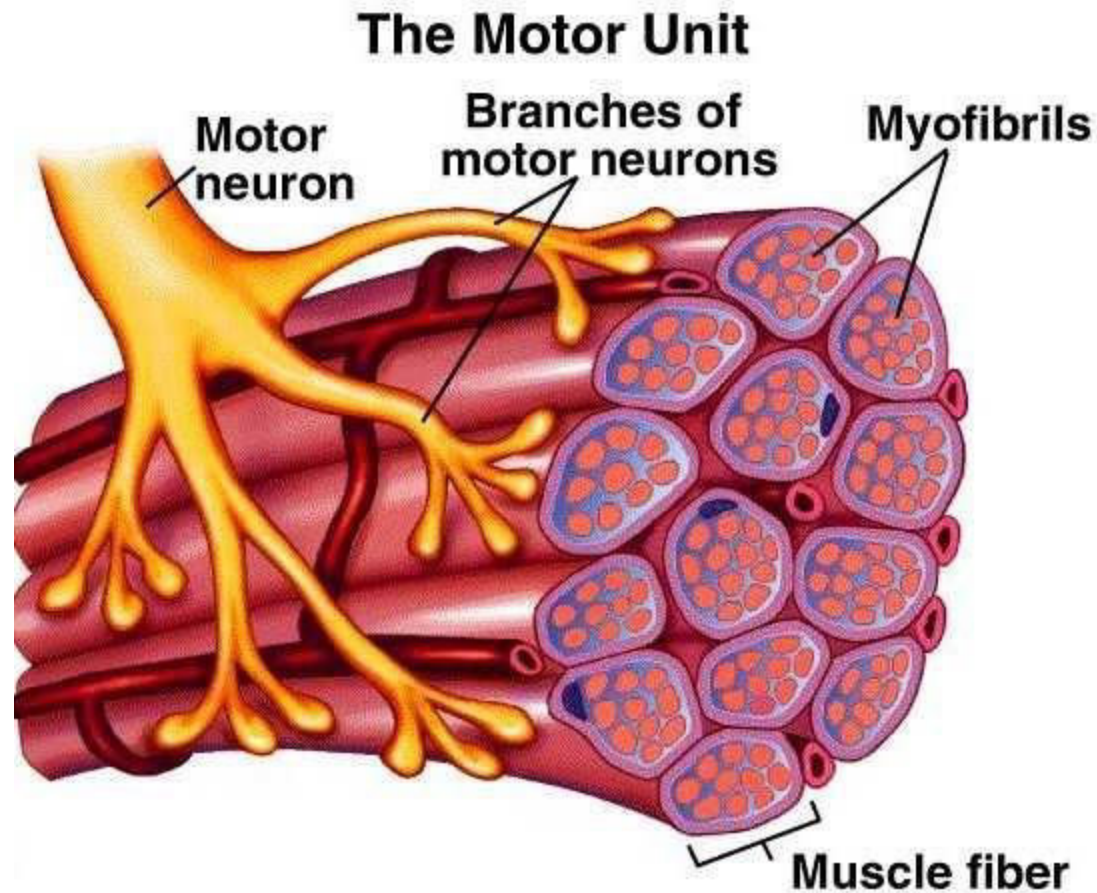


Fig. alpha motor neuron and the group of extrafusal muscle fibers it innervates.

Interneurons

- Diffused in the spinal cord are 30x as numerous as the anterior motoneurons
- found in all areas of the cord gray matter:
 - ✓ in the dorsal horns, the anterior horns, and the intermediate areas between them
- They are small and highly excitable
- Almost all signals from brain and peripheral sensory signals are transmitted first through interneurons, where they are appropriately processed!!!!
- The interconnections among the interneurons and anterior motor neurons are responsible for most of the integrative functions of the cord

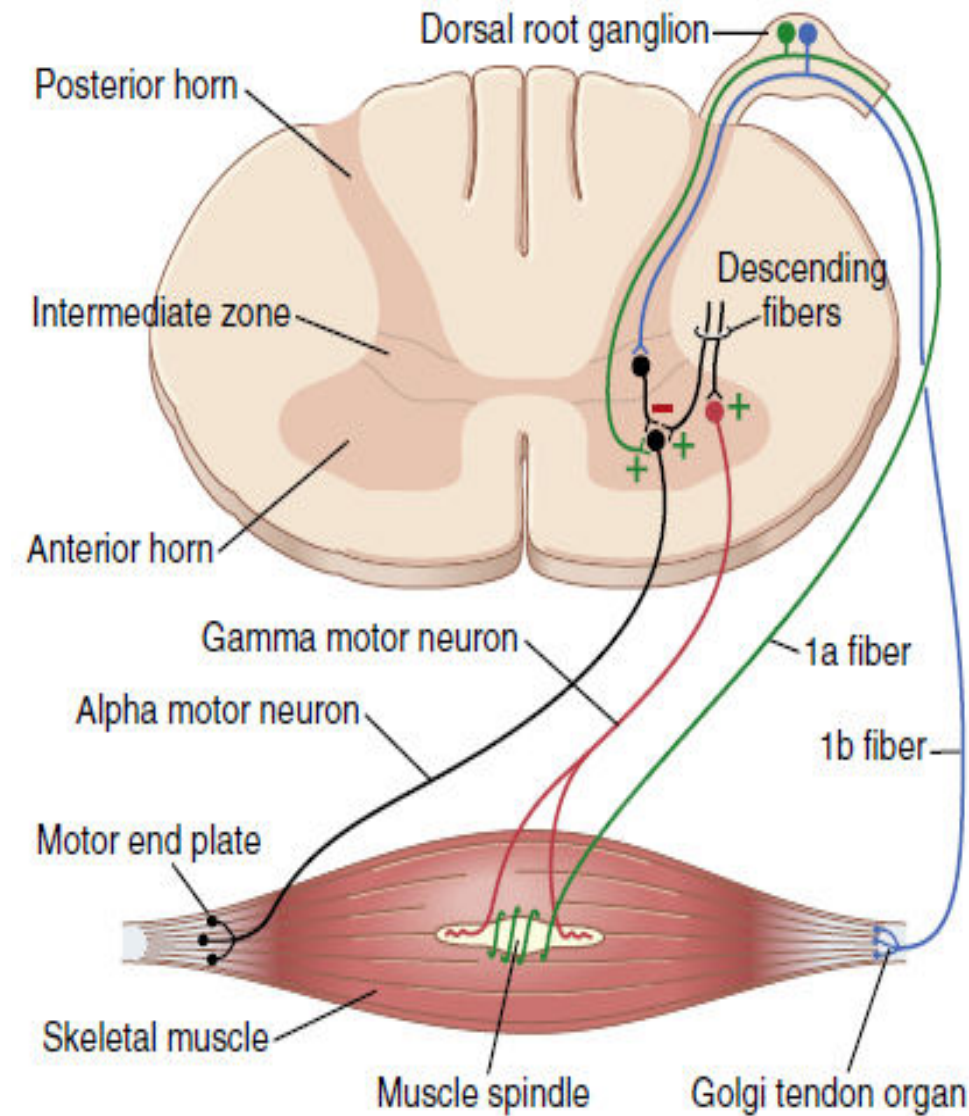
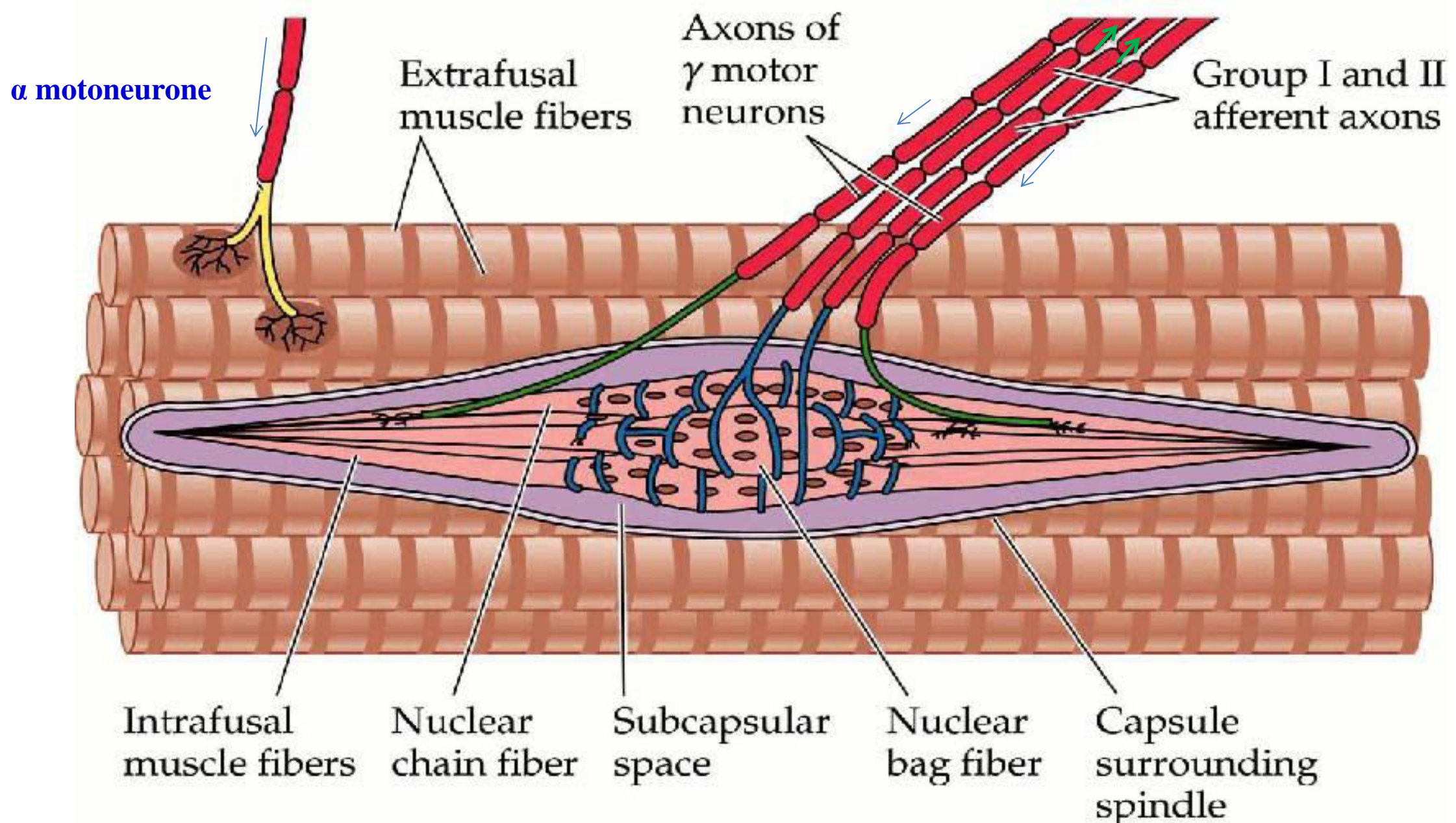


Figure 55-2. Peripheral sensory fibers and anterior motor neurons innervating skeletal muscle.

The Muscle Receptors: muscle Spindle and golgi tendon organ

1. The Spindle: located deep within the muscle mass, attached to tendon

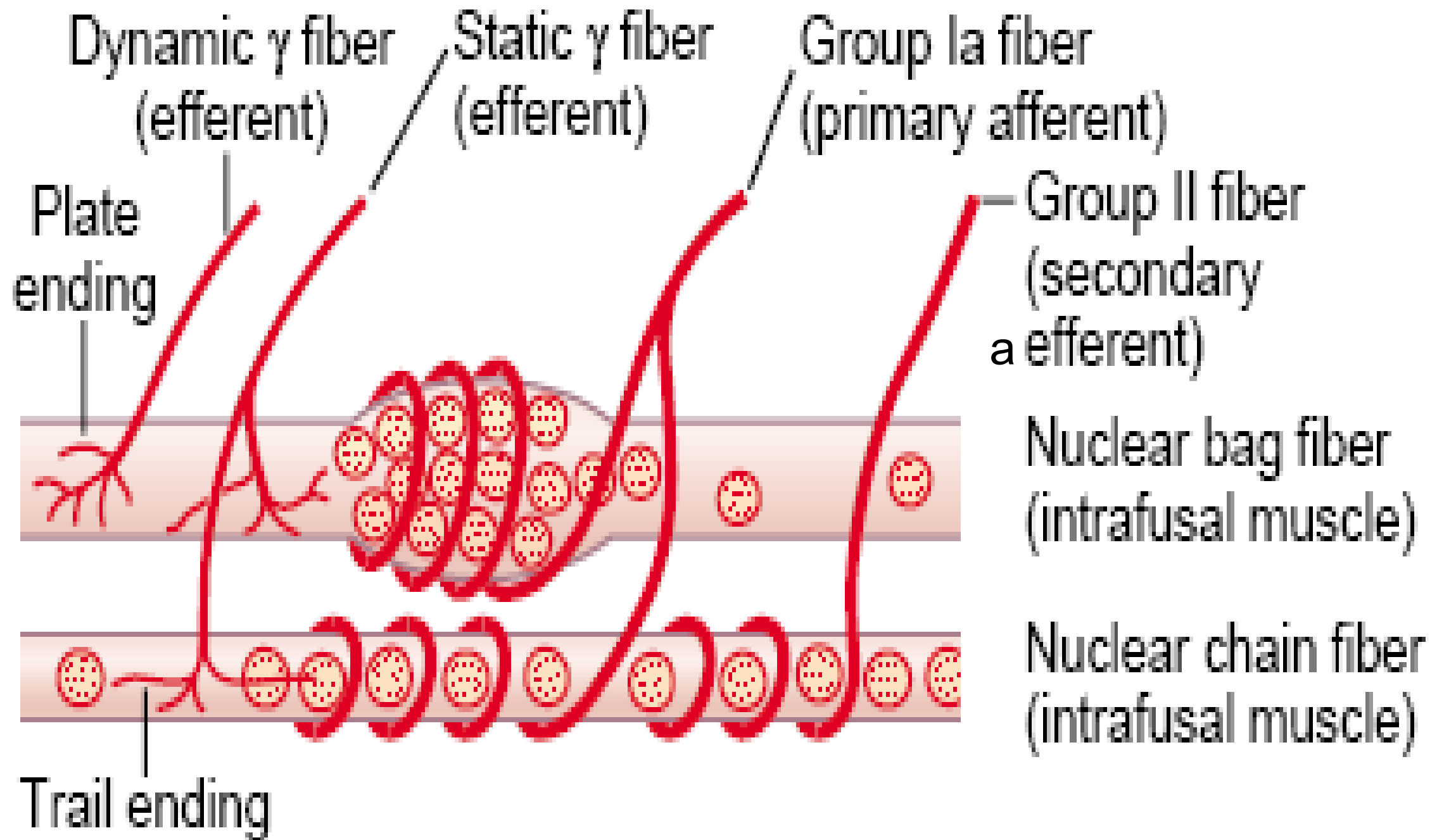
- send information to the CNS about **muscle length & rate of change of length**
- Each muscle spindle has **three** essential elements:
 - 1. **A group of specialized intrafusal muscle fibers** with contractile polar ends and a noncontractile center,
 - 2. **large diameter myelinated afferent nerves (types Ia and II)**
 - Originate from the noncontractile **central portion** of the intrafusal fibers
 - send information to the CNS about **muscle length & rate of change of length**
 - 3. **small diameter myelinated efferent nerves (Gamma motor neurons)**
 - supply the **polar contractile** regions of the intrafusal fibers



- ❖ Intrafusal muscle fibers are walled off from the rest of the muscle by an outer connective tissue sheath.
- ❖ This sheath has a spindle or "fusiform" shape, hence the name spindle and "intrafusal"
- ❖ Both ends of the intrafusal fibers contract, but the central region only stretches and does not contract.

Types of intrafusal muscle fibers

- **are** skeletal muscle fibers that serve as specialized sensory organs (proprioceptors).
 - ✓ They detect the amount and rate of change in length of a muscle
 - ✓ are innervated by both sensory (afferent) and motor (efferent) fibers
- **The muscle spindle contains two types of intrafusal muscle fibers:-**
 - ❖ **1. *Nuclear bag fibers* :** 1 -3 fibers per spindle
 - ✓ are thicker and longer than the nuclear chain fibers
 - ✓ **a large number of muscle fiber nuclei** are congregated in expanded “bags” in the central portion of the receptor area
 - ❖ **2. *Nuclear chain fibers*:** 3 -7 fibers in each spindle
 - ✓ are about half in diameter and **length than NBF**
 - ✓ **have nuclei** aligned in a chain throughout the receptor area



Sensory Innervation of the Muscle Spindle

- The receptor portion of the muscle spindle is its central portion
 - In this area, intrafusal fibers have no contractile elements: no myosin & actin
- **Two types of sensory endings in the central receptor area of the spindle:**
 1. **primary ending (annulospiral ending)**
 - ✓ is a **type Ia fiber** averaging 17 μm in diameter,
 - ✓ transmits sensory signals to spinal cord at a velocity of 70 -120 m/sec
 - ✓ excited by both the **nuclear bag fibers & the nuclear chain fibers**.
 - ✓ only one group Ia afferent fiber per spindle
 2. **secondary ending (Flower spray endings)- group II afferent fibers ,**
 - ✓ Are **type II nerve fibers** with an average diameter of 8 μm (30-60m/s)
 - ✓ usually **excited only by nuclear chain fibers**.

- **The muscle spindle receptors can be excited in two ways:**
 - **1. Lengthening the whole muscle** stretches the midportion and, therefore, excites the receptor.
 - **2. contraction of the end portions** of the spindle's intrafusal fibers
 - ✓ contraction of the polar ends of the intrafusal fibers stretches the midportion
 - ✓ this contraction is stimulated by gamma motoneurons
 - ✓ Occurs even if the length of the entire muscle does not change

Function of Intrafusal muscle fibers

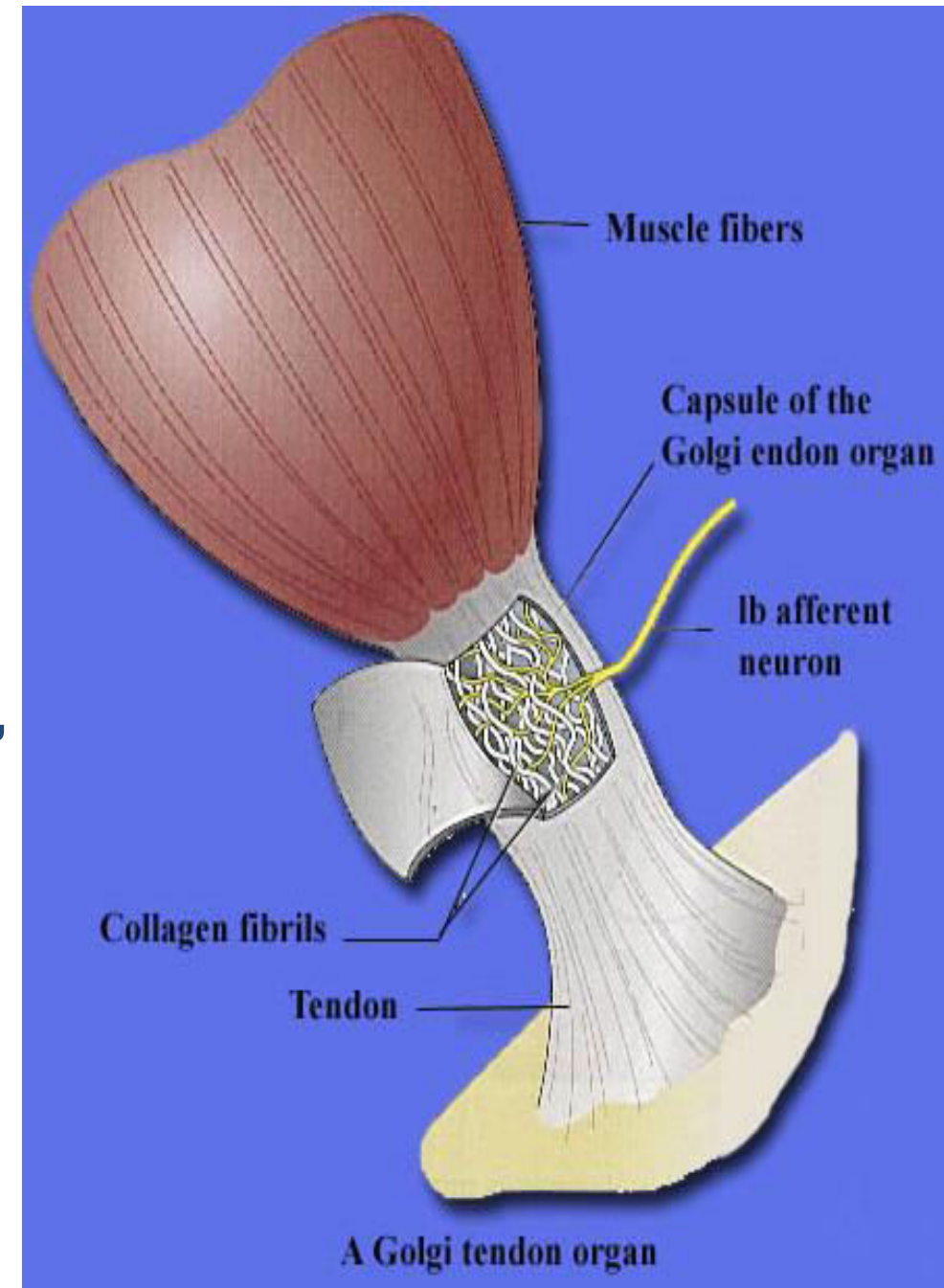
- ✓ To detect the amount and rate of change in muscle length, control muscle movements
- ✓ Position sensation(proprioception)

The Golgi Tendon Organ

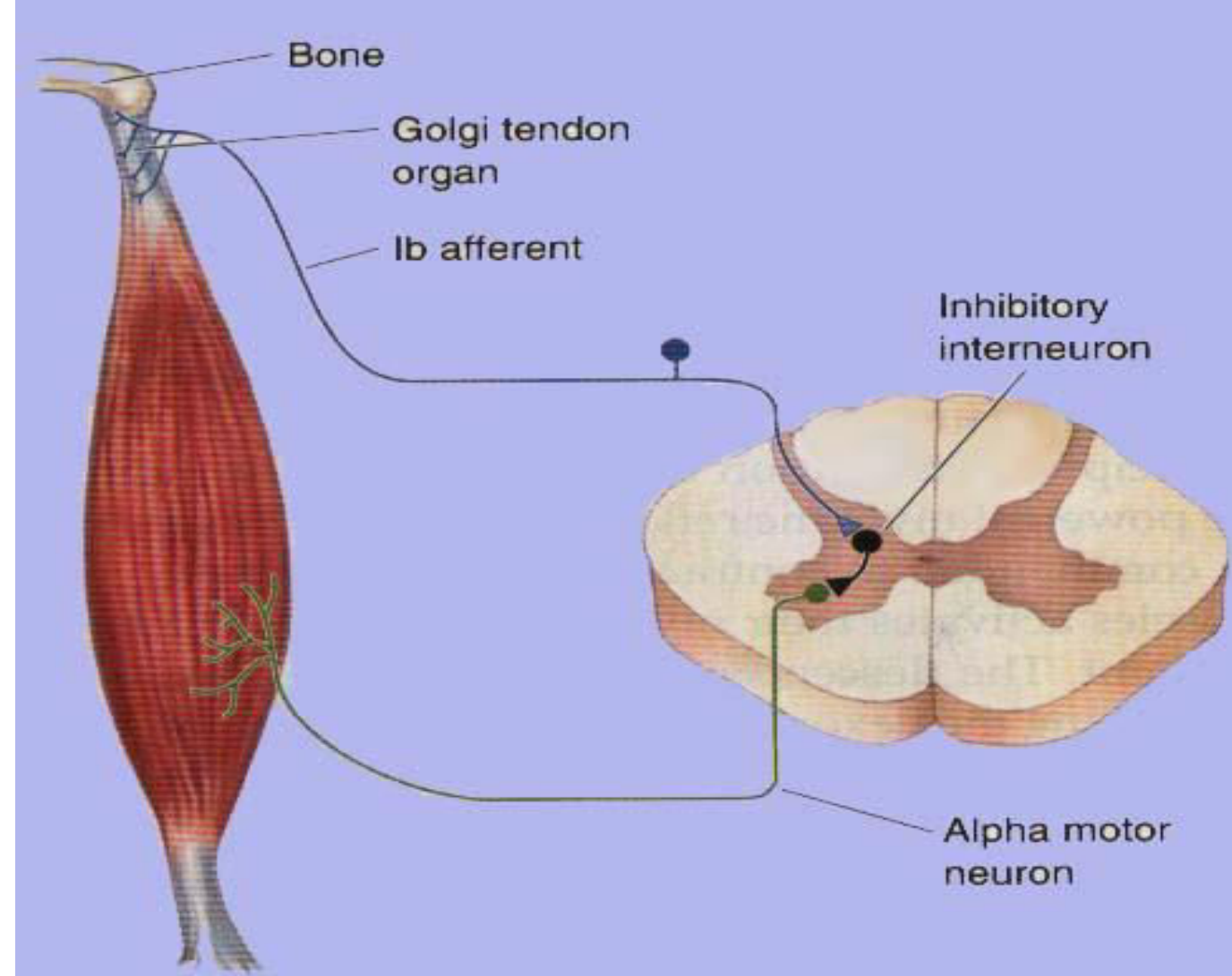
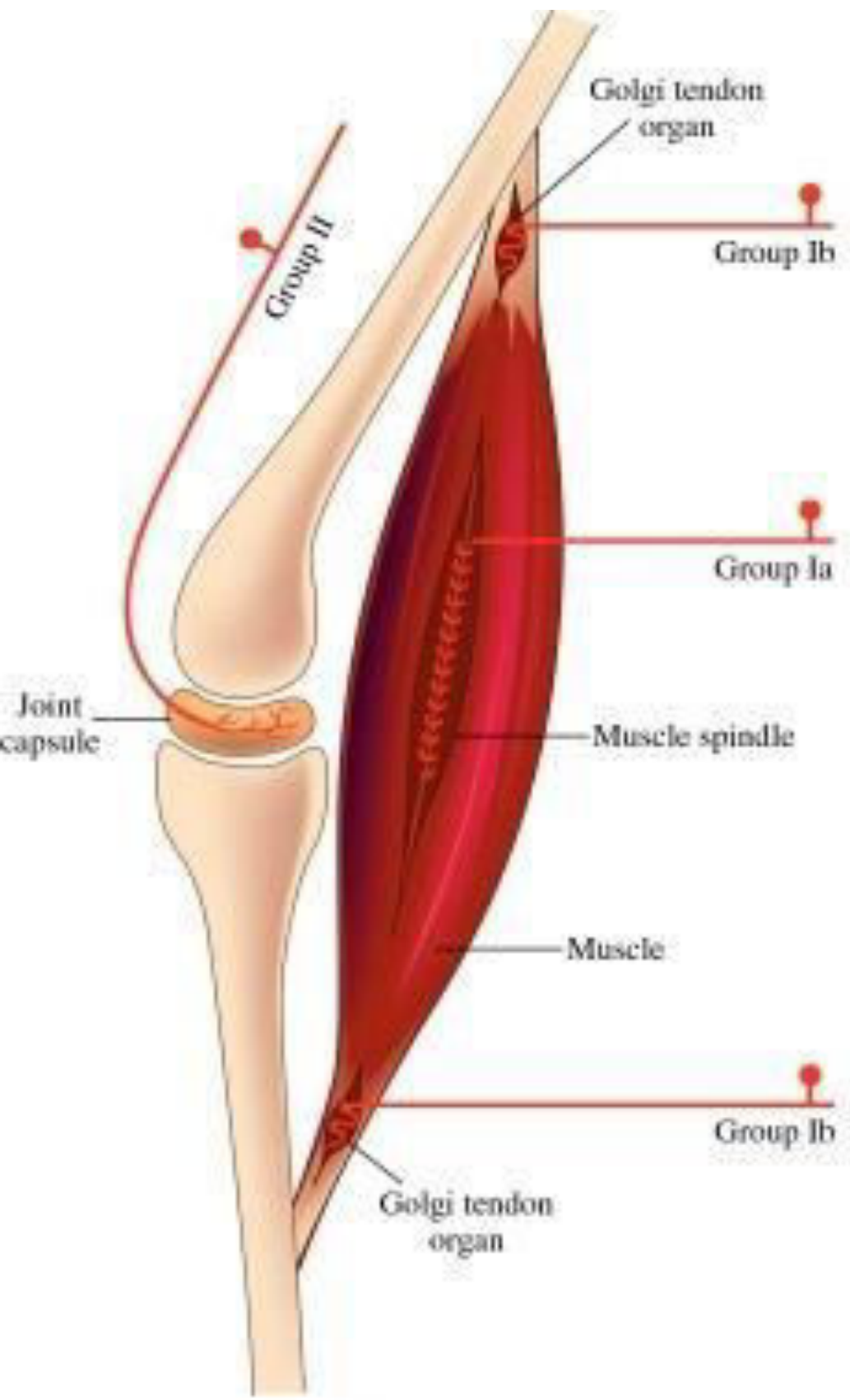
- ❖ located in the muscle tendons
- ❖ transmit information about **tendon tension or rate of change of tension**



- ❖ GTOs provide the NS with instantaneous information on the degree of tension in each muscle.
- ❖ Signals from the GTO are transmitted through **large, rapidly conducting type Ib nerve fibers**
 - ✓ average diameter $16\mu\text{m}$
 - ✓ only slightly smaller than those from 1^o endings of the muscle spindle.



- The major difference between the function of the GTO and the muscle spindle
 - **the spindle detects relative muscle length**
 - **the Golgi tendon organ detects muscle tension**
 - ✓ The GTOs serve to inhibit contractile responses evoked by the muscle spindle
- Signals from GTOs are transmitted via large rapidly conducting type 1b fibers (slightly smaller than 1a fibers):
 - ↓
 - ✓ Excite inhibitory interneurone (Renshaw cell)
 - ↓
 - ✓ These in turn inhibit α -motoneurons to the same muscle $\Rightarrow$ **Protect tearing of muscles due to excessive contraction**
 - ❖ If large tension on the muscle $\Rightarrow$ stimulation of GTO $\Rightarrow$ **sudden relaxation of the entire muscle (lengthening reaction (clasp knife reaction))**
- *Hence*, the threshold of GTO is considerably higher than muscle spindle



Tension receptors in tendons

- ✓ Afferents are type 1b
- ✓ Stimulated by high tension
- ✓ Monitor tension in muscles
- ✓ Protect tearing of muscles

Muscle Stretch reflex

- **Def:** is reflex contraction of skeletal muscle when stretched
 - ✓ is a muscle contraction in response to stretching within the muscle.
 - ✓ Involuntary motor acts, rapidly executed
- *Reflex arc – is the neural pathway for reflex action*
 - ❖ **Receptor** ⇒ afferent neuron ⇒ synapses in the CNS ⇒ efferent neuron ⇒ **Effector**
 - ✓ One synapse : monosynaptic muscle stretch reflex
 - ✓ Polysynaptic or withdrawal reflex : More than one synapse (Have interneuron)

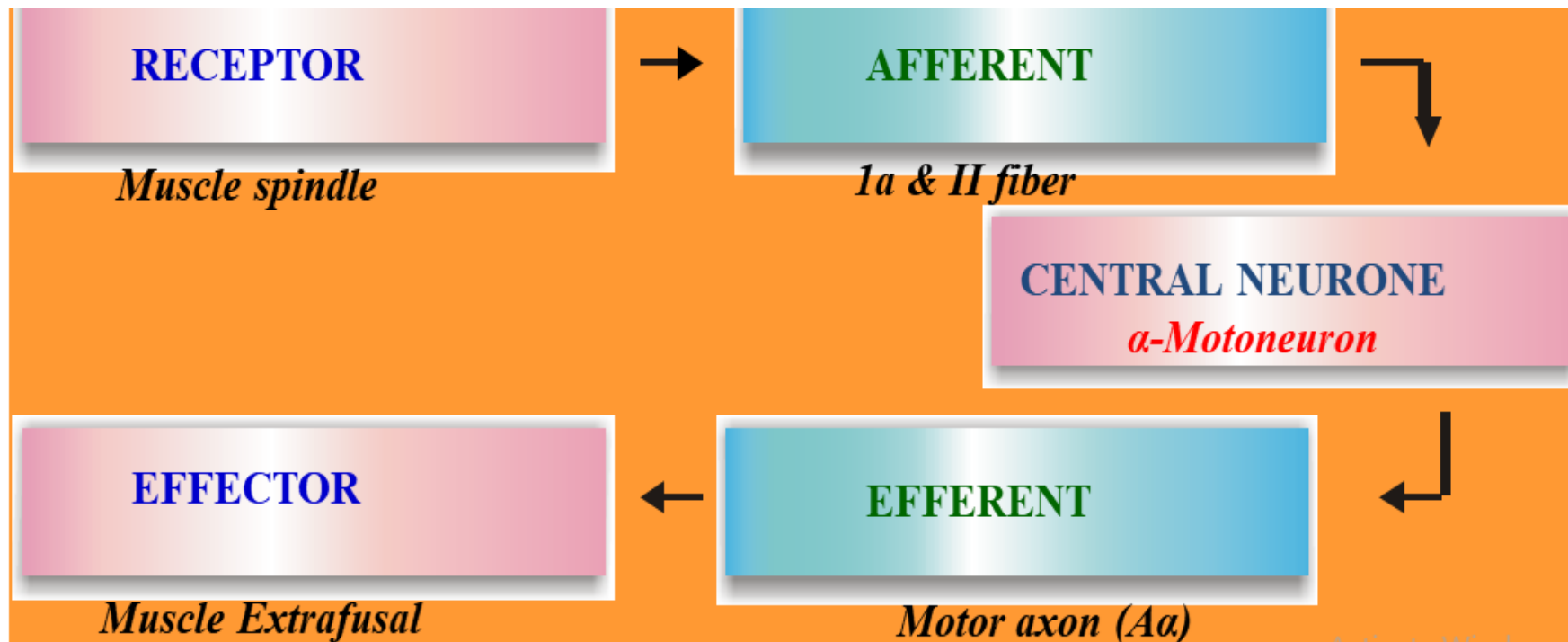


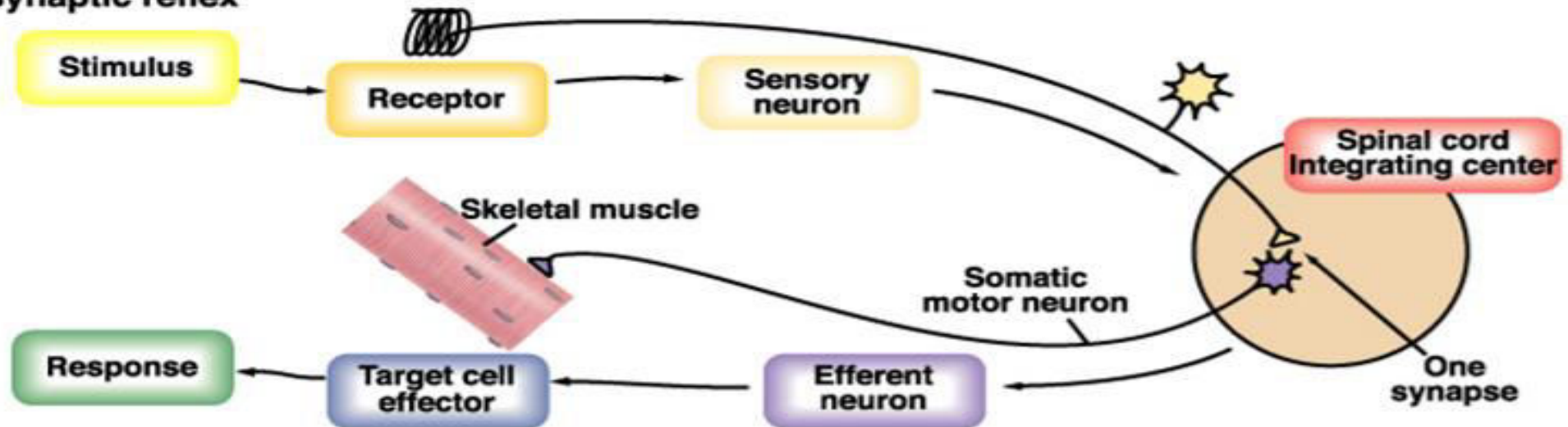
Fig. muscle stretch reflex

Neuronal Circuitry of the Reflex

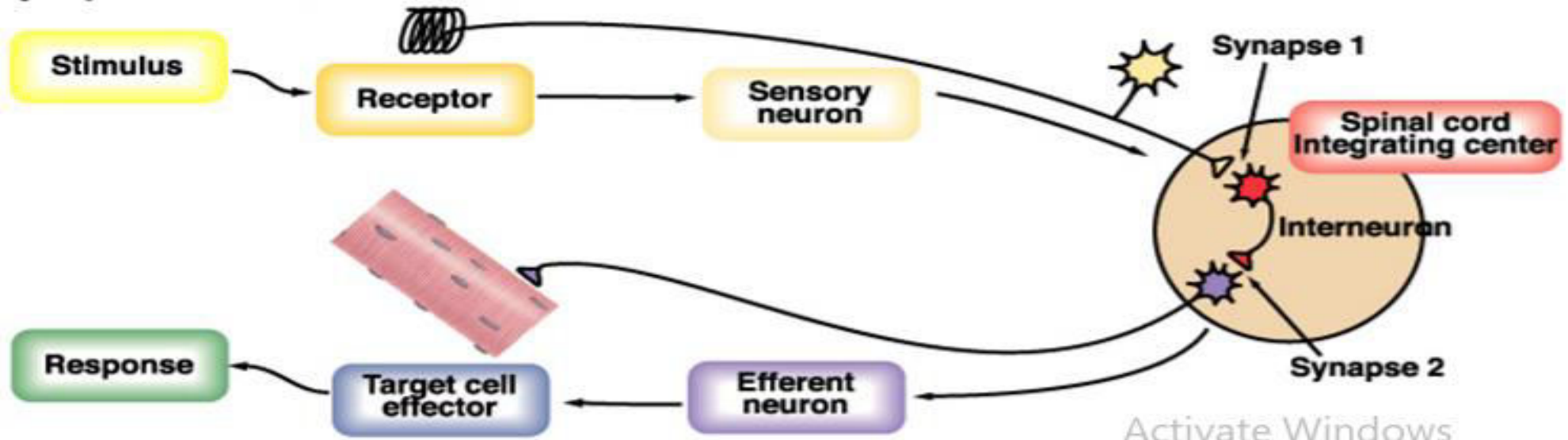
✓ Monosynaptic or stretch reflex

✓ Polysynaptic or withdrawal reflex

(a) Monosynaptic reflex



(b) Polysynaptic somatic motor reflex



Activate Windows

Clinical Applications of Stretch Reflex

- to determine how much background excitation, or “tone,” the brain is sending to the spinal cord
- can be elicited by simply striking the patellar tendon with a reflex hammer;

Mechanism of the knee jerk or patellar reflex:

Strike patellar tendon located just below the knee cap



stretch receptors (intrafusal fibers of the spindle) stimulated as the stimulus stretches the muscle



impulse carried by **Ia afferent fiber** and synapse with **α motoneuron**



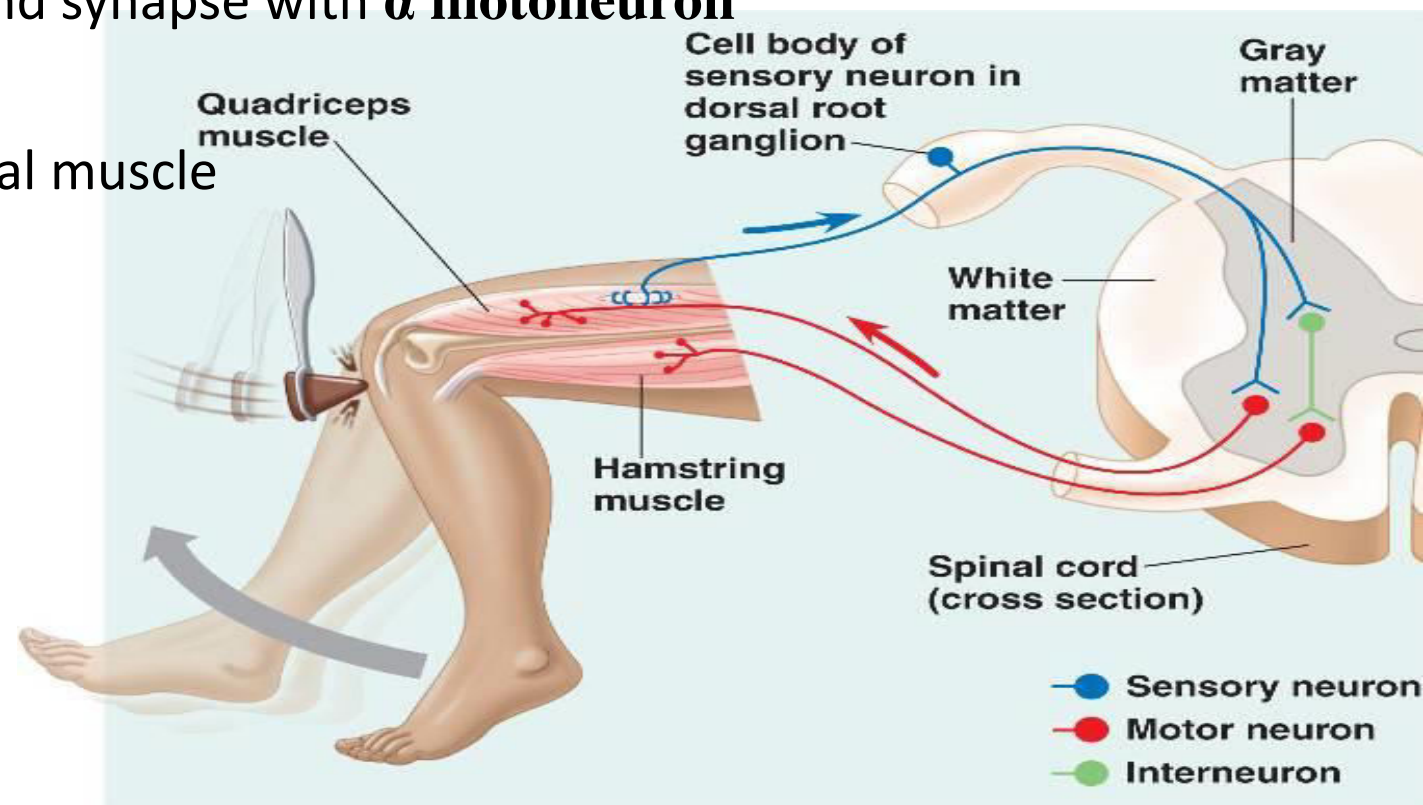
α motoneuron innervates the extrafusal muscle



contraction of quadriceps femoris and
relaxation of biceps femoris



extension of lower leg



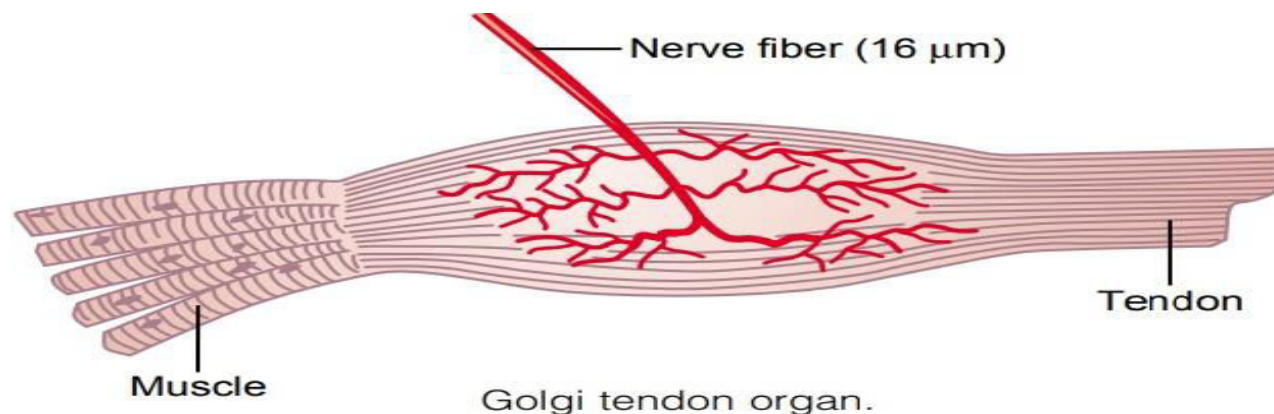
- The muscle jerks are used to assess the neurological abnormalities:
 - Normal: UMNs stimulate the inhibitory inter neurons
 - The interneurons via the inhibitory NTs to inhibit the alpha motor neurons-
 - Normal Reflex-2/4
 - UMN lesion eg. Due to stroke, the muscle jerks are greatly exaggerated, eg- exaggerated reflex :- 3/4 or exaggerated with clonus:- 4/4
 - LMN lesion eg. GBS, trauma etc affecting the motor neurons, reflexes are depressed or absent (0/4, or 1/4)

Grading of stretch reflexes upon tapping muscle tendon^[8]

Grade	Response	Significance
0	no response (areflexia)	always abnormal
1+	slight but definitely present response	may or may not be normal
2+	brisk physiologic response	normal
3+	very brisk response	may or may not be normal
4+	clonus (Brisk with clonus)	always abnormal

Golgi Tendon Reflex: It is also called a *lengthening reaction*

- **Golgi Tendon Organ - Helps to Control Muscle Tension**
- It is an encapsulated sensory receptor through which muscle tendon fibers pass
- Each GTO connects about **10 to 15 muscle fibers**
- stimulated by excessive **tension over the attached muscle fibers** $\Rightarrow \Rightarrow$ inhibition of alpha motor neurons to the muscle $\Rightarrow$ relaxation
 - This relaxation in response to strong stretch is also called the **inverse stretch reflex** or **autogenic inhibition**
- It is a protective mechanism
 - prevents the development of too much tension on the muscle
 - prevents tearing of the muscle or avulsion of the tendon from its attachments to the bone.



Withdrawal reflex Or Flexor Reflex

- Noxious stimuli to limb → flexor muscles of limb contract with relaxation of extensor muscles → withdrawal of limb from stimuli
- It is called withdrawal reflex or **nociceptive reflex, or pain reflex**
- it is especially applicable to the limbs because of their highly developed flexor reflexes
- If other part of the body is painfully stimulated, similarly be *withdrawn from the stimulus*

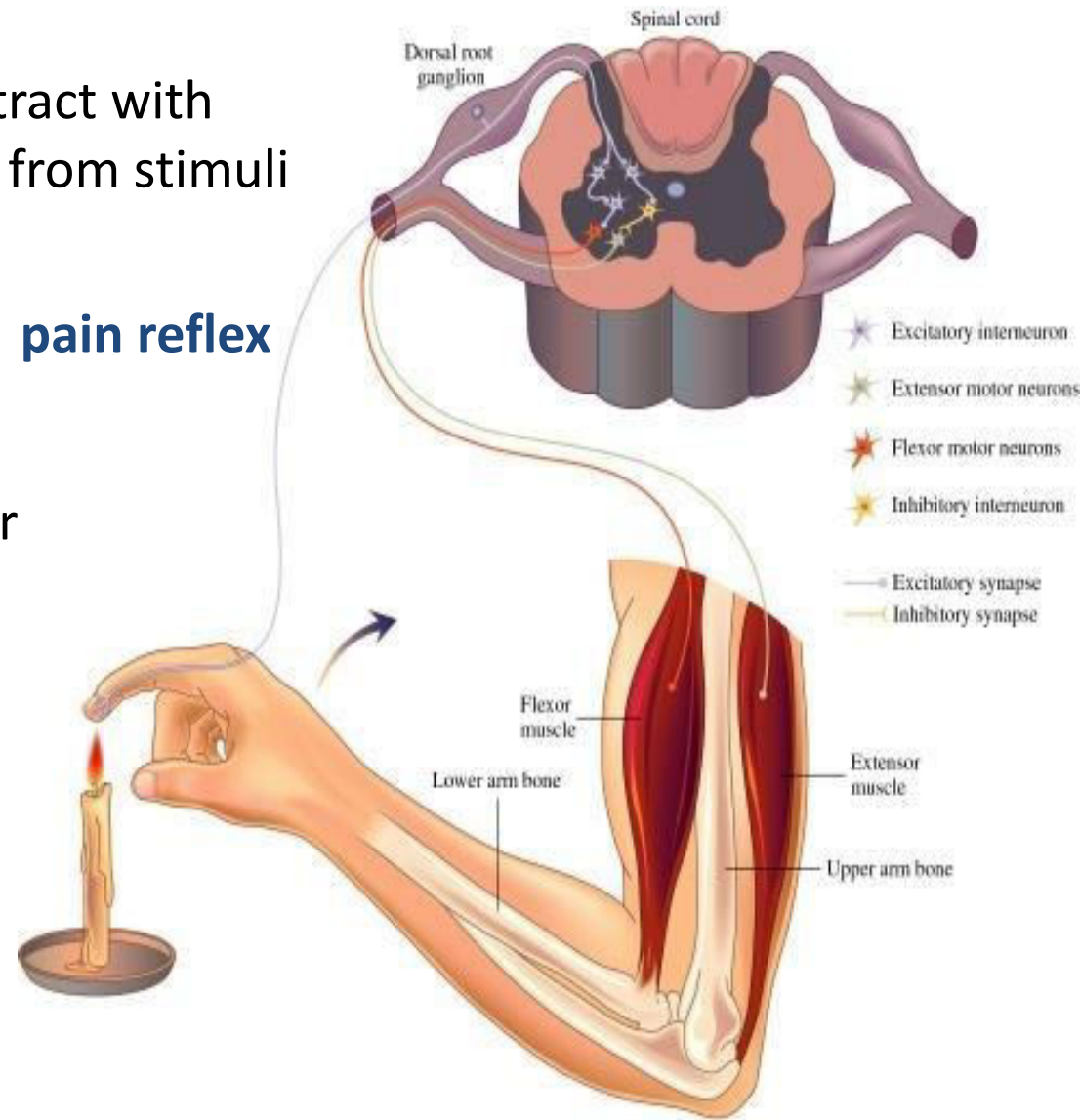


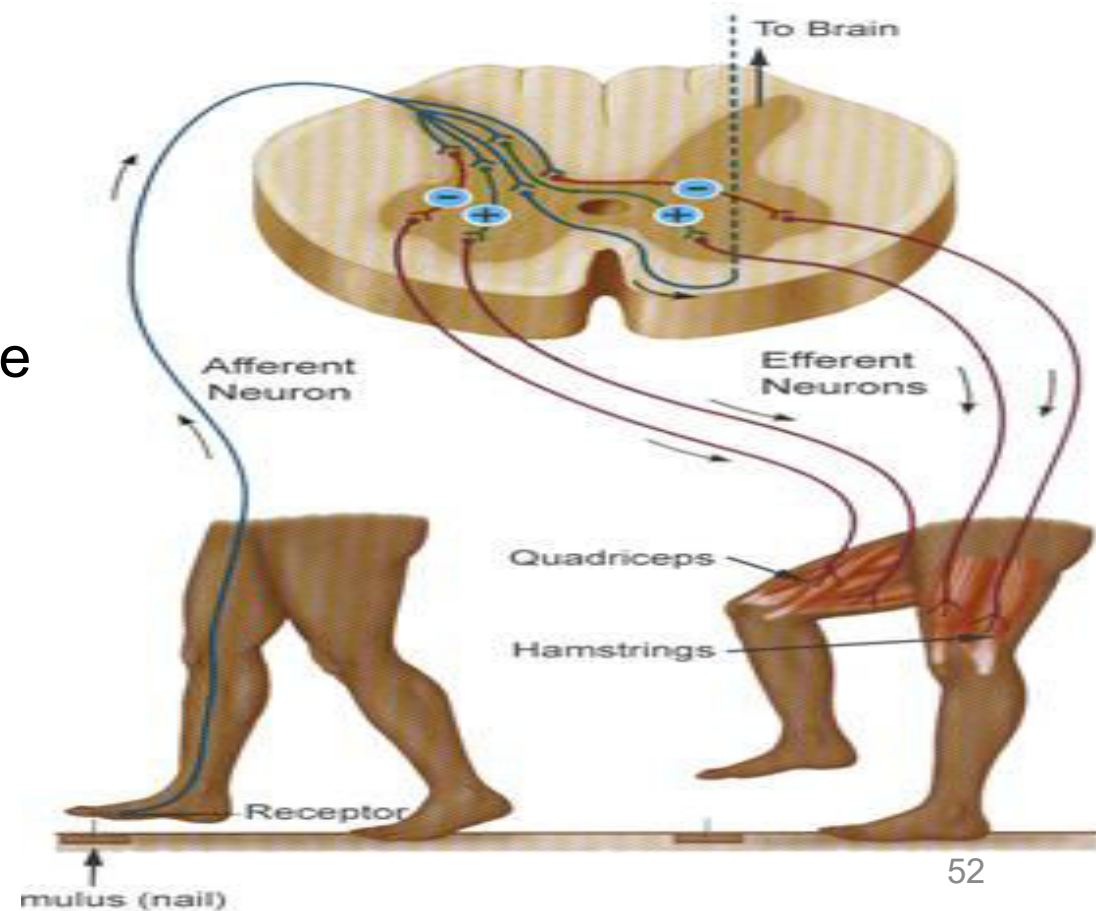
Fig. The flexor Reflex (The withdrawal reflex) or polysynaptic reflex

The Crossed Extensor Reflex

- ❖ is a reflex in which the contralateral limb compensates for loss of support when the ipsilateral limb withdraws from painful stimulus
 - ✓ Extension of the opposite limb can push the entire body away from the object causing the painful stimulus in the withdrawn limb
 - ✓ the opposite limb begins to extend 0.2 – 0.5 sec after a stimulus elicits a flexor reflex in one limb

❖ it is a polysynaptic reflex

❖ signals from sensory nerves cross to the opposite side of the cord to excite extensor muscles



- ❖ During a withdrawal reflex,
 - ✓ the flexors in the withdrawing limb contract and the extensors relax,
 - ✓ while in the other limb, the opposite occurs as part of the crossed extensor reflex

- ❖ When a flexor reflex excites one muscle, it often simultaneously inhibits the antagonist muscles.



this is called **reciprocal inhibition**

- ❖ The neuronal circuit that causes this reciprocal relation is called *reciprocal innervation*

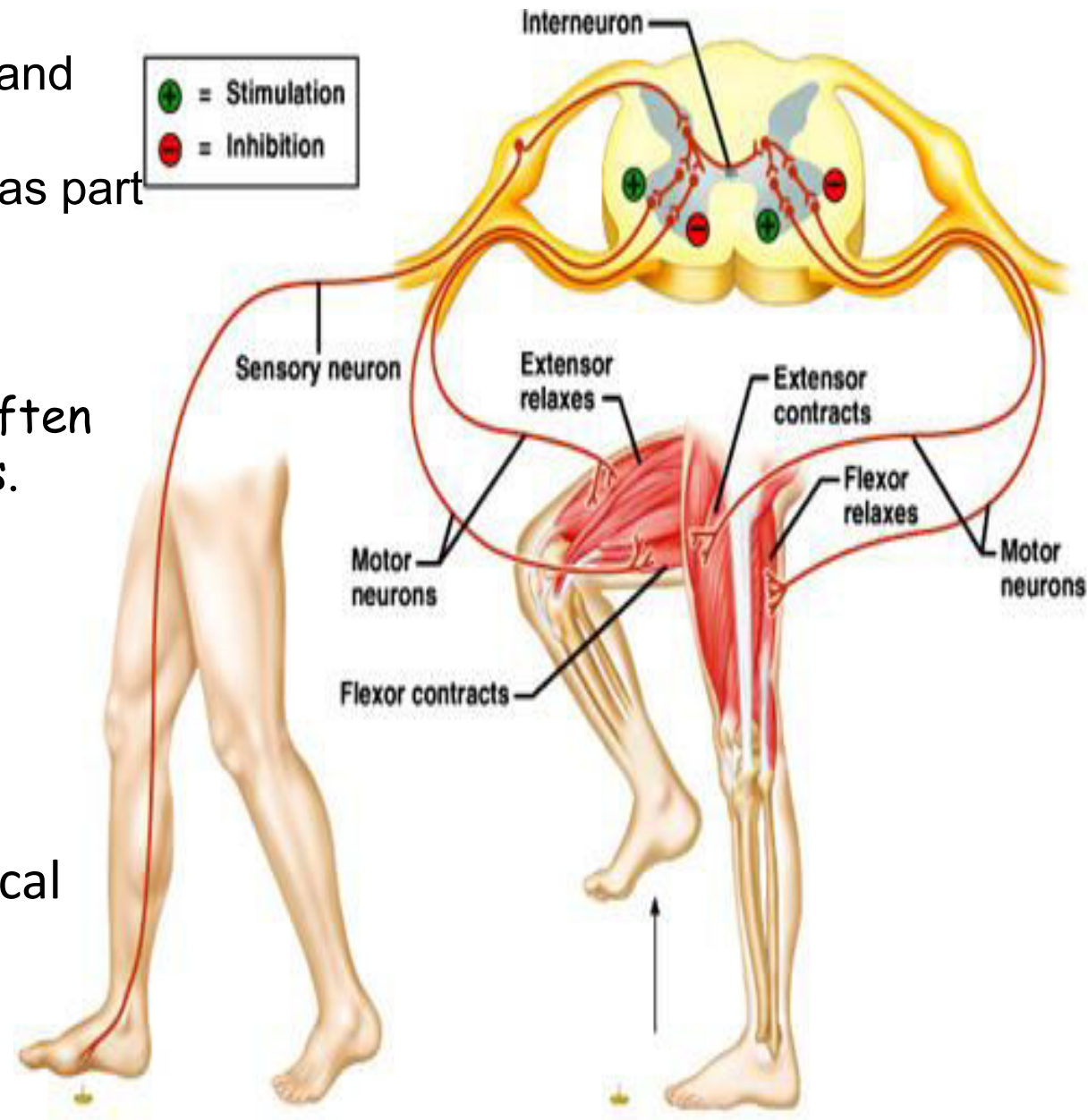


Fig. Flexor reflex, crossed extensor reflex, and reciprocal inhibition

Spinal Cord Transection and Spinal Shock

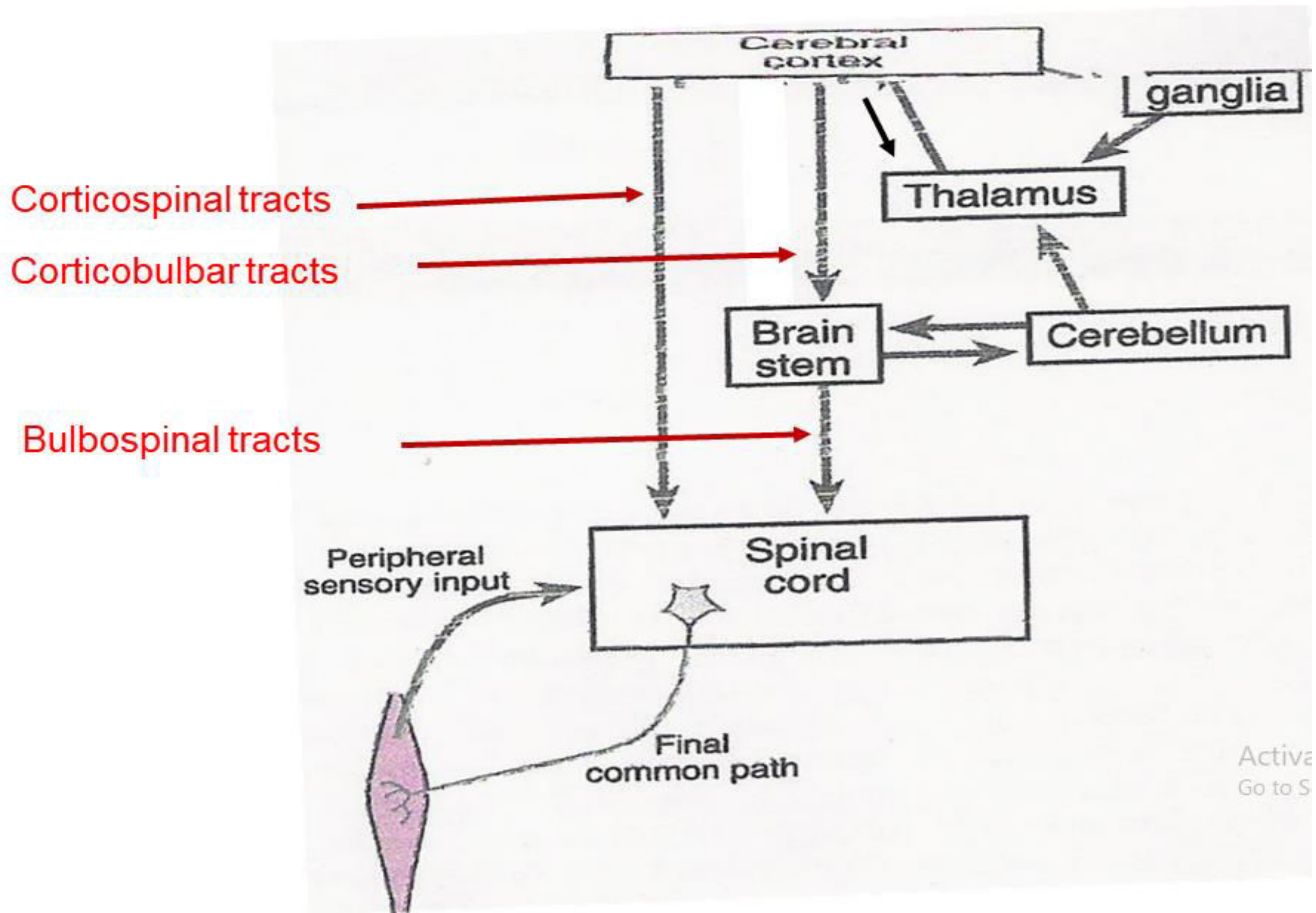
- When the spinal cord is suddenly transected in the upper neck,
 - Essentially all cord functions, including the cord reflexes, immediately become depressed to the point of total silence,
 - This reaction is called *spinal shock*.
- The reason for this reaction of spinal shock is that
 - normal activity of the cord neurons depend to a great extent on continual tonic excitation of nerve fibers entering the cord from higher centers,
 - ✓ particularly via the reticulospinal tracts, vestibulospinal tracts, and corticospinal tracts.
- After a few hrs to a few wks, the spinal neurons gradually regain their excitability.
- This phenomenon is natural characteristic of neurons everywhere in the CNS

- Some of the spinal functions specifically affected during or after spinal shock are:
 - At onset of spinal shock, the BP falls almost instantly and drastically
 - ✓ Due to block of SNS activities almost to extinction.
 - ✓ The BP ordinarily returns to normal within a few days
 - All skeletal muscle reflexes integrated in the spinal cord
 - ✓ are blocked during the initial stages of shock.
 - ✓ Requires 2 wks to several months to return
 - The sacral reflexes for control of bladder and colon evacuation
 - ✓ are suppressed in people for the first few weeks after cord transection,
 - ✓ but in most cases they eventually return.

Cortical and Brain Stem Control of Motor Function

- Most "voluntary" movements initiated by the cerebral cortex achieved when it activates "patterns" of function stored in lower motor centers:
 - the cord, brain stem, basal ganglia, and cerebellum
- Lower centers, in turn, send specific signals to the muscles: – tone , posture, equilibrium
- For a few types of movements the cortex has almost a direct pathway to the anterior motor neurons of the cord,
 - bypassing some motor centers on the way
 - Eg. For control of the **fine movements of the fingers and hands**

Overview of Motor System



MOTOR CORTEX and CORTICOSPINAL TRACT

- The motor cortex is divided into:
 1. the primary motor cortex (Brodmann area 4)
 2. the premotor area, and
 3. the supplementary motor area

(Brodmann area 6)
- Each has its own topographical representation of muscle groups and specific motor functions

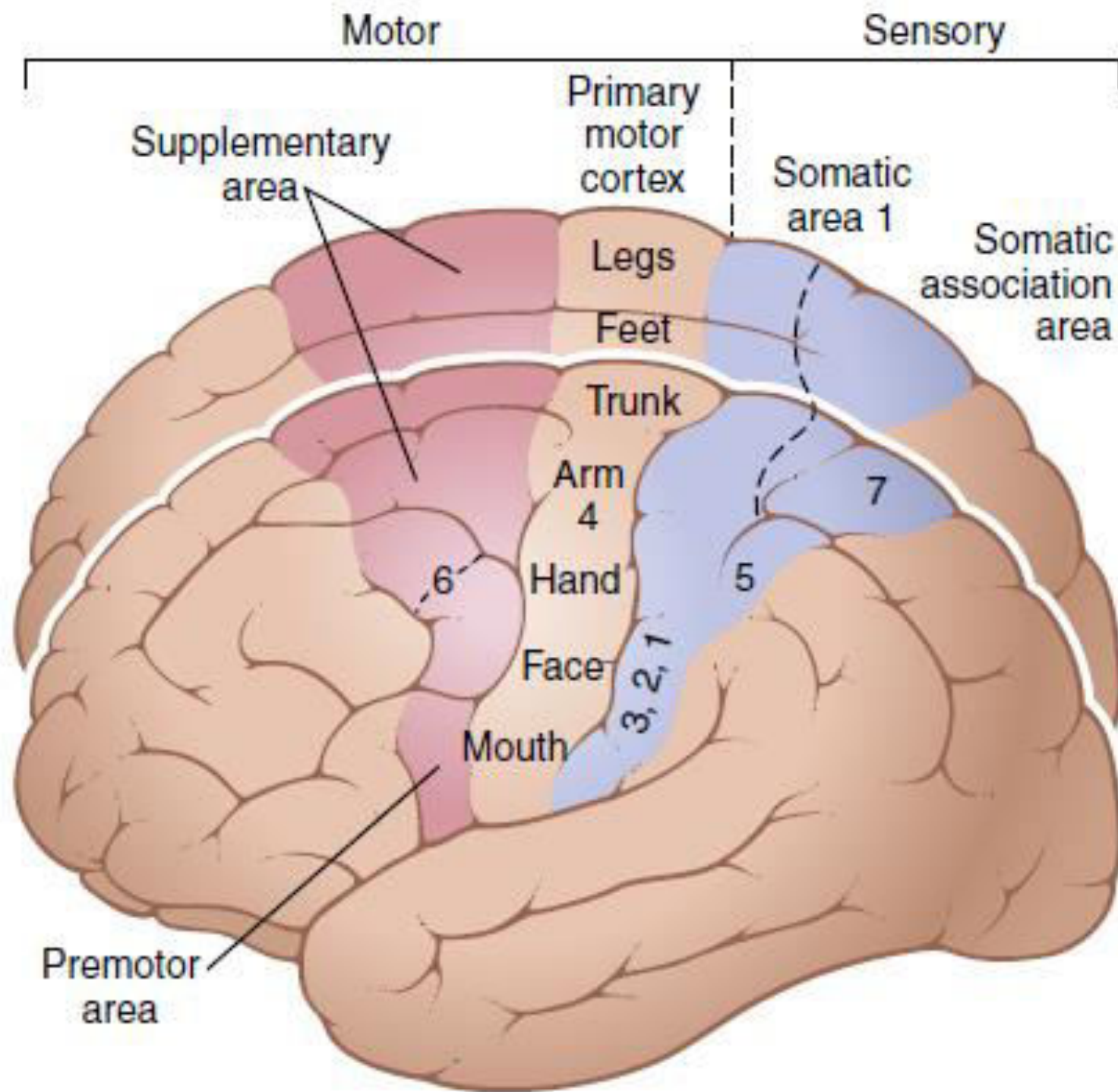


Fig. Motor and somatosensory functional areas of the cerebral cortex. The numbers 1, 2, 3, 4, 5, 6, and 7 are Brodmann's cortical areas

1. Primary motor Cortex (A4)

- Lies in the posterior portion of the frontal lobe (in precentral gyrus)
- Extends laterally to the sylvian fissure
- is the same as area 4 in Brodmann's classification of cortical area
- Has somatotopic representation of the different body parts in an arrangement called a **motor homunculus**
- The representation is inverted & crossed

Functions of primary motor cortex:

- ❖ Initiates voluntary, fine, & discrete movement of distal parts
- ❖ send the impulse to lower areas of the brain
- ❖ *Modulation of stretch reflex*

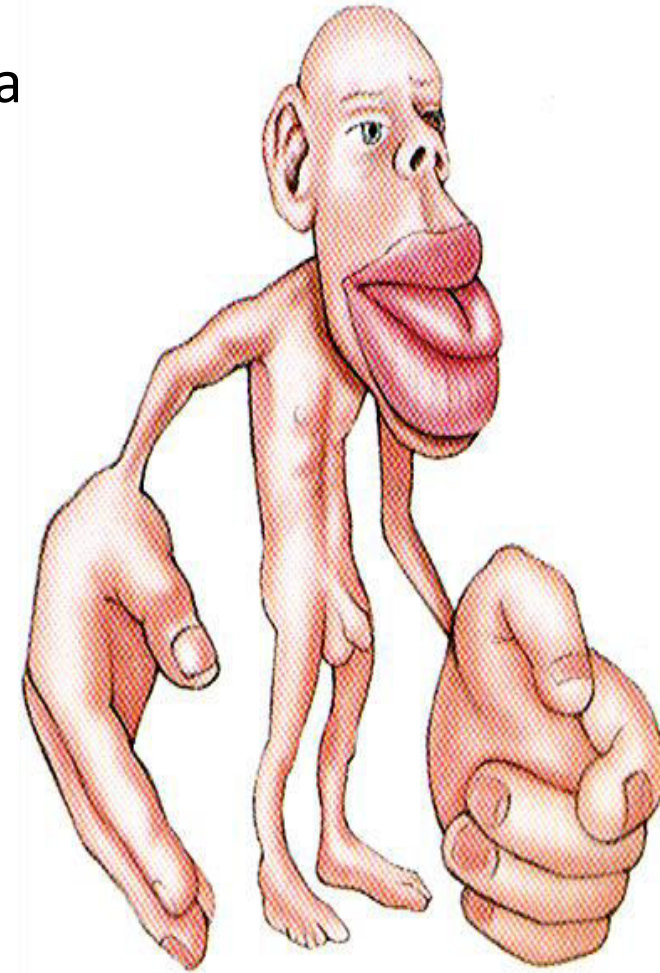


Fig motor homunculus

Arrangement of the 1^o motor cortex:
the representation is inverted & crossed

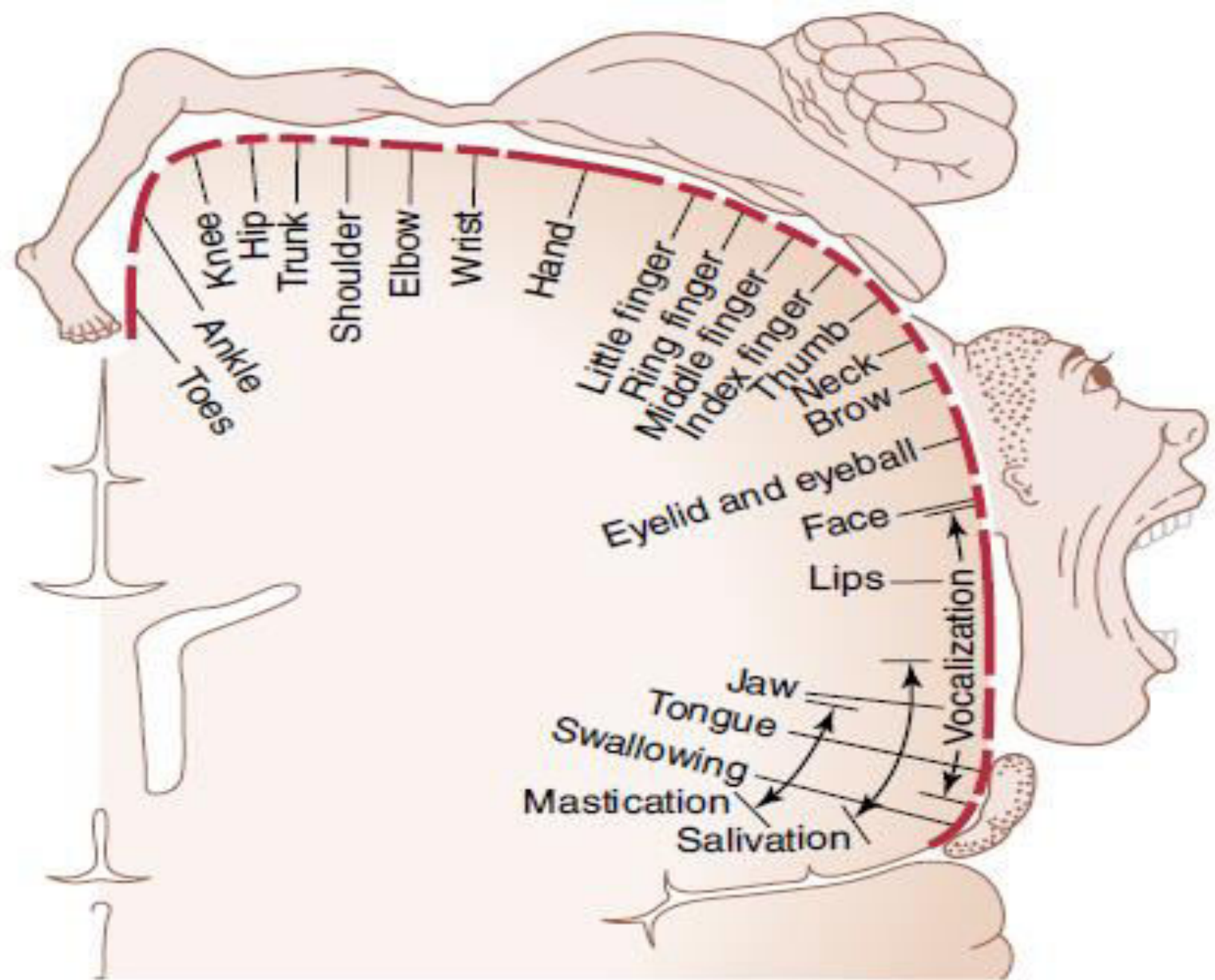
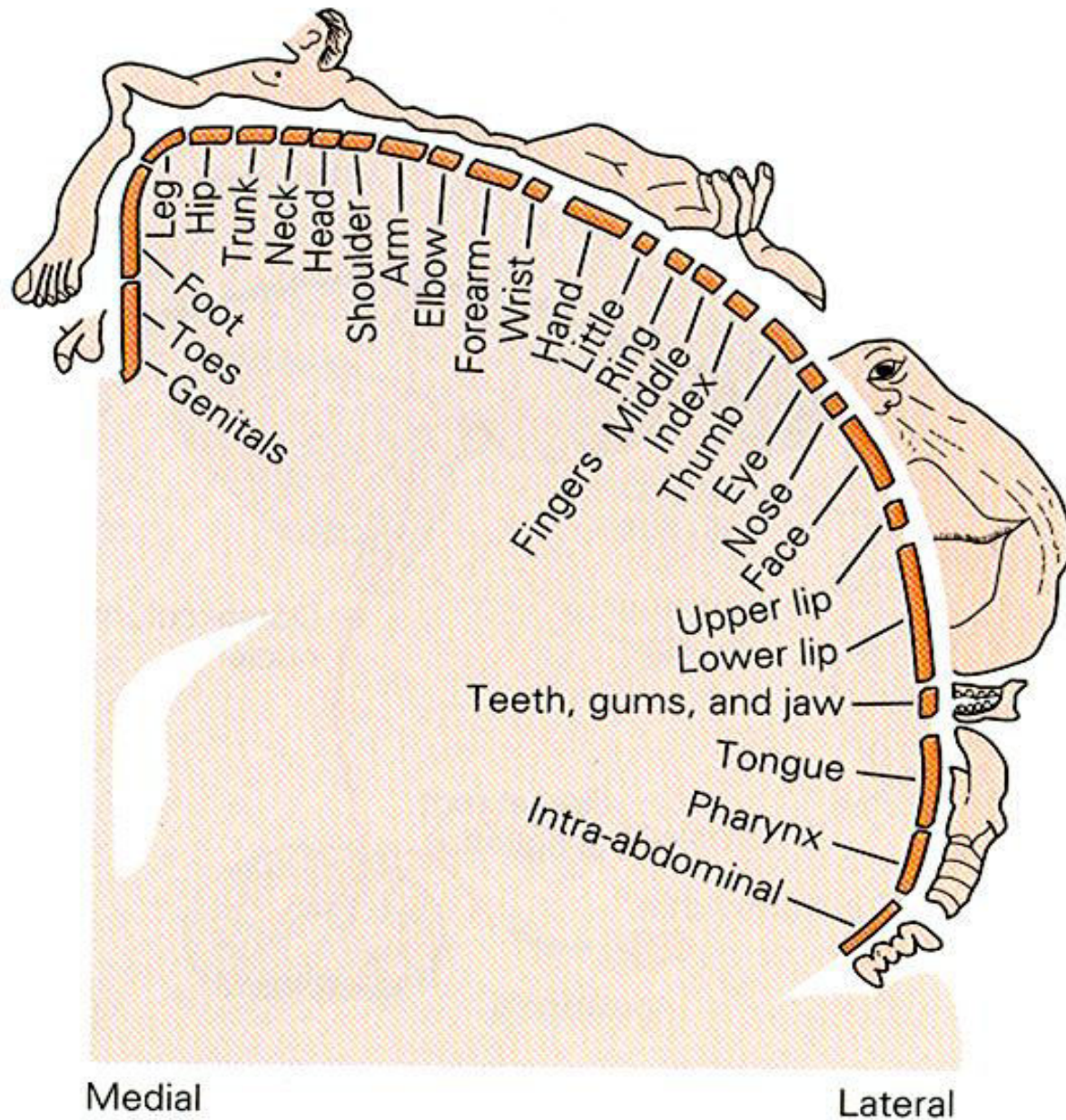


Fig. Degree of representation of the different muscles of the body in the primary motor cortex.

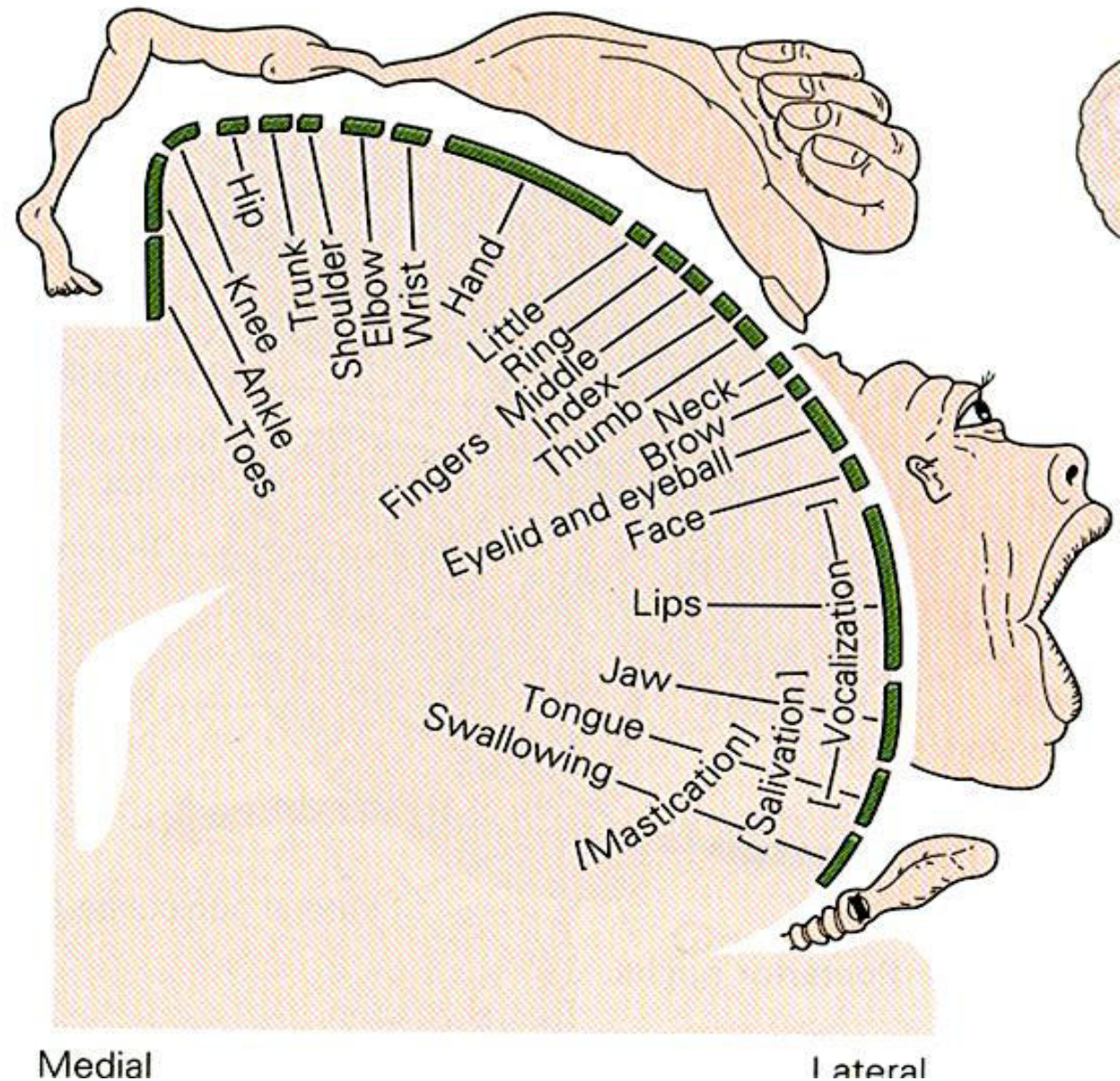
Note that more than 1/2 of the entire primary motor cortex is concerned with controlling the muscles of the **hands** and the **muscles of speech**.

Sensory and motor homunculus

A Sensory homunculus



B Motor homunculus



2. Pre-motor Area (A6, 8, 11)

- lies 1 to 3cm anterolateral to the primary motor cortex.
- causes much more **complex "patterns" of movement** than the discrete patterns of 1^o motor cortex
 - ✓ Involves basal ganglia, thalamus, and 1^o motor cortex for the control of complex patterns of coordinated muscle activity
- **Anterior premotor** (role in complex pattern of movt):
 - ✓ **develops a "motor image"** of the total muscle movement that is to be performed
- **Posterior premotor:**
 - ✓ **excites each successive pattern of muscle** activity required to achieve the image
 - ✓ Sends its signals:
 - either directly to 1^o motor (A4) to excite specific muscles
 - or
 - by way of basal ganglia and thalamus back to the 1^o motor cortex
- Its topographical organization is roughly the same as that of A4

3. The supplementary motor area

- located on the medial face of the hemisphere, just in front of 1^o motor cortex
- Contractions elicited by stimulating this area are often **bilateral rather than unilateral**
 - ✓ Eg. bilateral grasping movements of both hands simultaneously
- functions in concert with the premotor area to provide:
 - ✓ body-wide attitudinal movements,
 - ✓ fixation movements of the different segments of the body,
 - ✓ positional movements of the head and eyes etc, as back ground for finer motor control of arms and hands by premotor area and A4.
- In contrast to premotor, the SMA is involved in performance of a **sequence of movements from memory** (as opposed to movements guided by a visual cue)

- **Some Specialized Areas of Motor Control in the Motor Cortex:**
 - *control specific motor functions*, most in the pre-motor area and also in SMA
 - *localized either by electrical stimulation or by noting the loss of motor function when destructive lesions occur in specific cortical areas.*
 - **Broca's area(A44)** – *word program for verbalization*
 - **Located in** Premotor area for “word formation”
 - Lesioned (eg. By Stroke): difficulty of speech or word formation[Broca's aphasia]
 - **Eye field (A8)/“Voluntary” Eye Movement Field**– *directing eyes, blinking,...*
 - In the premotor area immediately above Broca's area
 - is a locus for controlling **voluntary eye movements**
 - **If Damaged:** No *voluntarily moving the eyes* toward different objects
 - Instead, the eyes tend to lock involuntarily onto specific objects

- ***head rotation area (part of A6)*** – *directing head*
 - Location: slightly higher in the motor association area
 - it directs the head toward different objects
 - This area is closely associated with the eye movement field
- ***hand skills areas (part of A6)*** – *skilled hand movements*
 - In the premotor area immediately anterior to the primary motor cortex for the hands and fingers
 - If lesioned or its circuit defected ⇒ loss of ability to execute skilled hand movements
 - ✓ This condition is called ***motor apraxia***

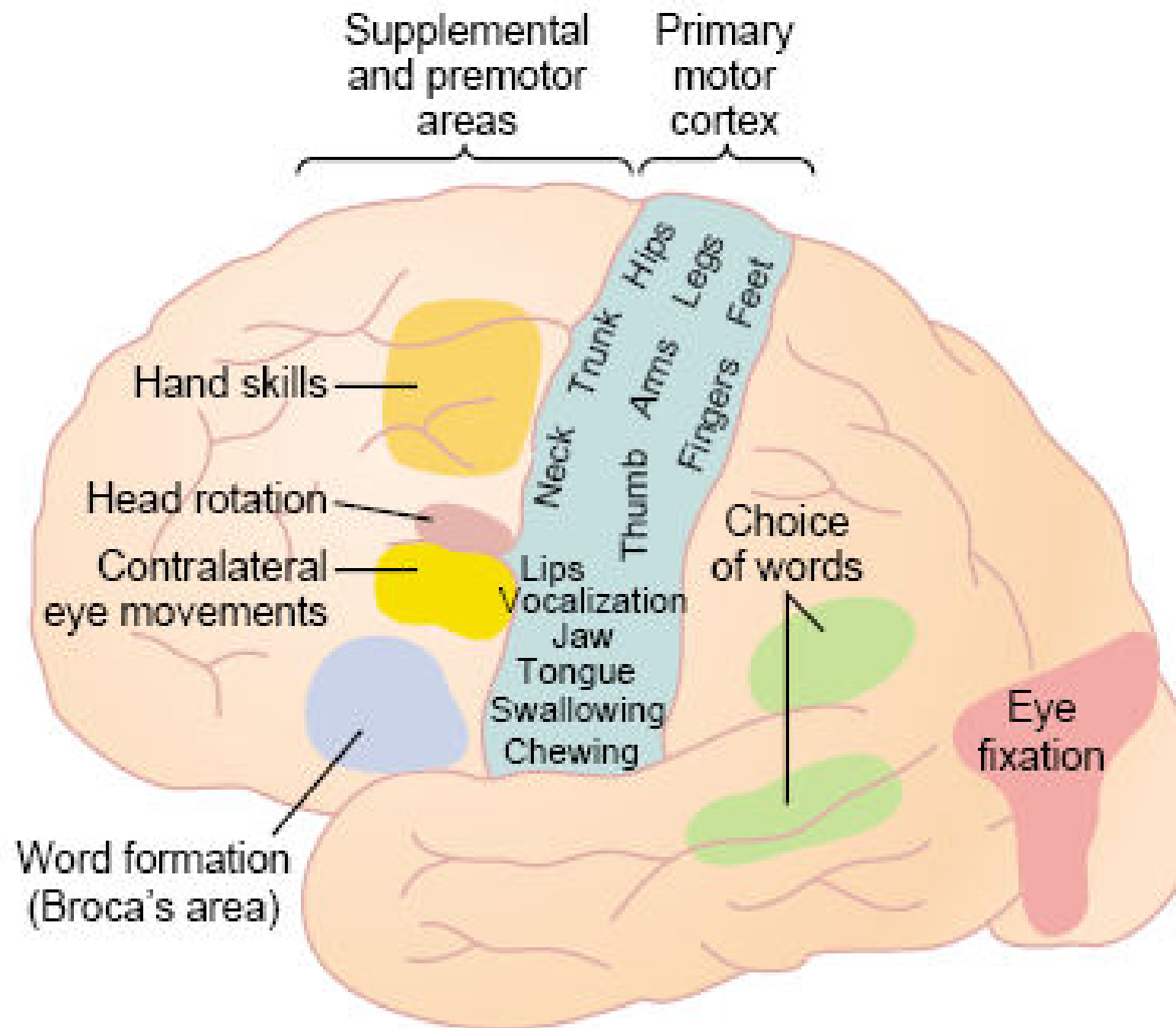
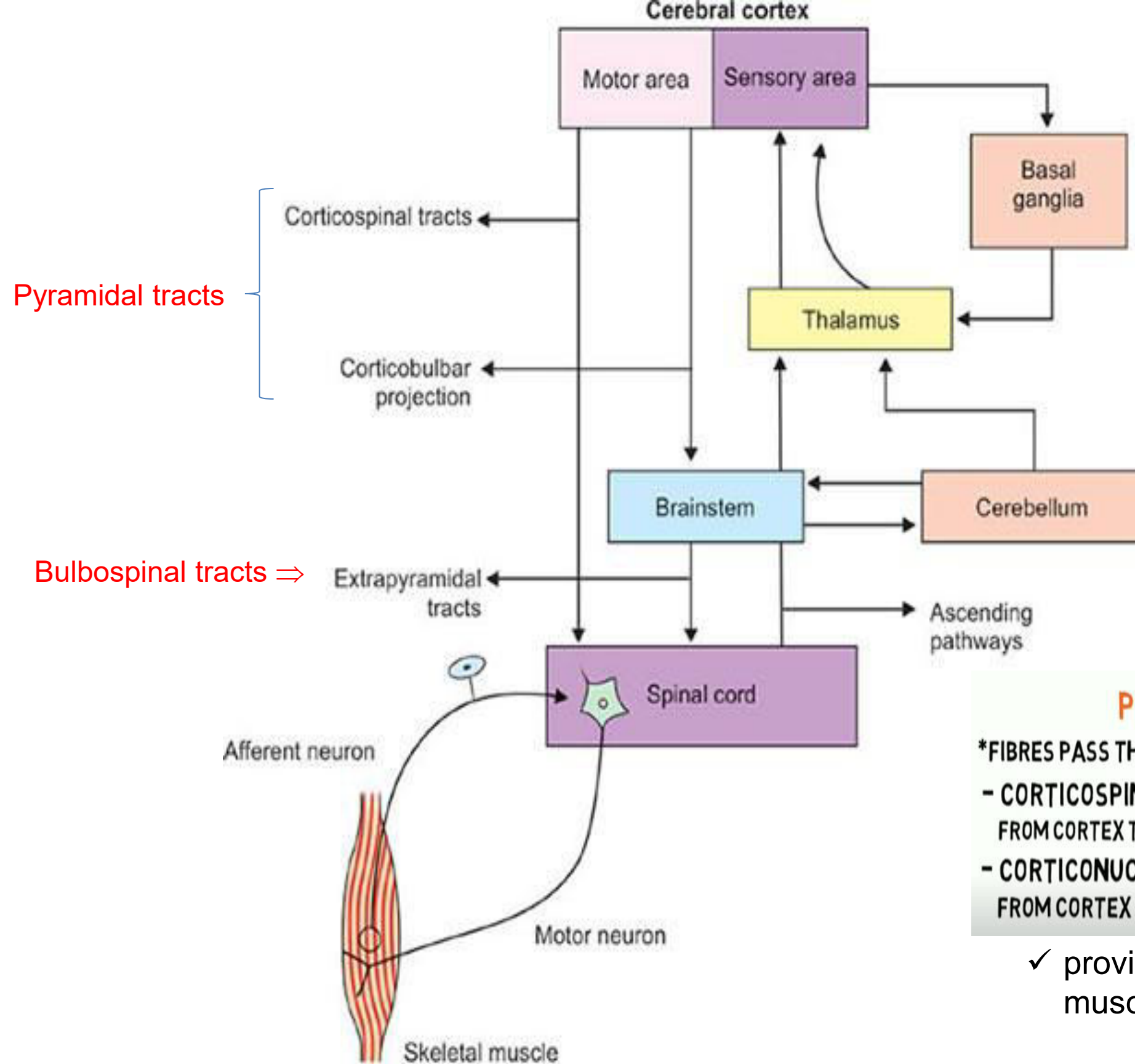


Fig. Representation of the different muscles of the body in the motor cortex and location of other cortical areas responsible for specific types of motor movements.

Transmission of Signals from Motor Cortex to the Muscles

- ❖ Motor signals are transmitted from cortex to spinal cord:
 - **1. Directly through Corticospinal (Pyramidal) Tract**
 - The direct pathway from cortex to the spinal cord
 - Concerned with **discrete and detailed movements**,
 - Movement of distal segments of the limbs, particularly hands and fingers.
 - **2. Indirectly through multiple accessory Pathways**
 - To the basal ganglia
 - To the brain stem , including the red nucleus of midbrain
 - To the cerebellum
- ❖ Incoming Fiber Pathways to the Motor Cortex mainly from the **somatosensory cortex**



PYRAMIDAL TRACTS

*FIBRES PASS THROUGH PYRAMID OF MEDULLA

- CORTICOSPINAL TRACT :-

FROM CORTEX TO ANTERIOR HORN CELLS OF SPINAL CORD

- CORTICONUCLEAR (CORTICOBULBAR) TRACT :-

FROM CORTEX TO CRANIAL NERVE NUCLEI IN BRAINSTEM

✓ provide voluntary control of muscular movements

Corticospinal (Pyramidal) Tract

- The most important output pathway from the motor cortex to the cord originates:
 - about **30% from primary motor or A4**,
 - about **30% From premotor & supplementary Cortices &**
 - about **40% From somatic sensory areas** posterior to the central sulcus
- Projects from cerebral cortex to spinal cord without interruption
↓
- Synapses with:
 - **1. interneurons controlling motor neurons (MOST)**
 - **2. sensory relay neurons**
 - **3. motor neurons controlling distal muscles (α and γ motor neurons)**

- After leaving the cortex, it passes through the **posterior limb of internal capsule**
 - **internal capsule passes b/n caudate nucleus & putamen of the basal ganglia**

↓
- then downward through the brain stem, **forming the *pyramids of the medulla***
 - The **majority** of the pyramidal fibers (80%) then **decussate in the lower medulla**

↓

 - descend into the ***lateral corticospinal tracts of the cord***

↓

 - *Most terminate on interneurons* in the intermediate cord gray matter;
 - a few terminate on **sensory relay neurons** in the dorsal horn, and
 - a very few terminate directly on **anterior motor neurons** that cause muscle contraction
- A few of the fibers do not decussate in the medulla (20% of Corticospinal tracts)
 - but pass ipsilaterally down the cord in the ***ventral corticospinal tracts***
 - **Most of them decussate in the cord** either in the neck or in the upper thoracic region
 - concerned with control of the movements of axial muscles (of the trunk).

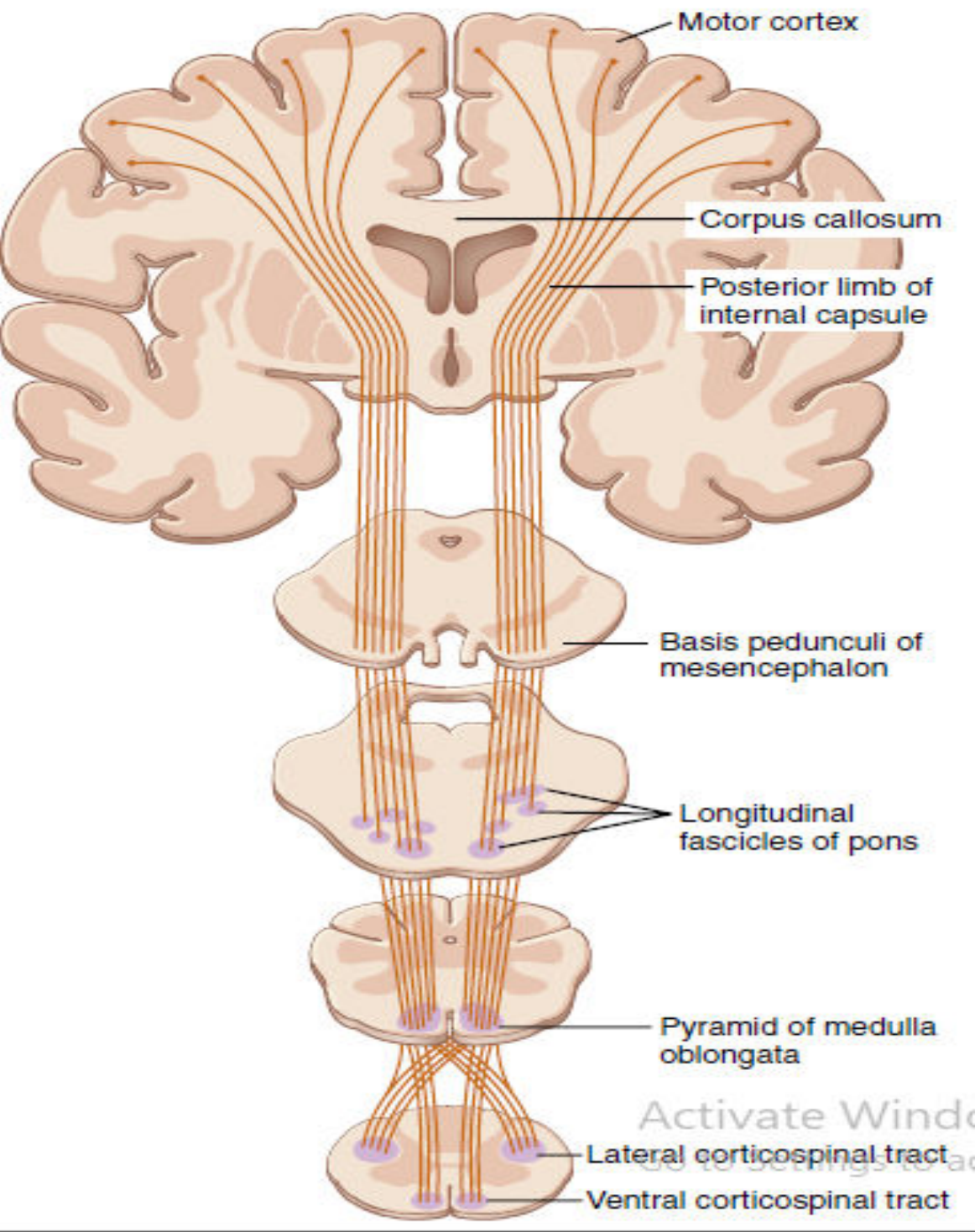
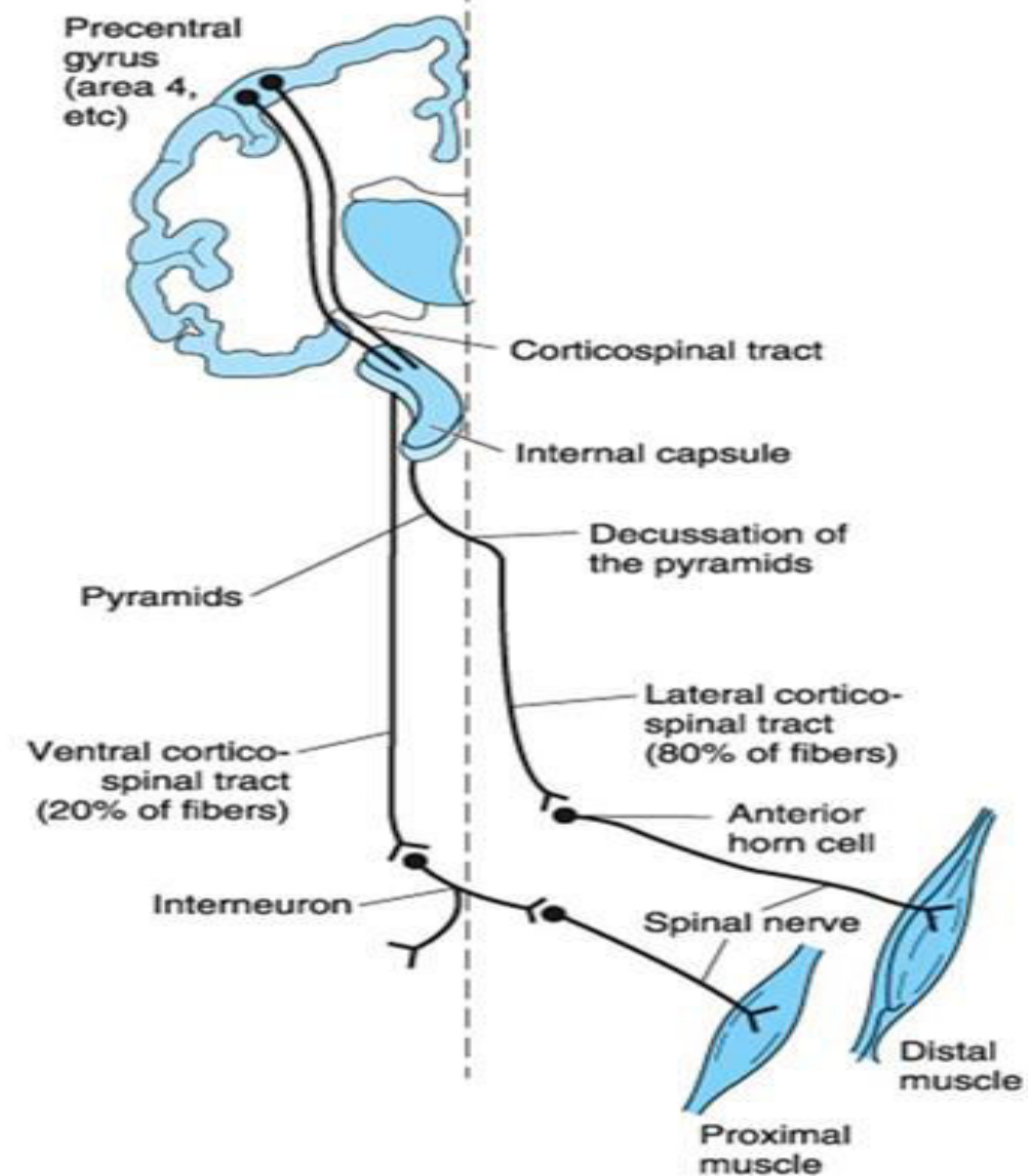
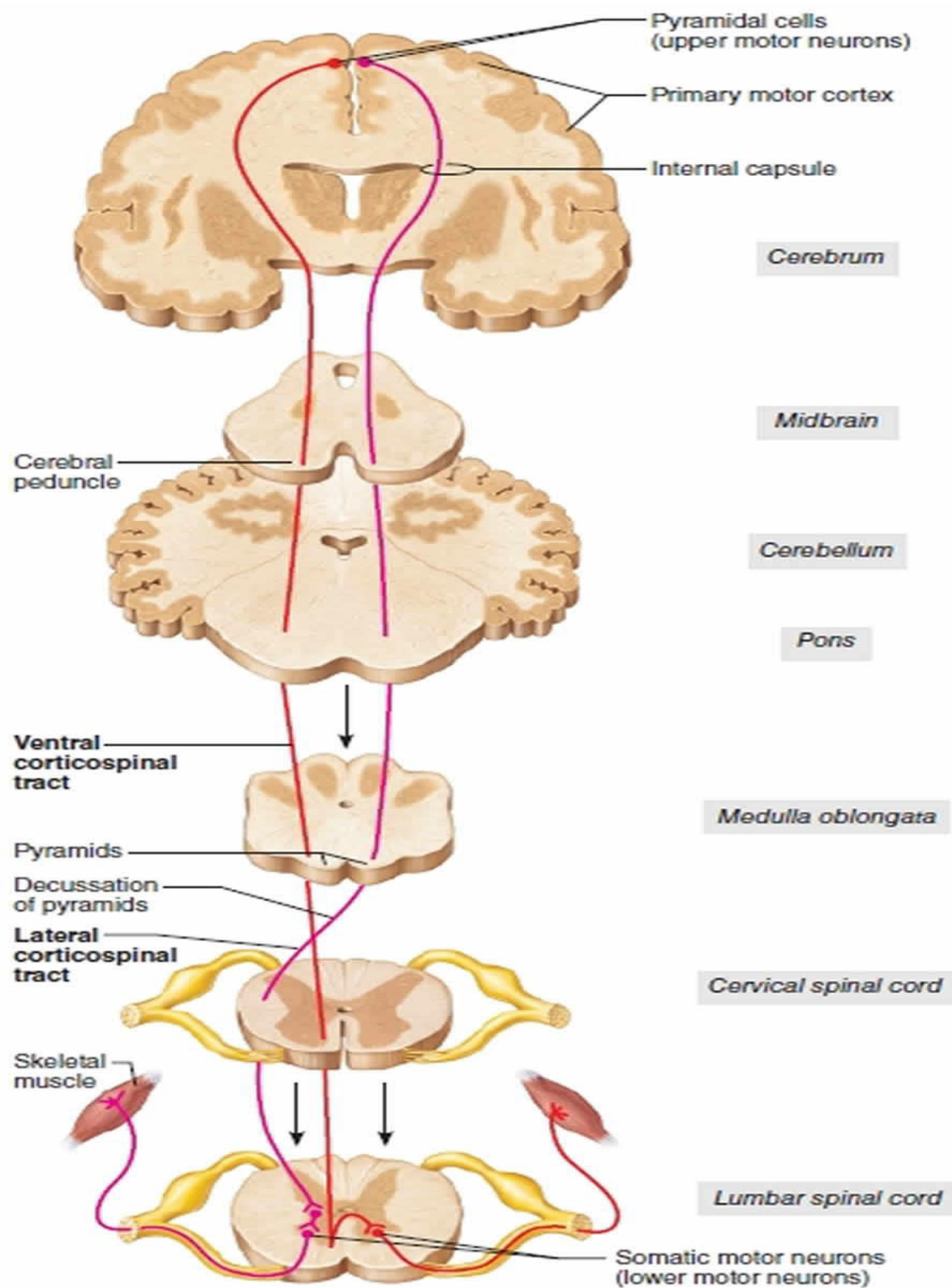


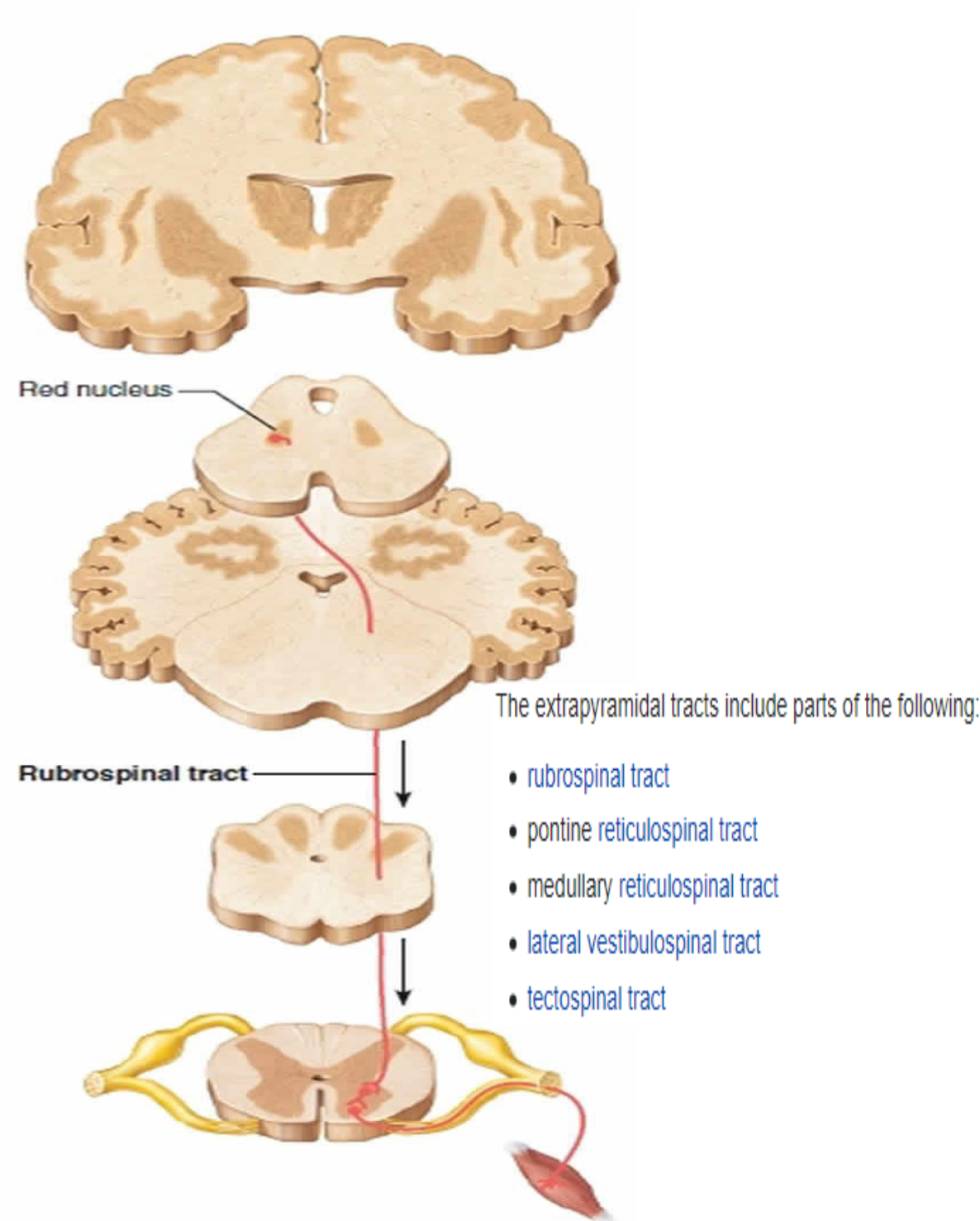
Fig. Corticospinal (pyramidal) tract



- ❖ *lateral corticospinal tracts synapse on lower motor neurons supplying the distal muscles*
- ❖ *Anterior corticospinal tracts synapse on lower motor neurons supplying the proximal muscles of trunk, limb and neck*



(a) **Pyramidal (lateral and ventral corticospinal) pathways**

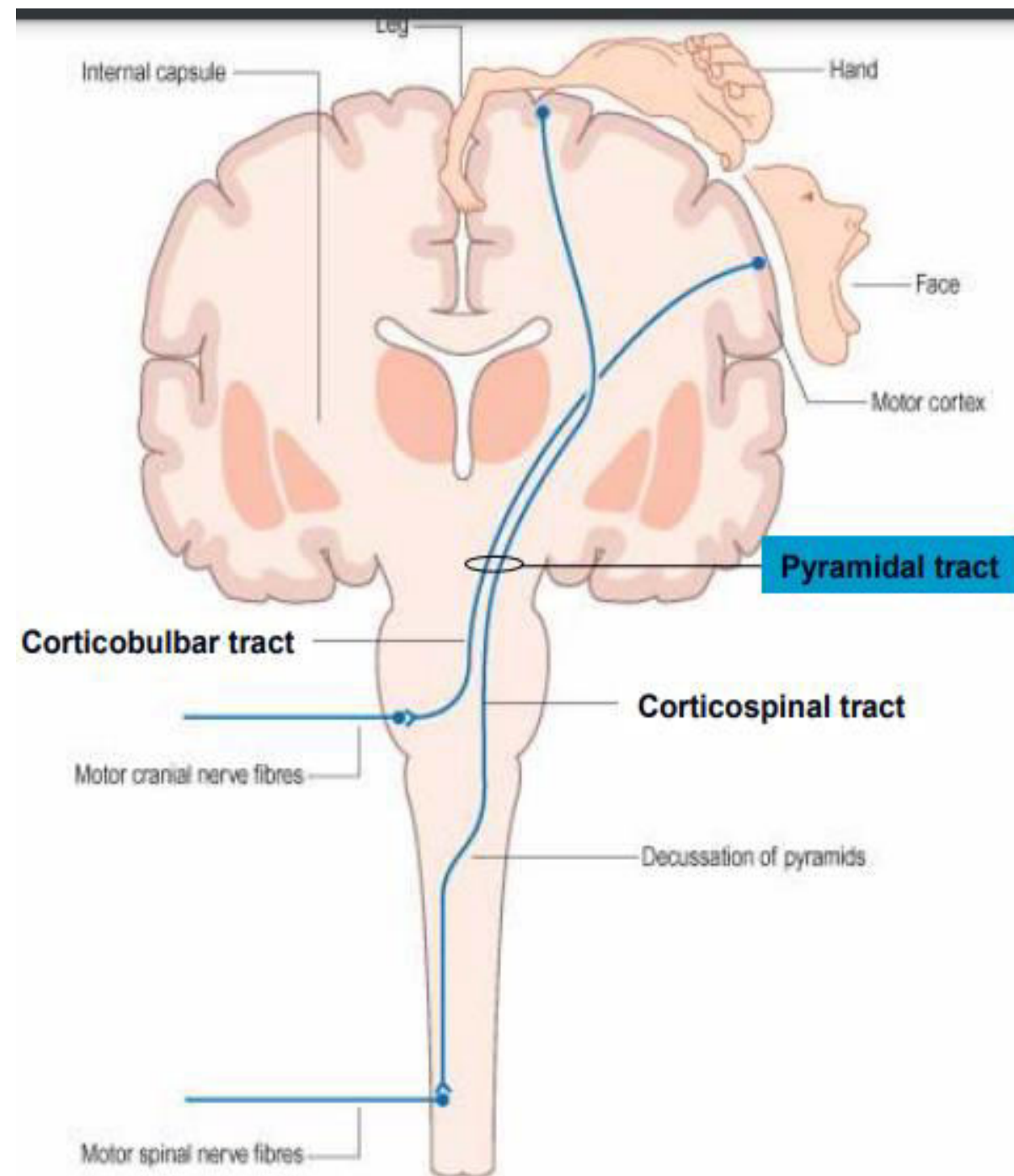


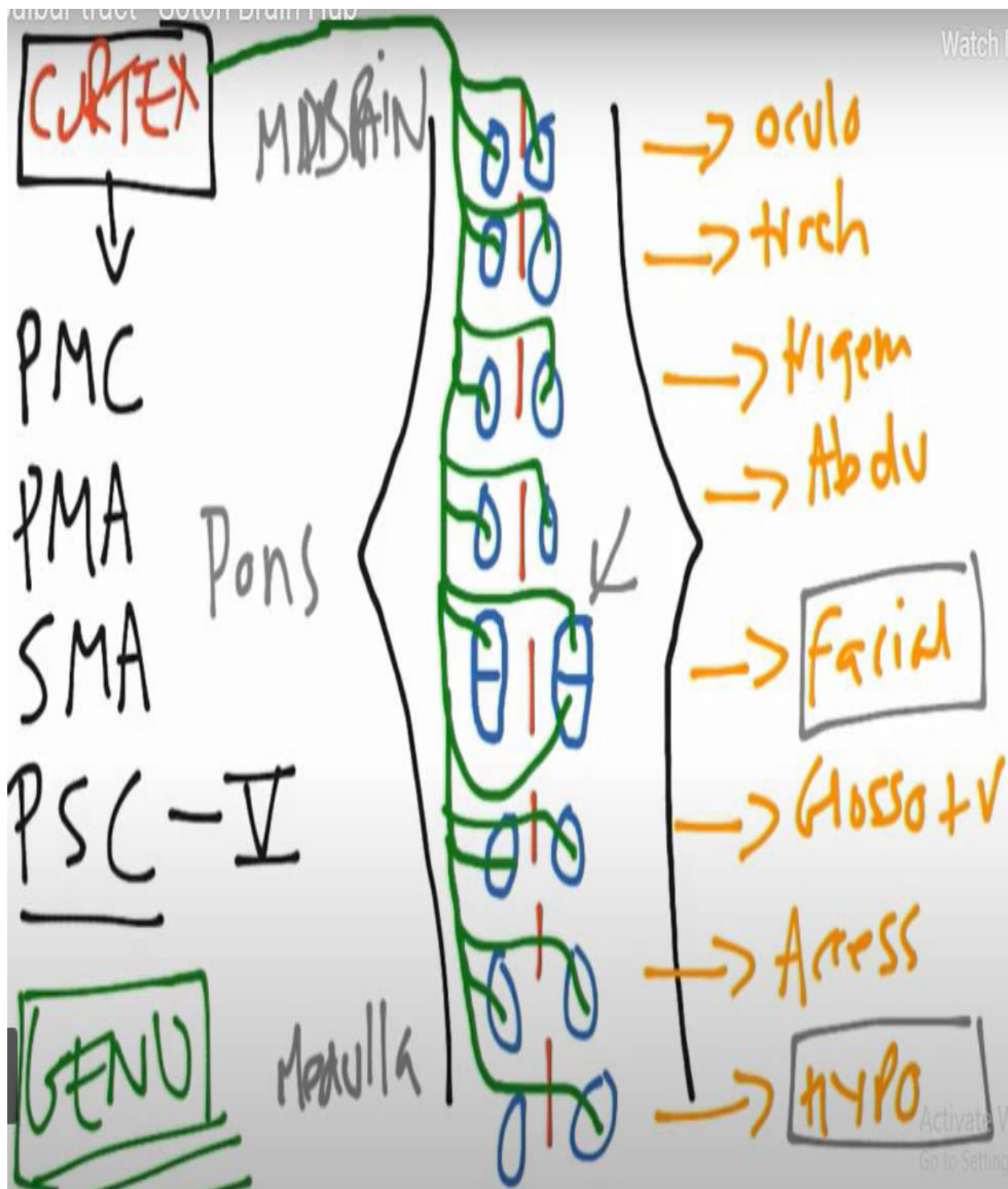
(b) **Rubrospinal tract (Extrapyramidal tract)**

- **3% of pyramidal fibers are large myelinated** (diameter of 16 μm)
 - These fibers originate from giant pyramidal cells, called **Betz cells**,
 - Betz cells are found only in the primary motor cortex
 - transmit signals to the spinal cord at a velocity of about 70 m/sec,
 - the most rapid rate of transmission of any signals from the brain to the cord.
- **97% of pyramidal fibers are small** (smaller than 4 μm in diameter)
 - conduct background tonic signals to the motor areas of the cord

Corticobulbar Tract- aka corticonuclear

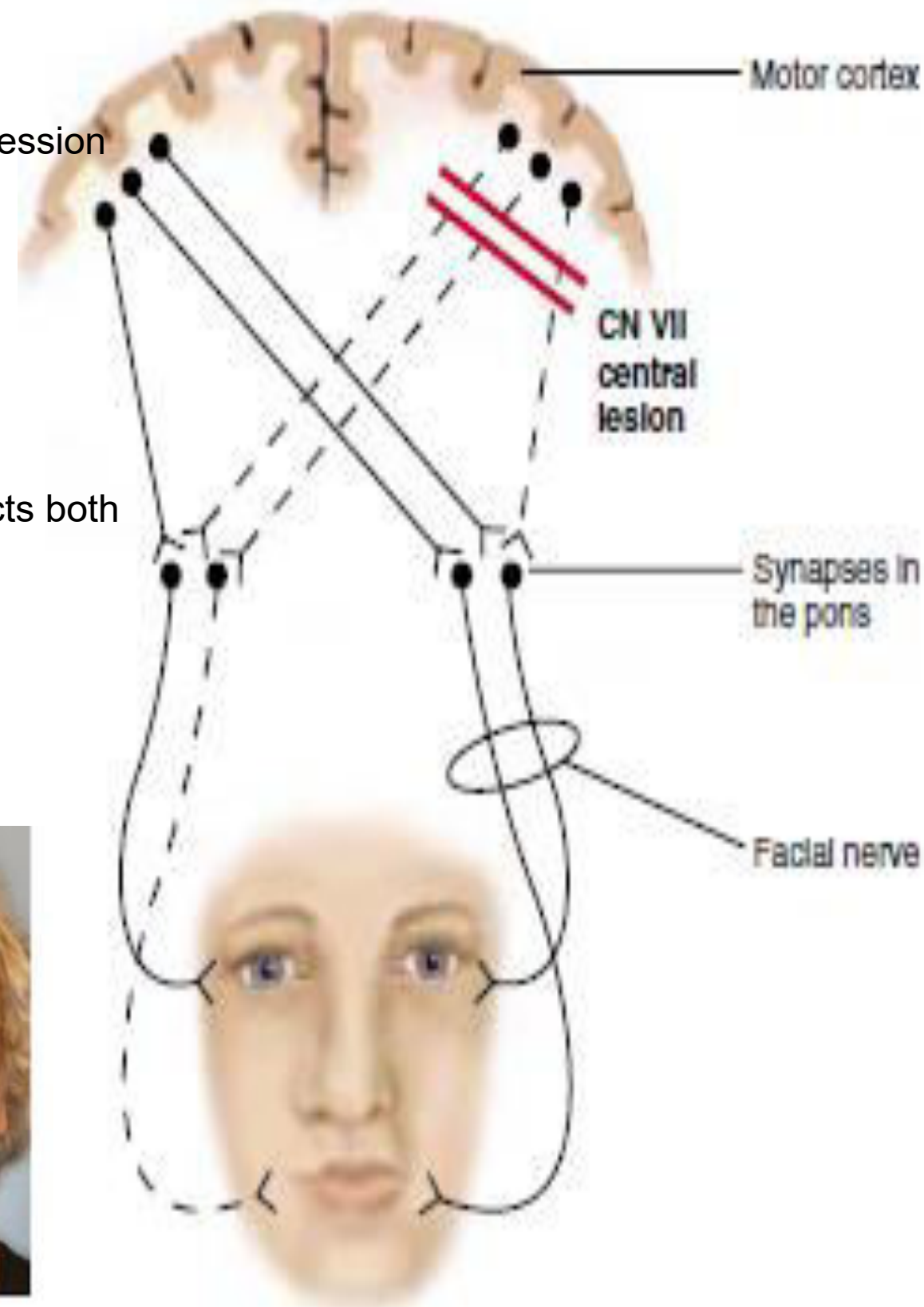
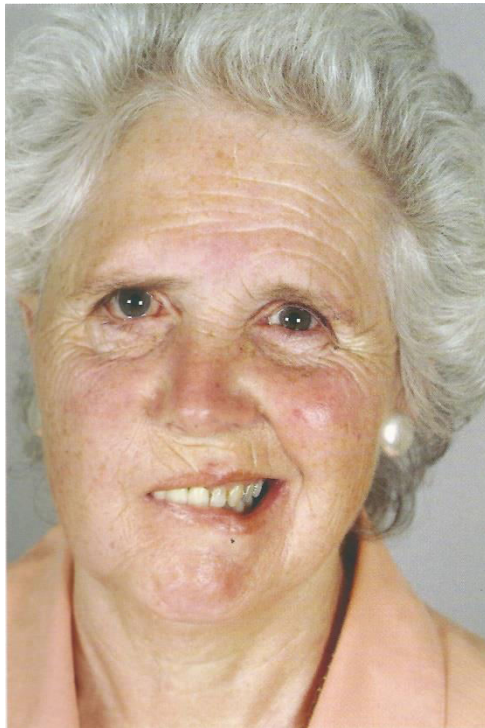
- The corticobulbar pathway serves for transmission of relatively discrete signals from the motor cortex to the α -MNs in the brainstem.
- descends through the genu of internal capsule with a few fibers in the posterior limb of the capsule
 - terminates within brainstem motor nuclei on the alpha motor neurons or interneurons innervating α -MNs
 - control the muscles of the face, head and neck, in contrast to the corticospinal tract that controls movement of the torso, upper and lower limbs.
 - Bilateral innervation except VII and XII





Facial palsy

- ❖ Fibers from the motor division supply muscles of facial expression
- ❖ UMN from cortex synapsing with LMN in the pons
(Note the crossed and uncrossed path ways)
- ❖ A central lesion(UMNL) affects mainly the lower face
 - ❖ Supranuclear facial palsy Eg. Stroke
- ❖ A peripheral injury to CN VII(LMNL), as in Bell's palsy, affects both the upper and the lower face;
 - ❖ Infranuclear facial palsy: Stroke, Bell's palsy



“Extrapyramidal” System

- The term *extrapyramidal motor system* denote all those portions of the cortex and brain stem that contribute to motor control but not part of the direct corticospinal-pyramidal system.
 - they modulate motor activity without directly innervating motor neurons.
- These include pathways through:
 - the basal ganglia,
 - the reticular formation of the brain stem,
 - the vestibular nuclei, and
 - often the red nuclei.

The extrapyramidal tracts include parts of the following:

- rubrospinal tract
- pontine reticulospinal tract
- medullary reticulospinal tract
- lateral vestibulospinal tract
- tectospinal tract

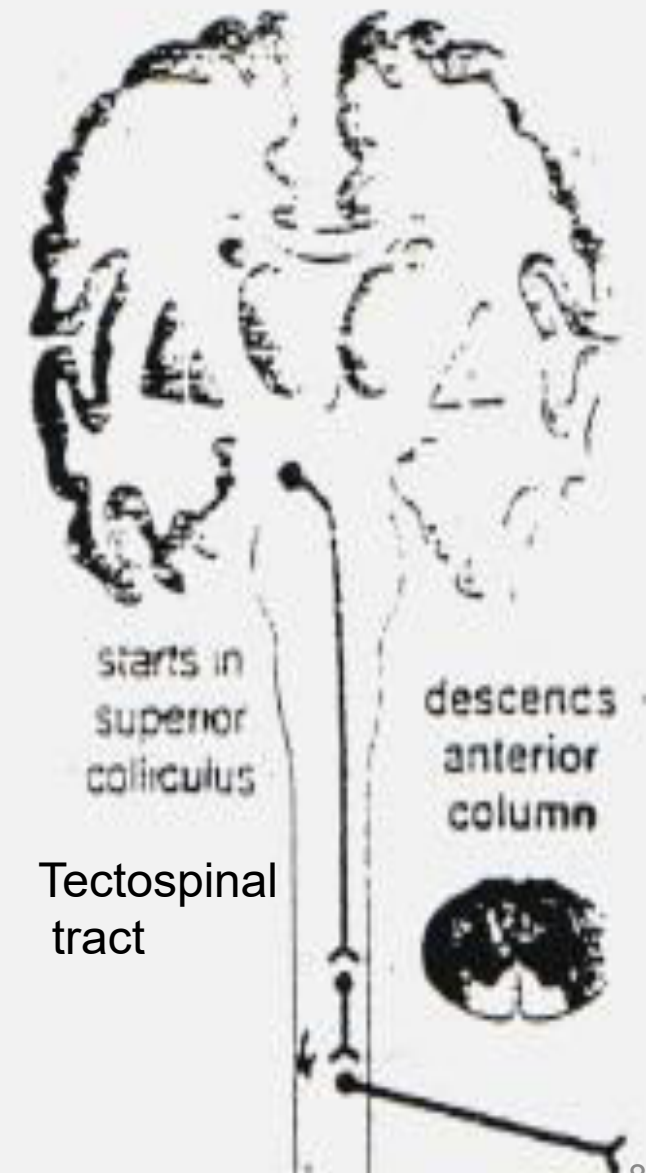
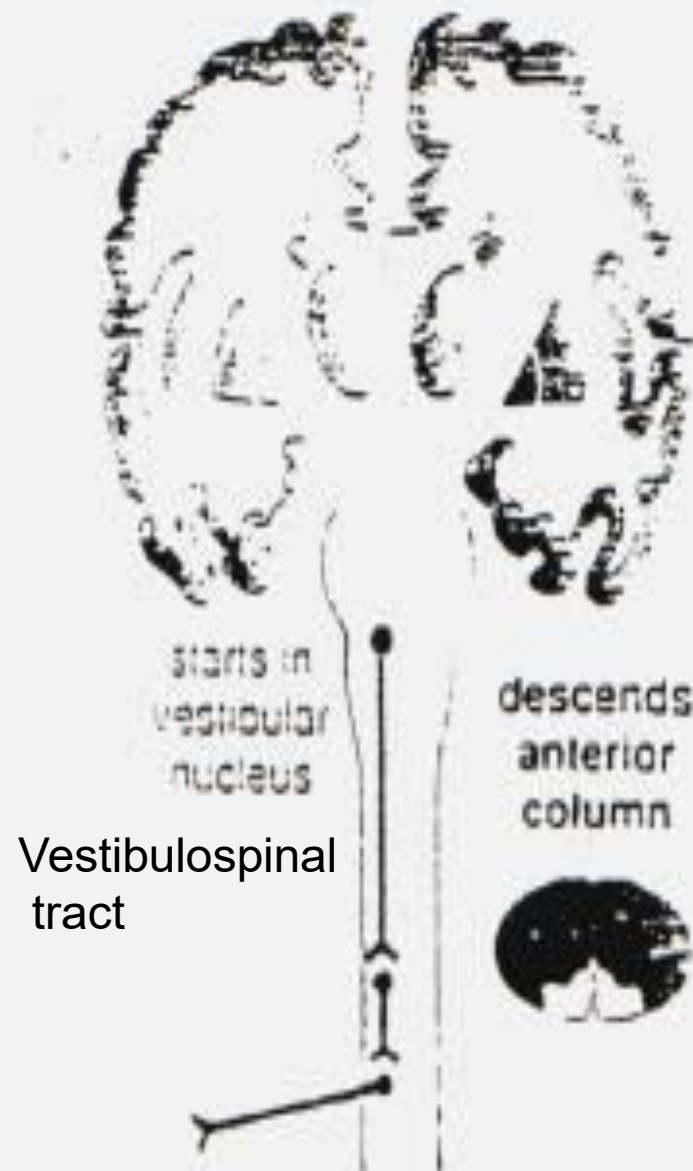
Indirect (Extrapyramidal) Pathways :

Subconscious Control of Skeletal Muscles

precise movements

balance

head movements



Divisions of the motor Neurons

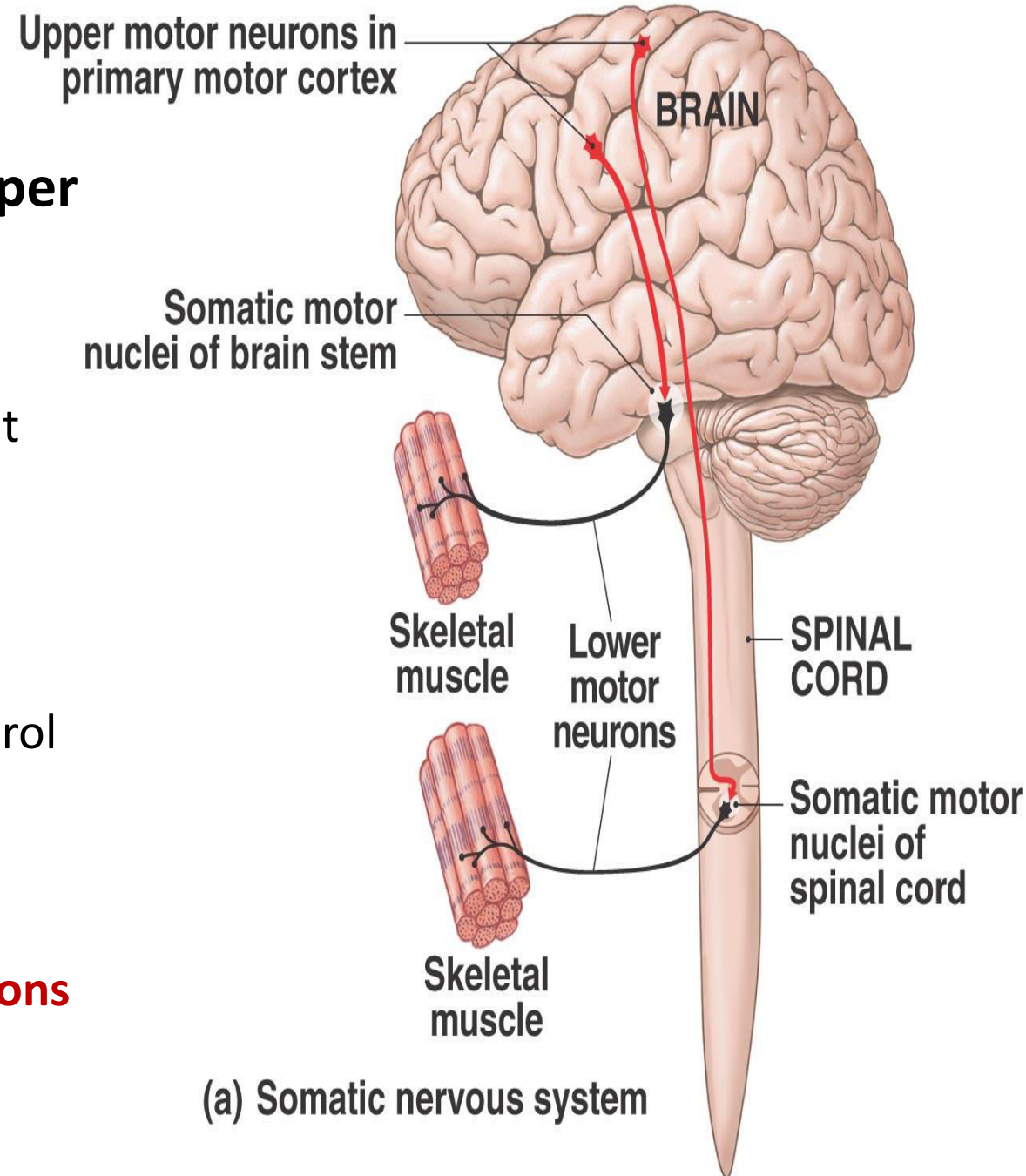
- The motor system can be divided into upper and lower motor neurons.

❖ Upper motor neurons

- ✓ are those in the cortex and brain stem that **activate the lower motor neurons.**
- ✓ refer to **corticospinal tract neurons** that innervate spinal motor neurons,
- ✓ also include **brain stem neurons** that control spinal motor neurons

❖ Lower motor neurons

- ✓ refer to the spinal and cranial **motor neurons that directly innervate skeletal muscles.**
- ✓ are those whose axons terminate on skeletal muscles



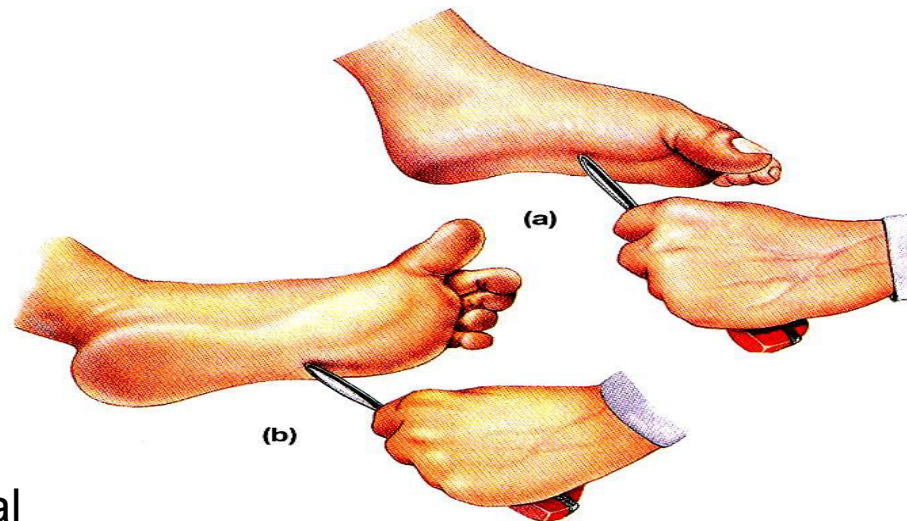
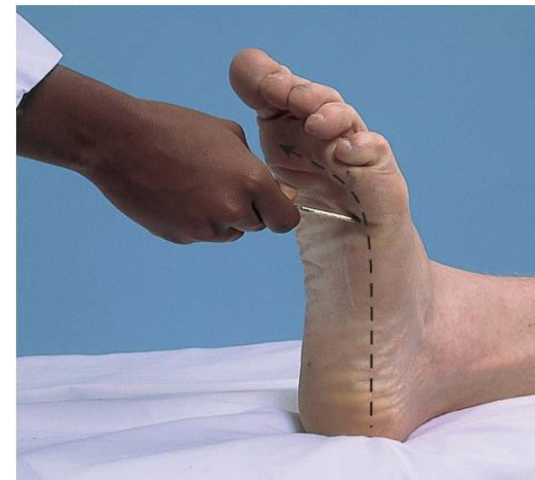
Upper and lower Neuronal lesions

- UMN lesion is a lesion in motor pathway from cerebral cortex to just before the ventral horn
- **Upper Motor Neuron Lesion (UMNL) can be due to:**
 - Stroke: Ischemic stroke, Hemorrhagic stroke
 - brain tumors: Glioblastoma, Primary CNS lymphoma.....
 - Trauma
 - Infections: CNS Toxo, Tuberculoma, Brain abscess.....
- **Clinical Feature of UMNL: spastic paralysis, exaggerated reflex, Babinski-positive**
 - **Primary motor area (A4) and its fibers (unilateral)**
 - ✓ Thrombosis of lenticulostriatal artery
 - ✓ Damage to the cortex itself: crossed monoplegia and/ or facial paralysis
 - **Corona radiata:** Crossed monoplegia/hemiplegia
 - **Corona radiata is a** White matter core of the cerebral hemispheres.
 - Contains afferent and efferent fibers to and from the cerebral cortex

- **Lower Motor Neuron Lesion, causes can be**
 - Peripheral neuropathy(eg. GBS), thrombosis of spinal artery, Infection(TB)
 - Autoimmune diseases: Transverse myelitis, SLE myelitis...
 - Accidental transection of spinal cord: Trauma to spinal cord
- **Duration of spinal shock (2-4 weeks)**
 - Failure of spinal centers below the damage with flaccid paralysis
(withdrawal of facilitatory higher centers)
 - Loss of all sensations: cutaneous, deep, visceral
 - Loss or depressed reflexes: superficial, deep and visceral reflexes
 - Flaccid paralysis of skeletal muscles and loss of muscle tone
 - Loss of vasomotor tone (vasodilatation and hypotension)

Plantar reflex/ Babiniski sign

- ❖ Is a cutaneous, superficial reflex
- ❖ Elicited by stroking, lateral surface of the sole of the foot.
- ❖ Normal response-plantar flexion of the toes
 - ✓ plantar flexion (Curling) of all toes
- ❖ **Extensor plantar response (Positive Babiniski sign)**
 - ✓ Dorsiflexion of big toe and fanning of the other toes
 - ✓ Normal in infants but abnormal in adults
 - ✓ **common cause: Corticospinal tract lesion(UMNL)**
 - ✓ Causes of Transiently positive **Babiniski sign**
 - Drug or alcohol Intoxication
 - Postictal period



a. Normal
b. Abnormal



UMN AND LMN lesion

Criteria	UMNL	LMNL
Cause	Pyramidal tract lesion	Anterior horn lesion
Paralysis ✓ Type ✓ Extent	Spastic Widespread	Flaccid Localized
Reflexes ✓ Cutaneous ✓ Babinski sign ✓ Deep reflex	Lost Present Exaggerated(with clonus)	hyporeflexia or areflexia: Lost Absent Absent
✓ Muscle tone ✓ wasting (Atrophy)	Increased (hypertonia) Minimal(Disuse Atrophy)	Decreased (hypotonia) Marked /Severe Atrophy (denervation atrophy)
✓ Fasciculation	No Fasciculation	Fasciculation
Response to electric stimulation	Normal	Reaction of degeneration (10-14 days after)

Role of the Brain Stem in Controlling Motor Function

- **Brain Stem** is an extension of the spinal cord upward into the cranial cavity
- consists of the *medulla*, *pons*, and *mesencephalon(midbrain)*
- contains motor and sensory nuclei
 - perform motor and sensory functions for the face and head regions
- serves as a way station for "command signals" from higher neural centers

- **also considered as its own master b/c it provides many special control functions:**
 - Control of respiration
 - Control of the cardiovascular system
 - Partial control of gastrointestinal function
 - Control of many stereotyped movements of the body
 - Control of equilibrium
 - Control of eye movements
- The important role of controlling whole-body movt and equilibrium.
 - Is due to brain stem's *reticular nuclei* and *vestibular nuclei*

Roles of the Reticular and Vestibular Nuclei

- These nuclei are important to support the body against gravity
- The reticular nuclei are divided into two major groups:
 - **1. pontine reticular nuclei,**
 - located in the pons and extending into the mesencephalon,
 - **2. medullary reticular nuclei**
 - extend through the entire medulla, lying ventrally and medially near the midline
- These two sets of nuclei function mainly antagonistically to each other, with
 - **the pontine reticular nuclei excite the antigravity muscles via reticulospinal tract**
 - the muscles of vertebral column & extensor muscles of the limbs
 - **the medullary reticular nuclei relax or inhibit the same muscles**
 - to counterbalance the excitatory signals from the pontine reticular system,
 - so that under normal conditions, the body muscles are not abnormally tense.
 - **NB: Can be “disinhibited” by higher brain when the brain wishes to excite the pontine system to cause standing**

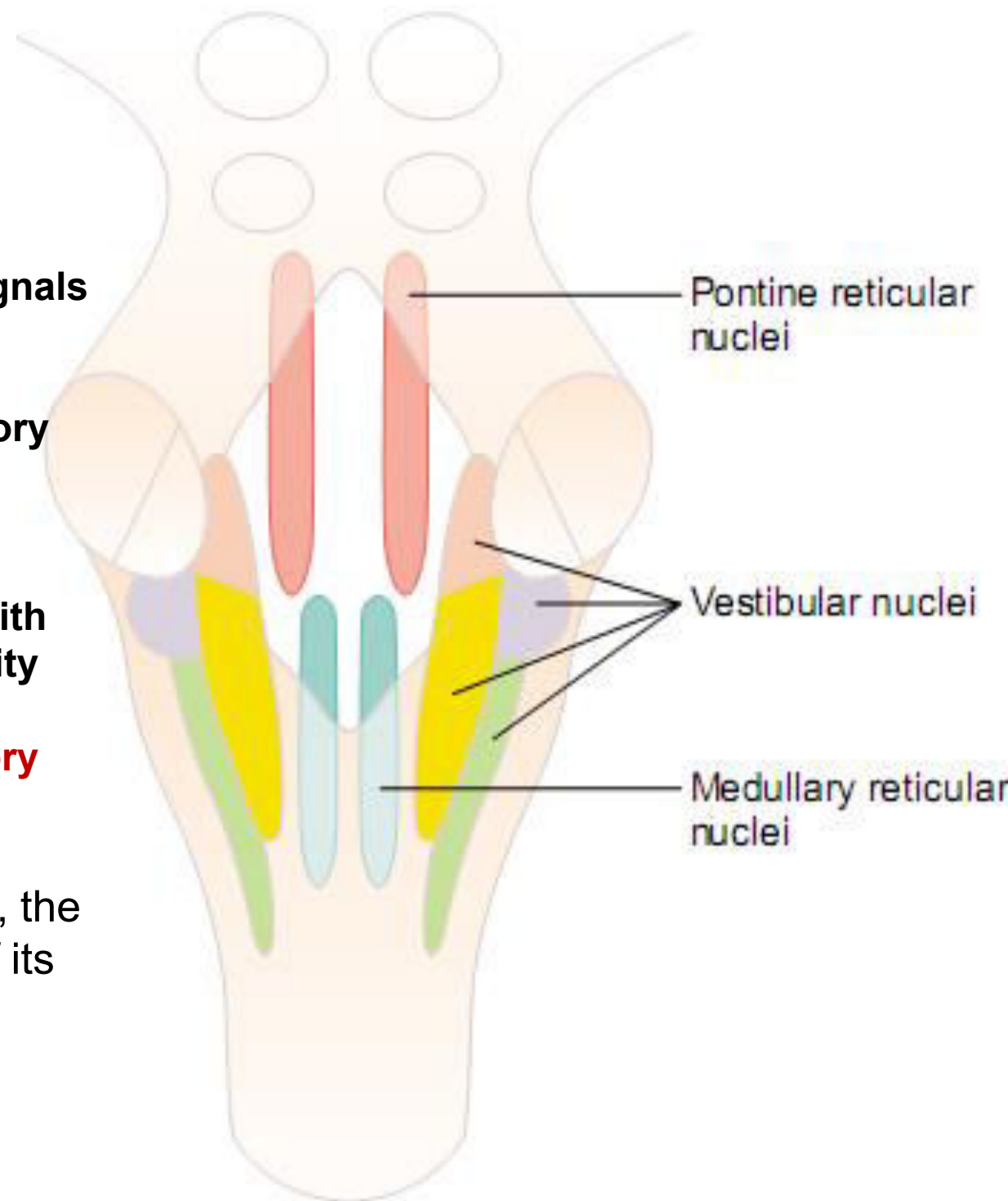
❖ **Pontine Reticular System Transmits Excitatory Signals to antigravity muscles**

❖ **Medullary Reticular System Transmit them Inhibitory Signals.**

❖ **All the vestibular nuclei, function in association with the pontine reticular nuclei to control the antigravity muscles**

✓ **The vestibular nuclei transmit strong excitatory signals to the antigravity muscles**

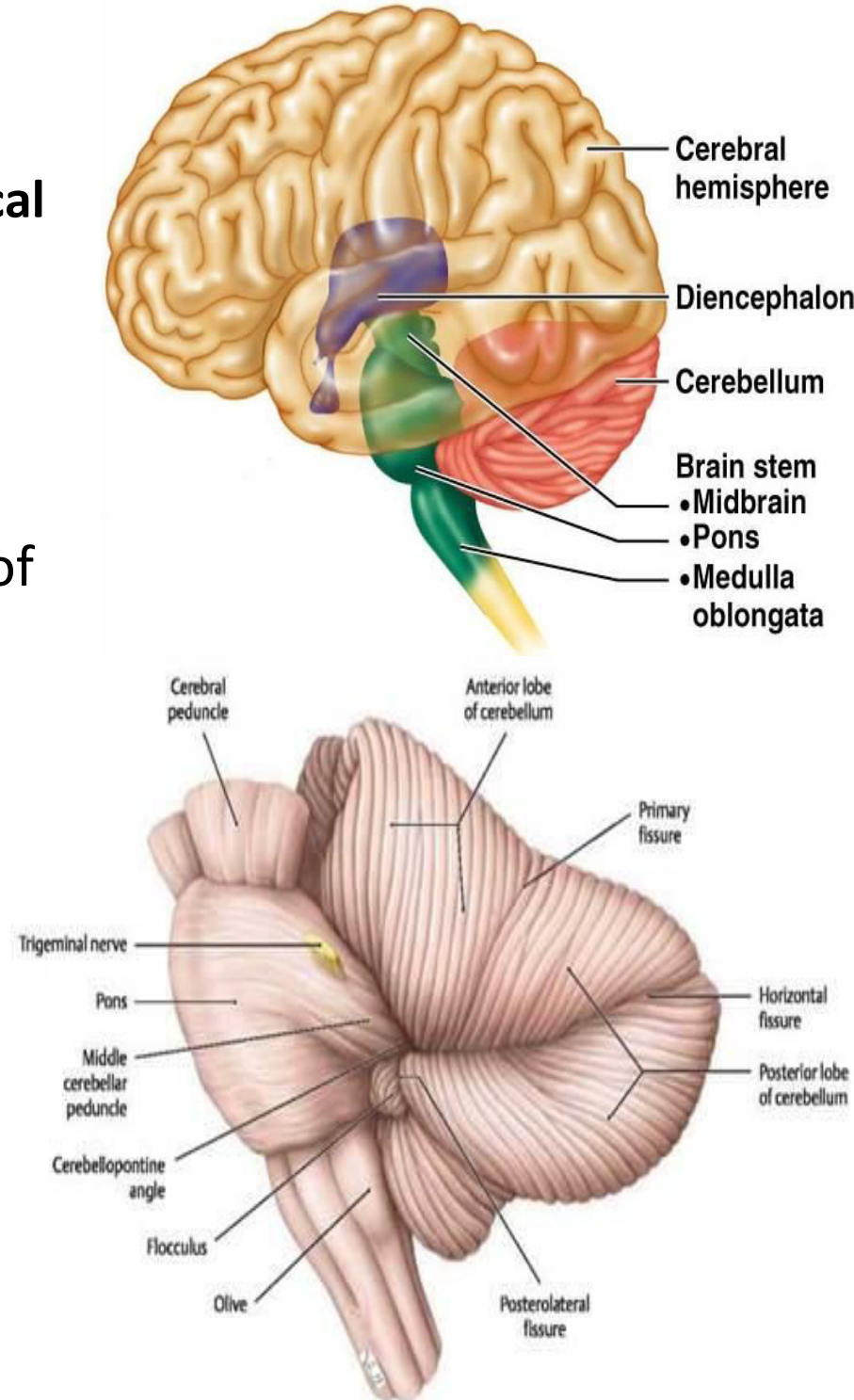
✓ Without this support of the vestibular nuclei, the pontine reticular system would lose much of its excitation of the axial antigravity muscles



**Contributions
of
Cerebellum and Basal Ganglia
to
Overall Motor Control**

Cerebellum (Latin for Little Brain)

- located behind the brain stem area
- long been called a silent area of the brain, because its electrical excitation
 - does not cause any conscious sensation and
 - doesn't initiate or cause movement
- however, its removal causes almost total incoordination of body movements
 - But its loss does not cause paralysis of any muscles
- **contributes to smooth coordination of motor activity**
- Cerebellum is especially vital during rapid motor activities such as running, typing, playing the piano, talking
 - Loss of cerebellum causes lack of coordination of these activities



How cerebellum is so important while no direct role to cause muscle contraction?

❖ The cerebellum receives continuously information

- about the desired sequence of muscle contractions from the motor control areas
- sensory information about the body position, rate and force of movement



– The cerebellum then *compares*

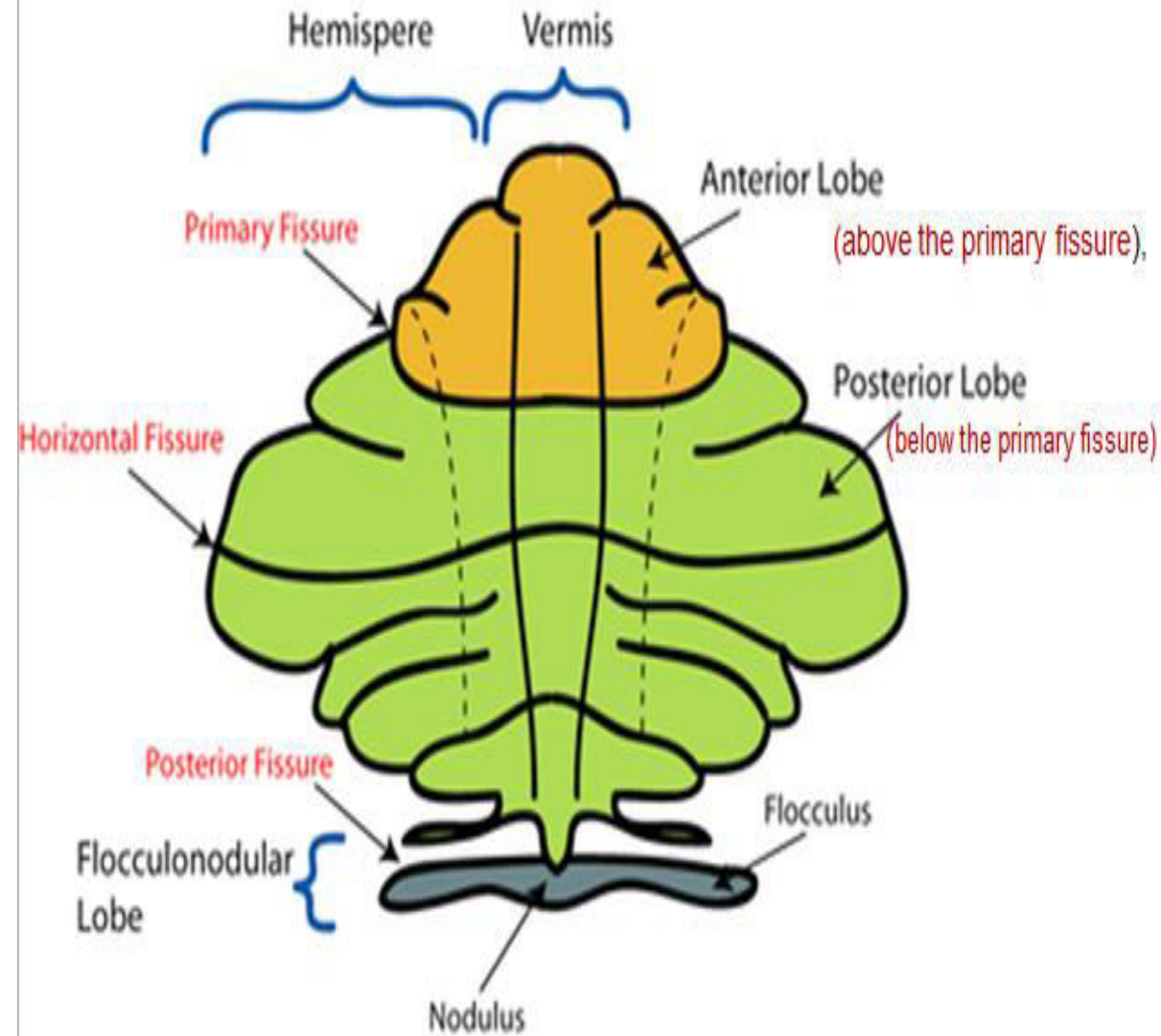
- the movements intended by the motor system.
- With the actual movements as depicted by the peripheral sensory feedback

❖ If the two do not compare favorably,

- It sends instantaneous subconscious corrective signals back into the motor system
- to increase or decrease the levels of activation of specific muscles.
- Hence, controls coordination of movements

Anatomic division of Cerebellum

- Anatomically, the cerebellum is divided into **three lobes** by two deep fissures
 - 1. the *anterior lobe*
 - 2. the *posterior lobe*
 - 3. the *flocculonodular lobe*
 - is the oldest of all portions of the cerebellum
 - it functions with the vestibular system in controlling body equilibrium
- Has three cerebellar peduncles which connect cerebellum to brain stem



Functional anatomy of Cerebellum

- shows the longitudinal divisions of the Anterior and Posterior Lobes.
 - **Vermis** = midline structures of cerebellum
 - separated from the remainder of the cerebellum by shallow grooves.
 - Responsible for control of muscle movements of *axial body*:- *neck, shoulders, and hips*
 - ***Its damage causes Axial ataxia or truncal ataxia***
 - Two cerebellar hemispheres: each with 2 zones:- lateral & intermediate zones
 - **The intermediate zone** is concerned with controlling distal muscle contractions
 - in the upper and lower limbs, especially the hands, fingers, feet, and toes.
 - **Its damage causes Appendicular Ataxia**
 - **The lateral zone**: operates at a much more remote level
 - Works with cerebral cortex in the overall planning of sequential motor movements.
 - Without this zone, most discrete motor activities of the body lose their appropriate timing and sequencing and therefore become uncoordinated,

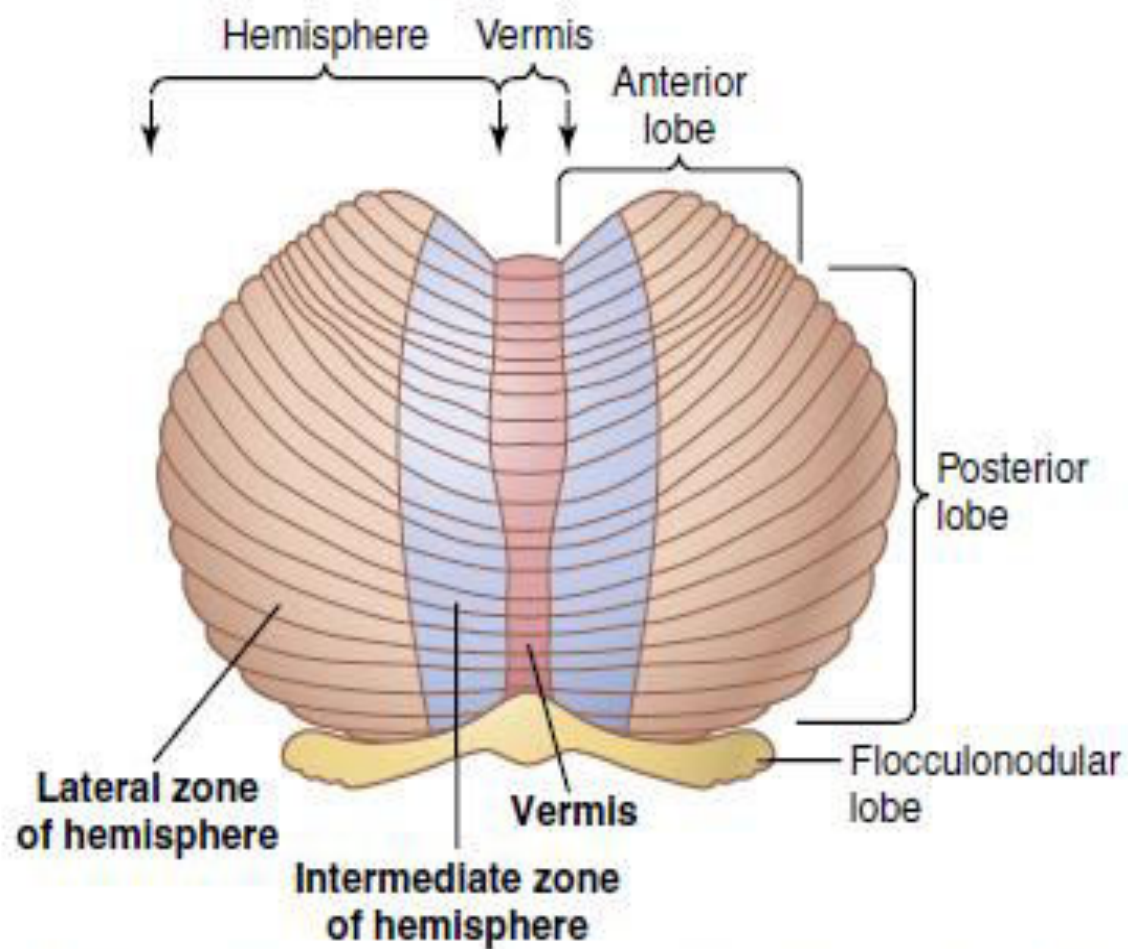
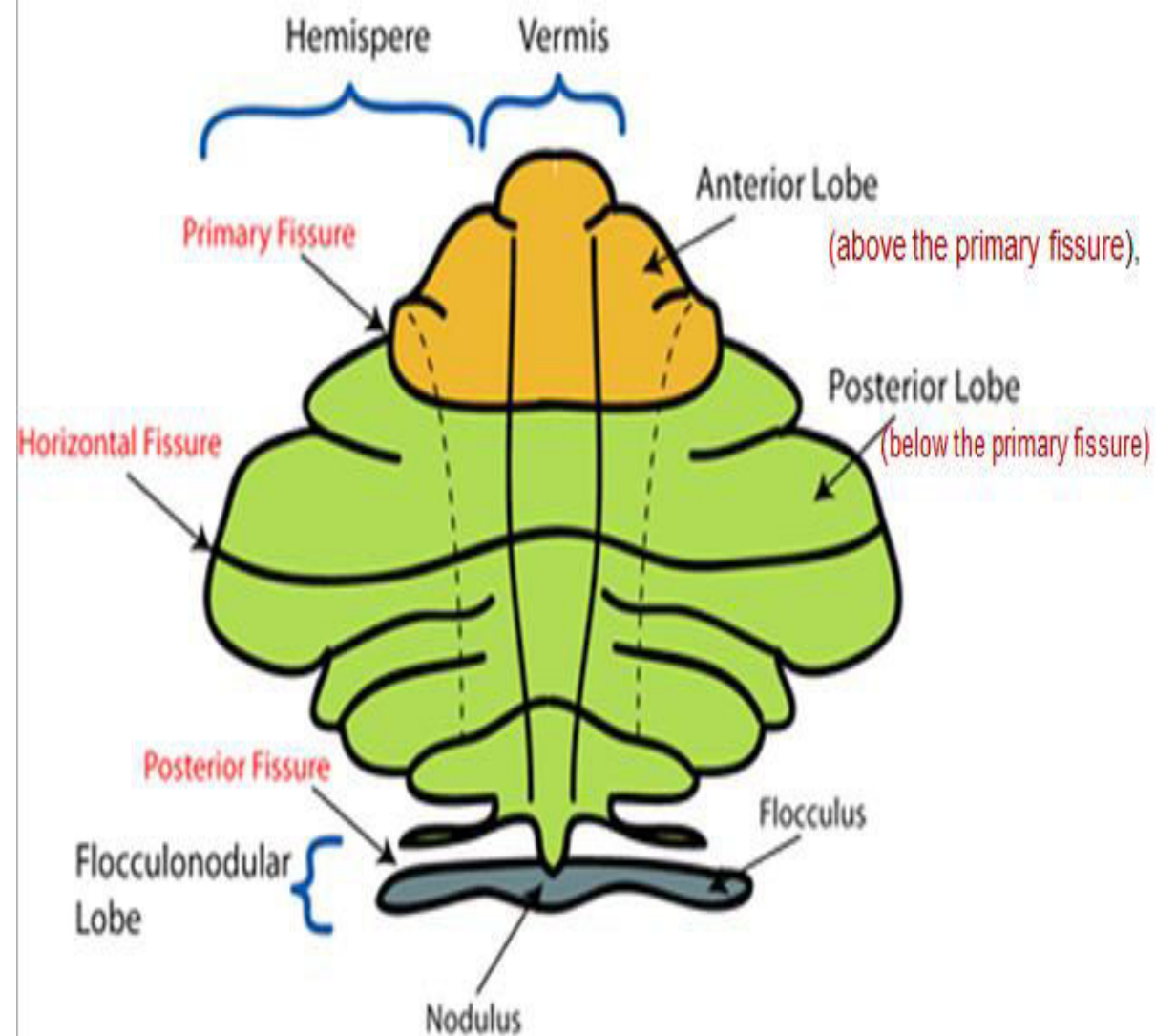


Figure 57-2 Functional parts of the cerebellum as seen from the posteroinferior view, with the inferiormost portion of the cerebellum rolled outward to flatten the surface.



Topographical representations of the cerebellum

- Vermis and intermediate zones have topographical representations of the different parts of the body
 - ❖ In the same manner that the
 - ✓ cerebral sensory cortex, motor cortex,
 - ✓ basal ganglia, red nuclei, and reticular formation
 - ❖ **the axial portions of the body lie in the vermis part**
 - ❖ **limbs and facial regions lie in the intermediate zones.**
 - ❖ These topographical representations receive
 - ✓ afferent nerve signals from all the respective parts of the body, as well as
 - ✓ from corresponding topographical motor areas in the cerebral cortex and brain stem.

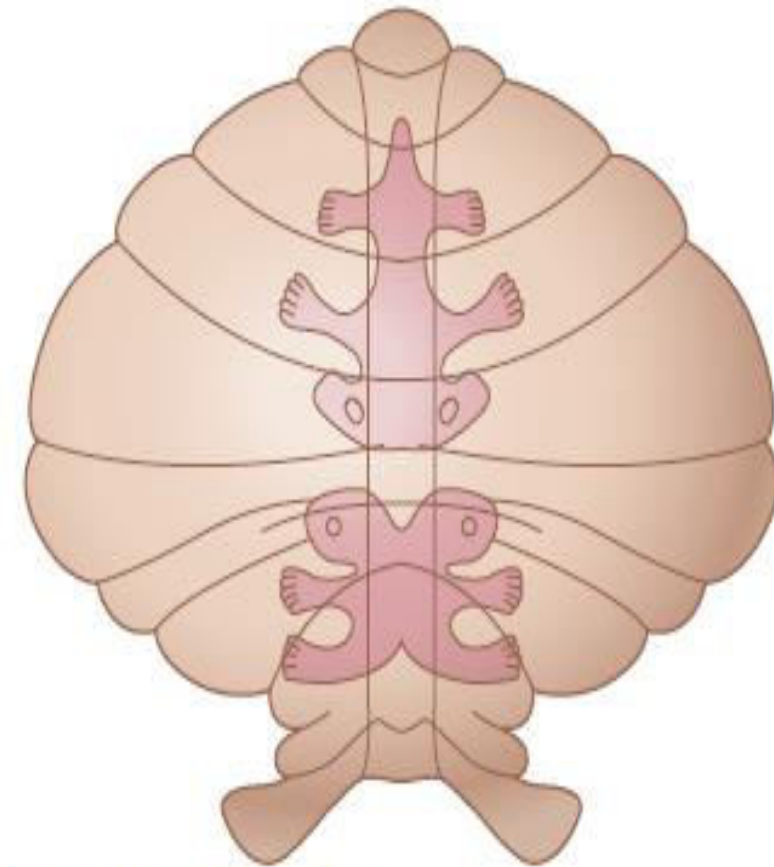


Figure 57-3 Somatosensory projection areas in the cerebellar cortex.

Layers of cerebellum

Cerebellum has two major parts

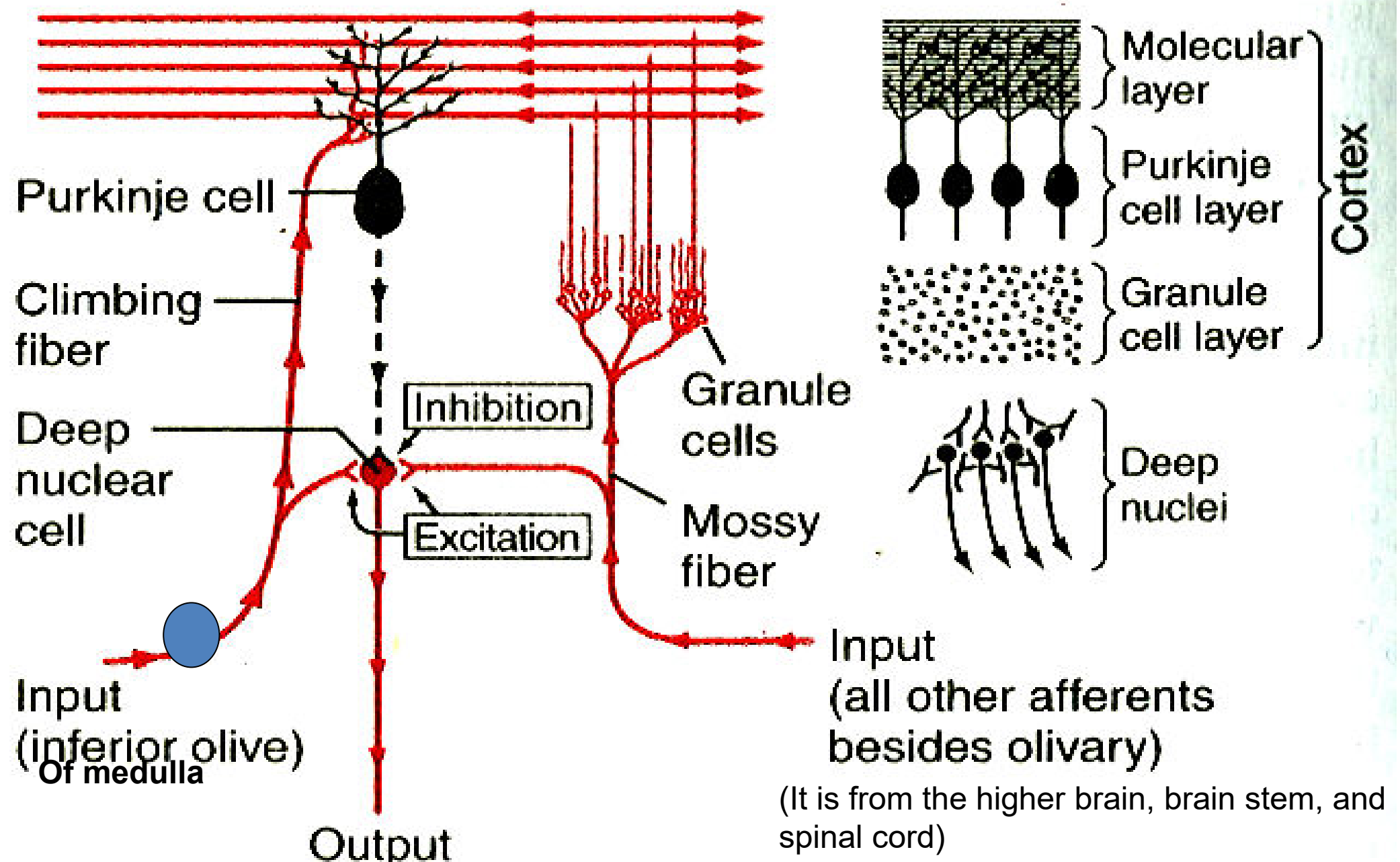
1. **Outer gray matter (cerebellar cortex): has three layers**

- Molecular layer = interconnecting fibers
- Purkinje cell layer = intermediate layer
- Inner granule cell layer = granule cells

2. **Deep white matter contains four nuclei: are beneath the cortex**

- **Dentate nucleus** ← Cerebellar hemisphere(lateral zone)
 - coordinate sequential motor activities initiated by the cerebral cortex
- **Nucleus interpositus (two)** ← Paravermal(intermediate zone)
 - helps to coordinate muscles in the limbs, especially in the hands, fingers, & thumbs.
- **Fastigial nucleus** ← Vermis + Flocculonodular lobe
 - Important for controlling posture and equilibrium

Cellular layers of the cerebellum



- ***Purkinje cell and deep nuclear cell form the functional unit of cerebellum***
 - The cerebellum has about 30 million nearly identical functional units
 - This functional unit centers on *Purkinje cell* and its corresponding *deep nuclear cell*.
- The afferent inputs to the cerebellum are mainly of two types,
 - one called the *climbing fiber type* and the other called the *mossy fiber type*.
 - The climbing fibers *originate from the inferior olives of the medulla*.
 - There is one climbing fiber for about 5 to 10 Purkinje cells.
 - The mossy fibers are from multiple sources: higher brain, brain stem, and spinal cord.
 - They then proceed to the granule cell layer of the cortex

- ❖ Thus, each time an input signal arrives in the cerebellum, it divides and goes in two directions:
 - (1) directly to one of the cerebellar deep nuclei and
 - (2) to a corresponding area of the cerebellar cortex overlying the deep nucleus.

↓ ↓
- ❖ Then, a fraction of a second later, the cerebellar cortex relays an *inhibitory* output signal back to the deep nucleus.
- ❖ Thus, all input signals that enter the cerebellum eventually end in the deep nuclei
 - in the form of initial excitatory signals
 - followed a fraction of a second later by inhibitory signals.
- ❖ From the deep nuclei, output signals leave the cerebellum

Cerebral and cerebellar control of voluntary movements

- **Cerebellar Input: cerebellum receives input**
 - from the **brain motor control areas** about the desired sequence of muscle contractions
 - sensory information about muscle activity from the peripheral parts of the body

↓
- **Integrates these inputs to fine tune motor activity**
 - The cerebellum compares the actual movements as depicted by the peripheral sensory feedback information with the movements intended by the motor system
 - If the two do not compare favorably, (if discrepancy exists)

↓
 - then **instantaneous subconscious corrective signals** are transmitted back into the motor system to increase or decrease the levels of activation of specific muscles

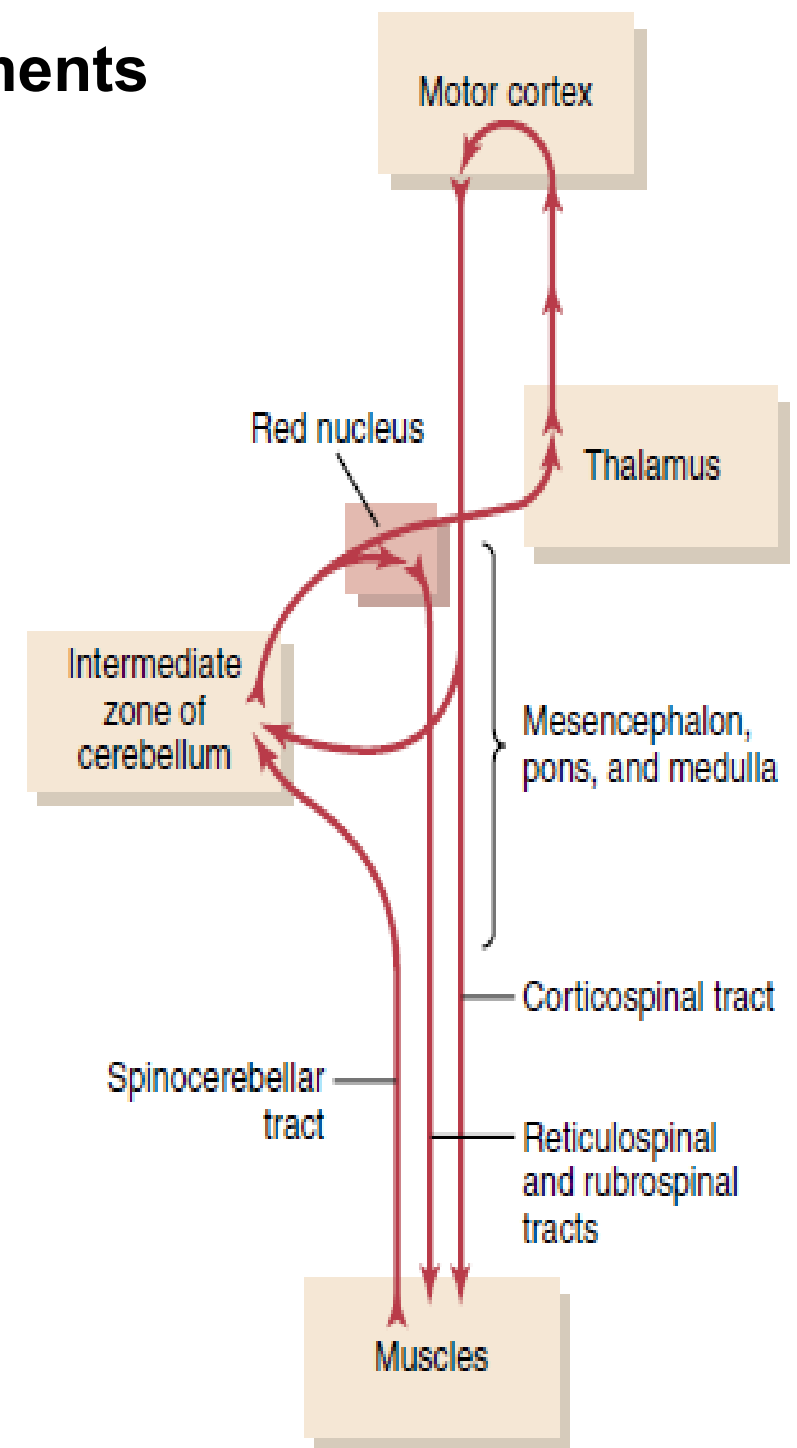


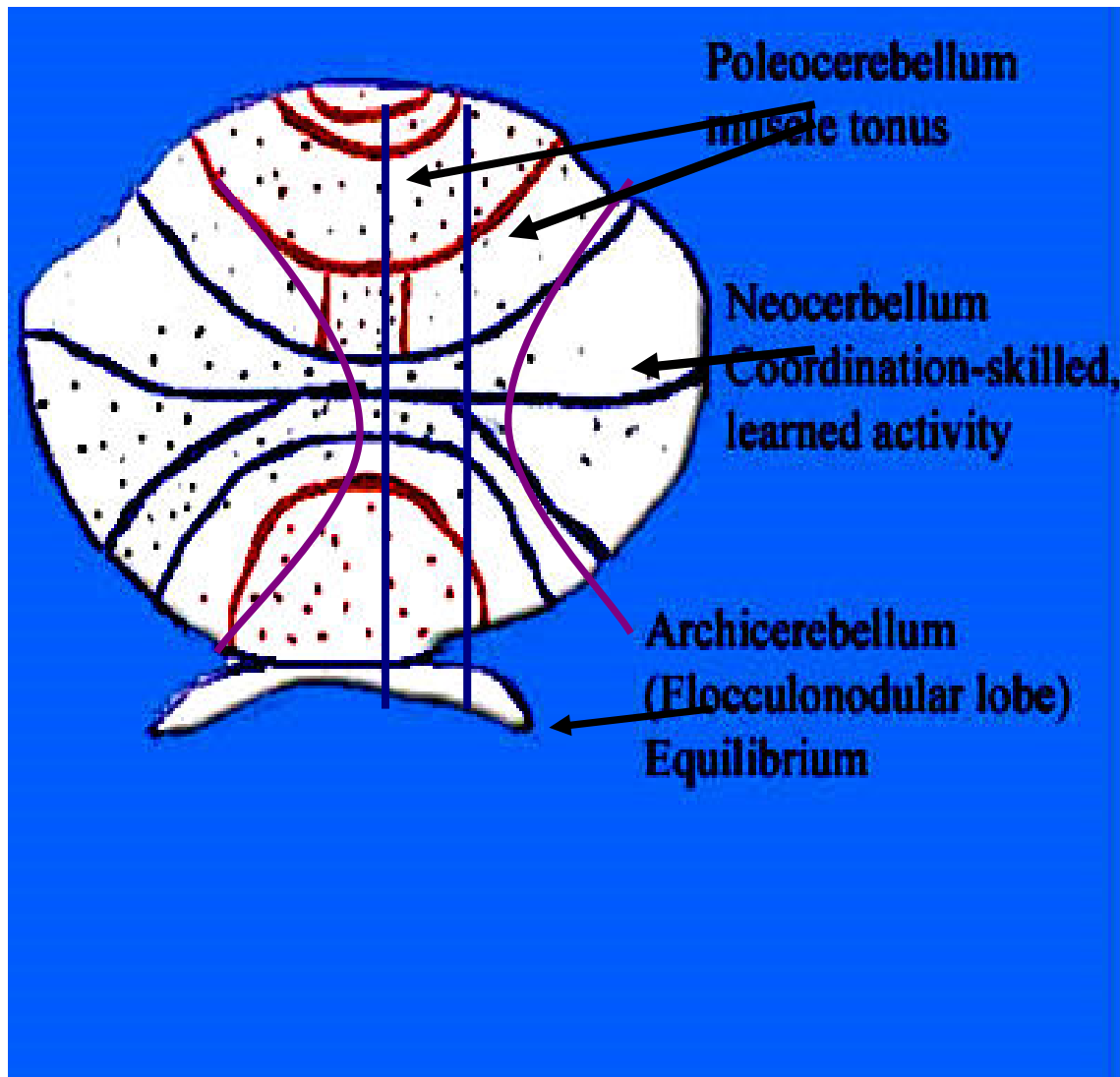
Figure 57-8 Cerebral and cerebellar control of voluntary movements, involving especially the intermediate zone of the cerebellum.

Physiological divisions of the cerebellum:

- ❖ The nervous system uses the cerebellum to coordinate motor functions at three levels
 - **1. Archicerebellum (vestibulocerebellum)**
 - consists principally of the small flocculonodular cerebellar lobes
 - **Connected to vestibular apparatus through vestibular nuclei**
 - **It provides neural circuits for maintenance of posture and equilibrium**
 - **2. Paleocerebellum (spinocerebellum) – connected to spinal cord**
 - **Consists of vermis and paravermal areas [Vermis + intermediate zones]**
 - **Receives signals from proprioceptors and also from cerebral cortex and midbrain**
 - ✓ comparing the intended movements with the actual movements
 - provides circuitry for coordinating movt of axial and appendicular muscles
 - ✓ Prevents dysmetria of voluntary movements
 - ✓ In its absence, the subconscious motor control system cannot predict how far movements will go

- **3. Neocerebellum (cerebocerebellum) – largest part**
 - Consists the large Lateral zone of cerebellar hemisphere
 - **This zone has no direct input of info from the peripheral parts of the body.**
 - receives all its input from primary motor cortex and PMA and somatosensory cortices
 - transmits its output information in the upward direction back to the brain
 - Functions in a feedback manner with cerebral cortex
 - to plan sequential voluntary body and limb movements.
 - To provide appropriate “timing” of the sequential movements.
 - Helps to Progress smoothly from one movt to the next in orderly succession.
- ❖ destruction of these zones along with their deep nuclei(the dentate nuclei),
 - can lead to extreme incoordination of complex purposeful movements of the hands, fingers, and feet and of the speech apparatus.

Fig. Physiological divisions of the cerebellum



- **Archicerebellum (vestibulocerebellum)**
 - Connected to vestibular apparatus through vestibular nuclei
 - Functions in association with brain stem, and spinal cord
 - For maintenance of balance and equilibrium
- **Paleocerebellum (spinocerebellum)**
 - Consists of vermis + paravermal areas
 - Receives signals from peripheral proprioceptors
 - Coordinates the axial and appendicular muscle contractions
- **Neocerebellum (cerebellocerebellum)**
 - Lateral part of cerebellar hemisphere
 - Connected to cerebral cortex
 - Involved in planning the sequential voluntary body movements.

Summary on Functions of the cerebellum

1. Control of posture and equilibrium

- Signals from vestibular apparatus and proprioceptives are compared to adjust posture and equilibrium
- These signals are used for predictive function during rapid motion that anticipate disturbances

2. Control of muscle tone

- Cerebellar damage may cause Hypotonia—Ipsilateral decreased Tone of the musculature
 - ✓ Due to loss of cerebellar facilitation of the motor cortex and brain stem motor nuclei by tonic signals from the deep cerebellar nuclei

3. Control of voluntary movement

- **Planning:** Efferent copy about motor intension and afferent copy about motor activity
- **Timing of movements:** determines the start and finish of any sequential movement
- **Joining of** succession of sequential movement

4. Predictive function:

- Knows rate and direction of any movement
- Eg. Thrust of goal keeper to catch a kicked ball

5. Servocomparator

- **Intension of cortex and muscles performance comparison**

6. Damping of movement

- If the cerebellum is intact, appropriately learned, subconscious signals stop the movement precisely at the intended point,
 - ✓ Ending movement abruptly without oscillation
 - ✓ **Lack of this leads to intension or kinetic tremor**

Overview of Clinical Abnormalities of Cerebellum

1. Ataxia: derived from the Greek word *taxis*, meaning “order”

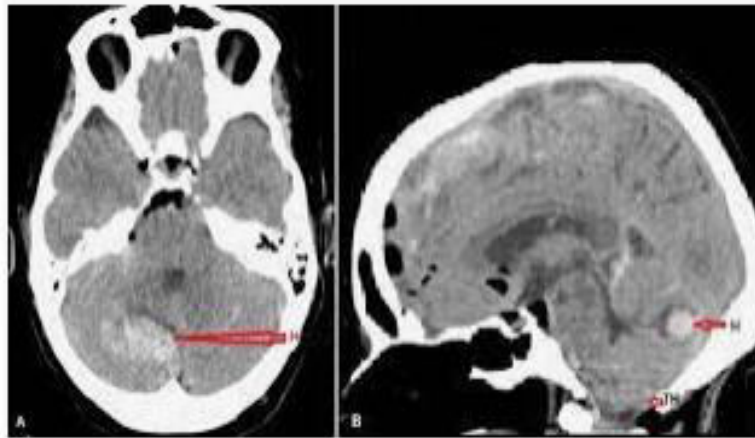
- Is a syndrome of imbalance and incoordination involving gait, limbs, and speech
 - Is usually results from a disorder of the cerebellum and/or its connections.
- True cerebellar ataxia must be distinguished from
 - **ataxia associated with vestibular nerve or labyrinthine disease,**
 - ✓ This ataxia has high degree of dizziness, light-headedness, or the perception of movt
 - ✓ True cerebellar ataxia is devoid of these vertiginous complaints
 - **sensory ataxia:** due to disturbance of proprioceptive input
 - ✓ caused by pathologies along the dorsal sensory pathways
 - ✓ The imbalance dramatically worsens when visual input is removed (Romberg sign +ve).
 - ✓ Eg. B12 deficiency, Tabs dorsalis of teritiary neurosyphilis, ...

- Generalized cerebellar lesions give rise to more symmetrical symptomatology.
- Localized cerebellar lesions cause ipsilateral symptoms and signs
- Acute focal ataxia commonly results from cerebrovascular Accident(CVA)
 - **usually ischemic infarction or cerebellar hemorrhage**
 - may be associated with an impaired level of consciousness
 - ✓ due to brainstem compression and increased intracranial pressure
 - ✓ true neurologic emergencies, if herniation occurs and is usually devastating

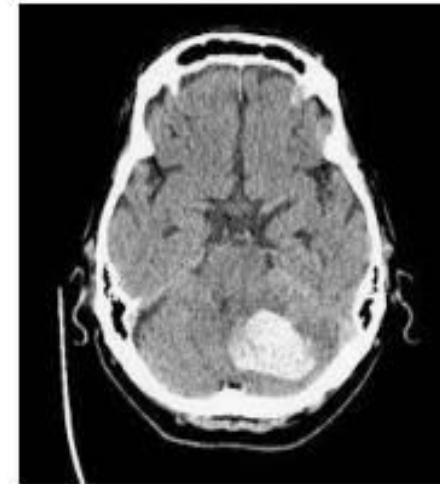
CT scan of Cerebellar hemorrhage



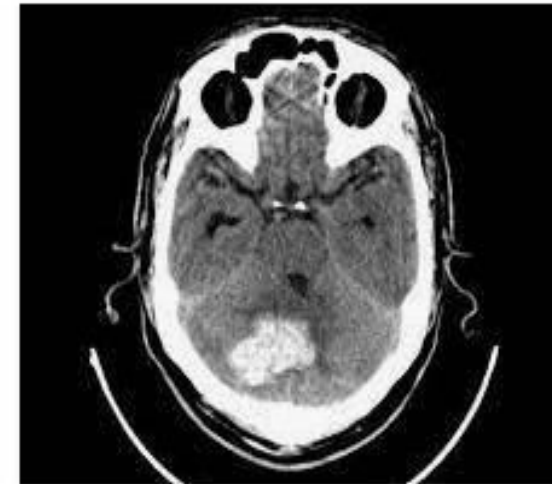
Cerebellar hemorrhage | Radiology Ref...
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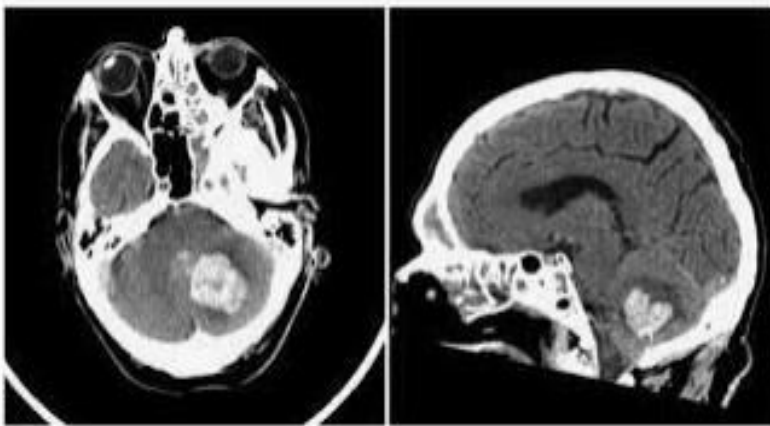
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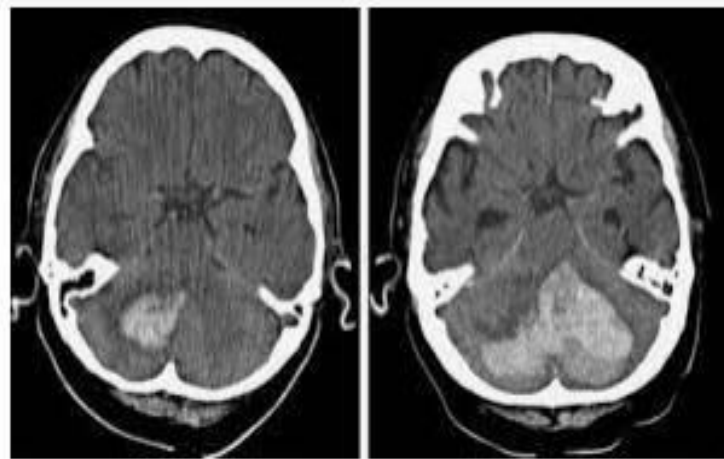
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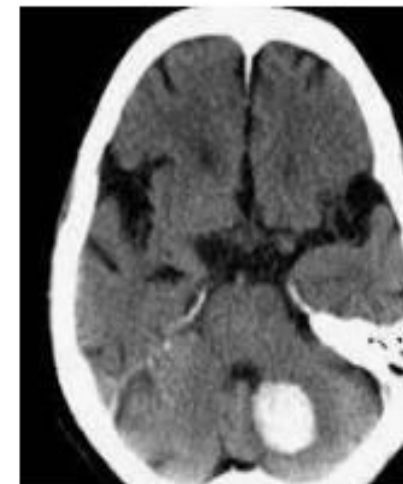
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operativeneurosurgery.com



Primary cerebellar haemorrhage: Complications, tre...
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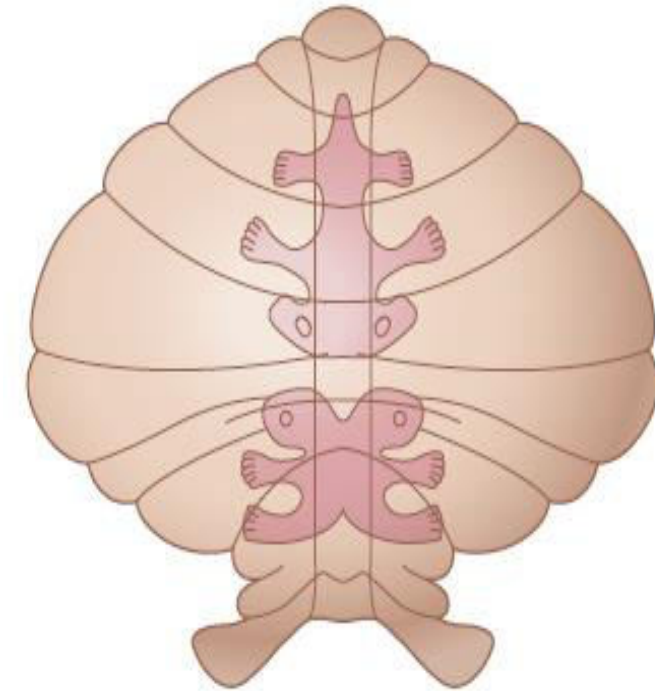
Cerebellar hemorrhage as an ...
ijidonline.com



Cerebellar Intracerebral Hem...
case.edu

NB: Cerebellar lesions typically produce cerebellar symptoms ipsilateral to the injured cerebellum

- **Truncal Ataxia:** Midline cerebellar lesions cause truncal ataxia.
 - Patients may experience head tremor and truncal instability
 - leading to oscillatory movts of the head and trunk while sitting or standing
 - They may need back support while sitting.
- **Limb Ataxia (Appendicular Ataxia)**
 - Ataxic diseases cause a variety of symptoms in the limbs,
 - resulting from incoordination and tremor
 - Clumsiness of movts during writing, picking up objects, and buttoning.
 - These symptoms are one-sided with lateralized lesions of the cerebellum.



2. Dysmetria

- refers to an inaccuracy of movement in which the desired target is either under-reached (hypometria) or over-reached (hypermetria).
- **In the absence of the cerebellum, the subconscious motor control system cannot predict how far movements will go**
 - It is a disturbance of the rate, range, and force of movement.

3. Kinetic Tremor or Intention Tremor.

- oscillations of the limb during a voluntary movt intended to reach a target;
- the tremor often increases in amplitude as the target is reached.
- **It is detected during the finger-to-nose test and the heel-to-shin test.**

4. Dysdiadochokinesia.

- refers to irregularity of the rhythm and amplitude of rapid alternating movements.
 - **Inability to perform rapid alternating movements**
- can be demonstrated by turning one hand upward and downward at a rapid rate.
 - Abnormal: The patient will have jumbled movements
 - normal: coordinated rhythmic upward and downward motions.

5. Dysarthria—Disorder of articulation and phonation

- Is difficulty of producing clear and normal speech (scanning speech)
- Normal word formation depends on rapid and orderly succession of individual muscle movements in the larynx, mouth, and respiratory system.
 - Dysarthria is a lack of coordination among these structures
 - Results in jumbled vocalization, with some syllables loud, some weak, some held for long intervals, and some for short intervals

6. Cerebellar Nystagmus—Tremor of the Eyeballs

- is tremor of the eyeballs when attempted to fixate the eyes to one side of the head.
 - Is a rapid, tremulous movements of the eyes rather than steady fixation
 - is another manifestation of the failure of damping by the cerebellum
 - **It occurs especially when the flocculonodular lobes of the cerebellum are damaged**
 - it is also associated with loss of equilibrium
 - ✓ because of dysfunction of the pathways of flocculonodular with semicircular ducts.

7. Hypotonia—Decreased Tone of the Musculature

- decreased tone of the musculature on the side of the cerebellar lesion.
 - Due to loss of the deep cerebellar nuclei,
 - **particularly of the dentate and interposed nuclei**

THE BASAL GANGLIA AND THEIR MOTOR FUNCTIONS

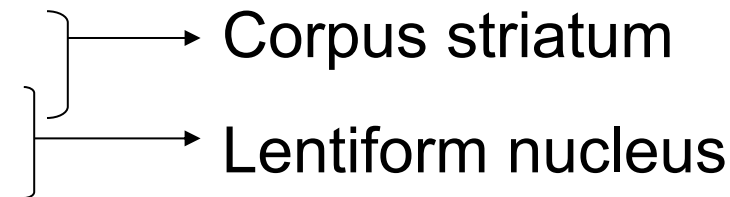
Basal Ganglia

- collection of nuclei beneath the white matter of the cerebral cortex
- Are also Subcortical masses of gray matter found lateral to the thalamus

- They include

- ❖ Three large nuclear masses

- ✓ Caudate nucleus(Paleostriatum)
 - ✓ Putamen (Neostriatum)
 - ✓ Globus pallidus (External & internal)



- ❖ Two functionally related nuclei

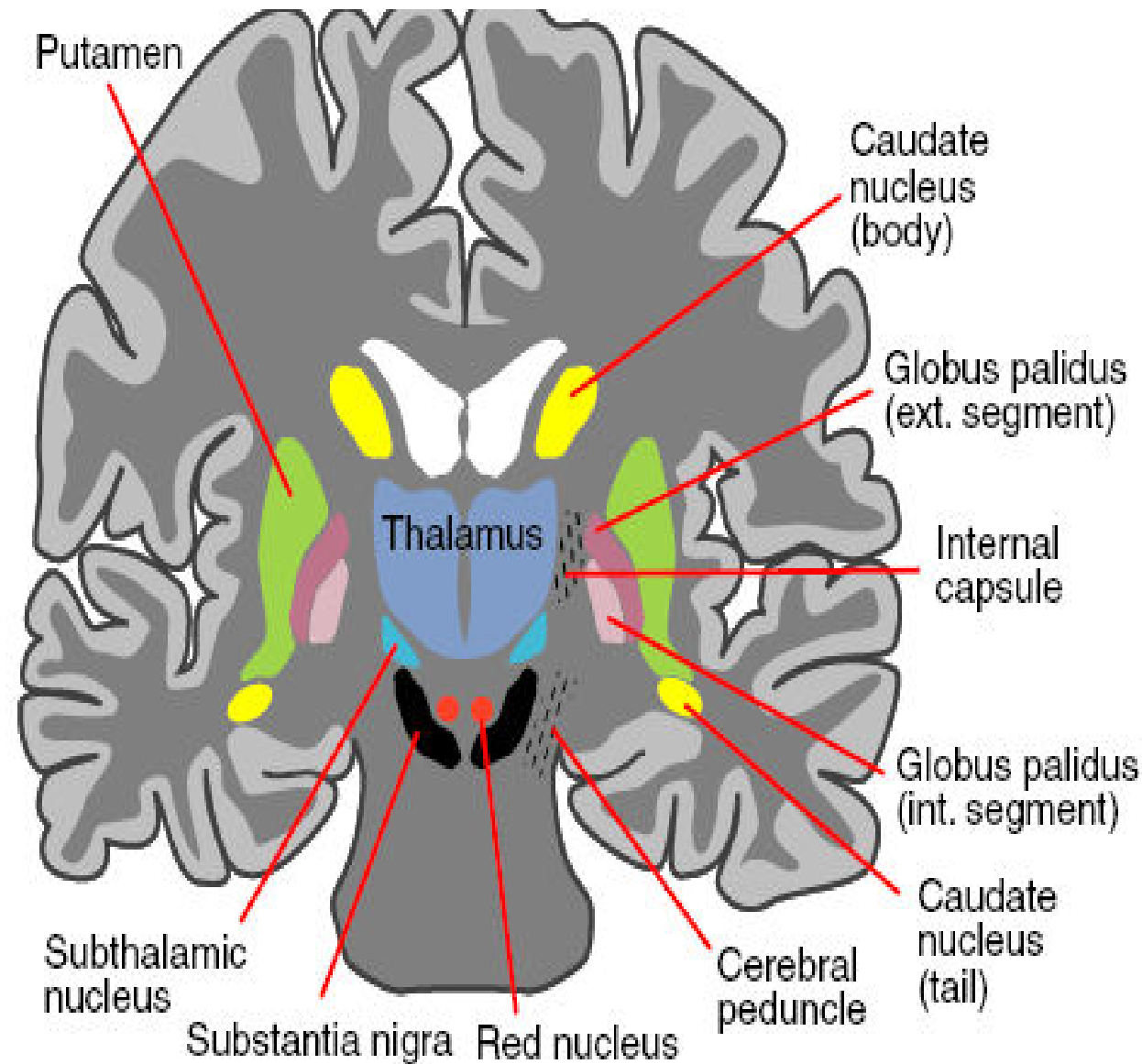
- ✓ Subthalamic nucleus
 - ✓ Substantia Nigra: pars compacta, pars reticulata

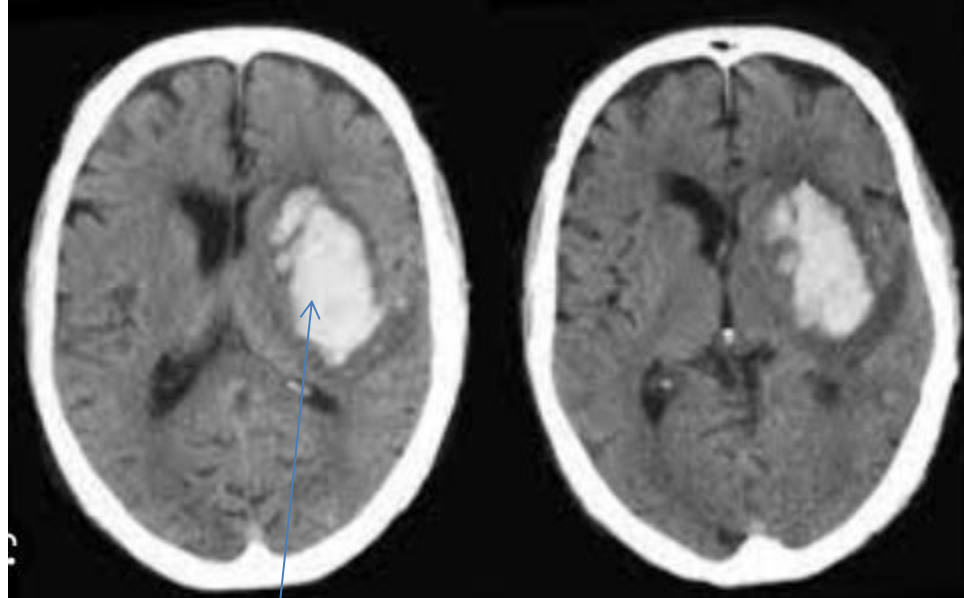
- INTERNAL CAPSULE SEPARATES CAUDATE NUCLEUS AND PUTAMEN.

- ✓ ALMOST ALL MOTOR AND SENSORY NERVE FIBERS CONNECTING THE CEREBRAL CORTEX AND SPINAL CORD PASS THROUGH THIS SPACE

Basal Nuclei

- ❖ like the cerebellum, constitute another *accessory motor system* that functions usually not by itself
- ❖ but in close association with the cerebral cortex and corticospinal motor control system
- ✓ **Don't initiate movements**
- ✓ but once movement is underway, they assist the pattern and rhythm (especially for trunk and proximal limb muscles)





(Putamen is the most common site of HTN hemorrhage)

INTERNAL CAPSULE

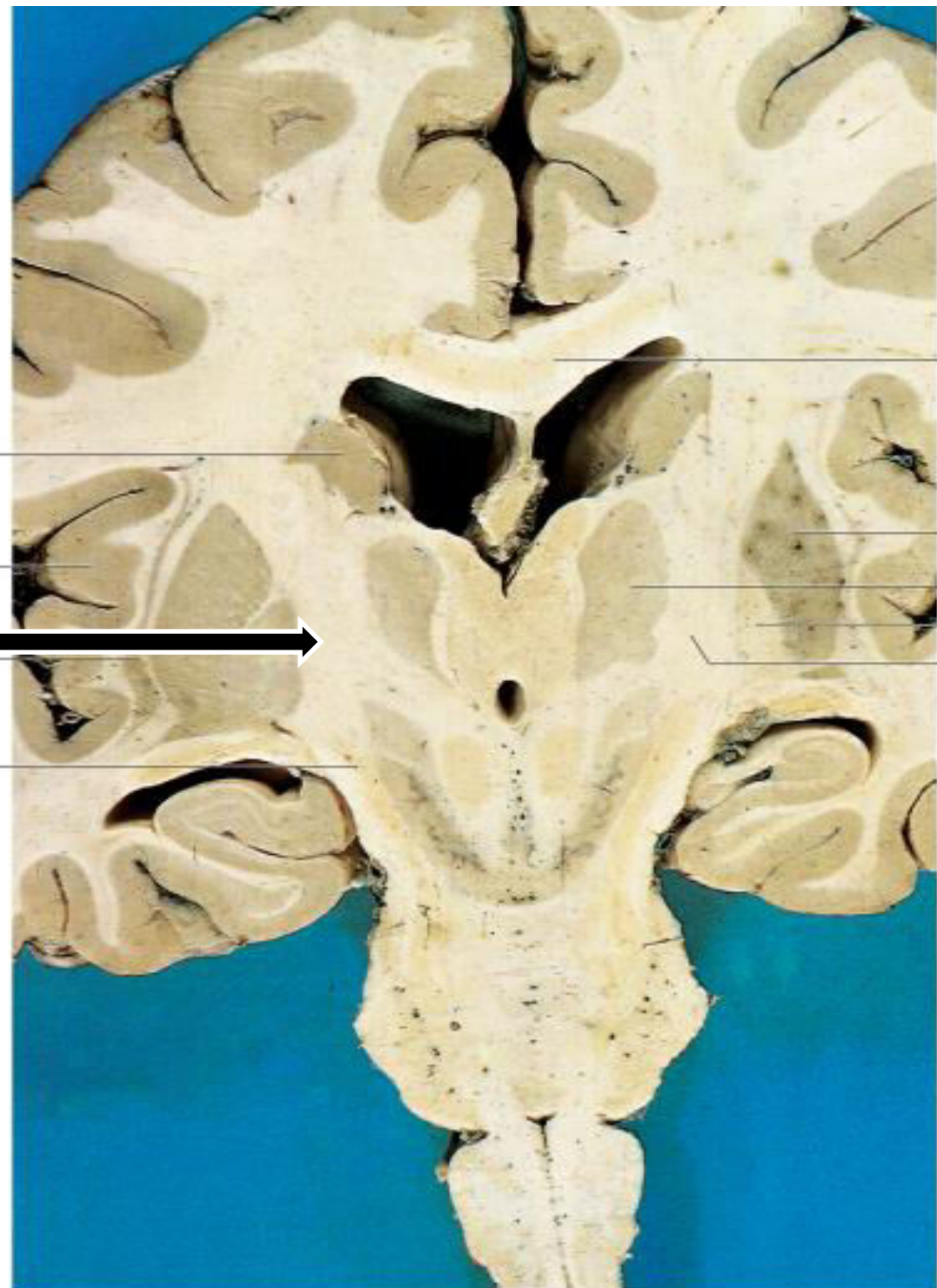
- ❖ INTERNAL CAPSULE SEPARATES CAUDATE NUCLEUS AND PUTAMEN.
- ❖ Almost all motor and sensory nerve fibers connecting the cerebral cortex and spinal cord pass through this space
- ❖ Thus, it is intimately associated with corticospinal motor control system
- ❖ Eg. Any injury in the putamen can affect the corticospinal tract at its proximity and hence weakness

Caudate nucleus

Insula

Caudate nucleus

Crus cerebri



Corpus callosum

Putamen

Thalamus

Globus pallidus

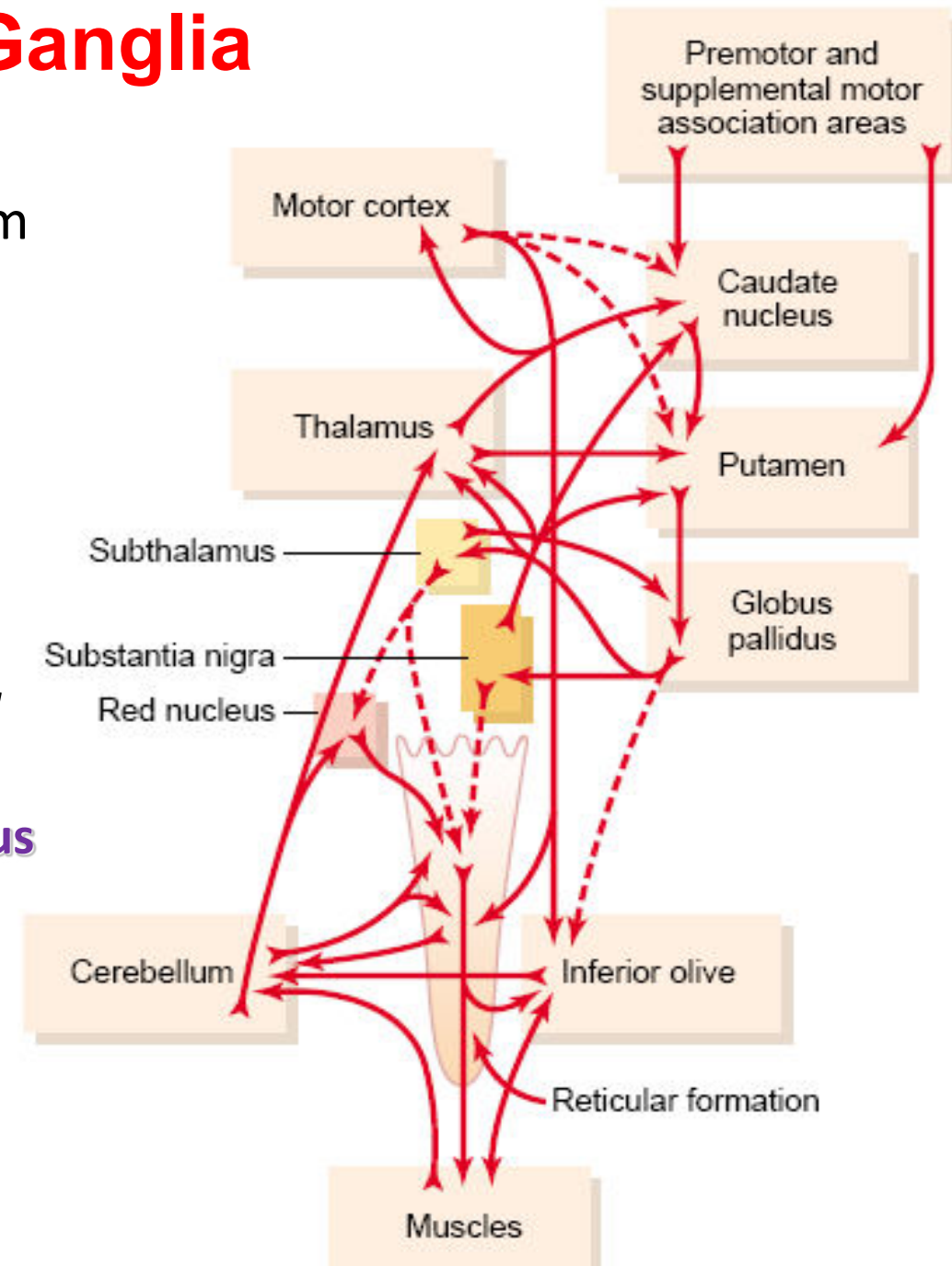
Internal capsule

Figure. Oblique coronal section of the brain

Activa

Neuronal Circuitry of the Basal Ganglia

- Basal ganglia receive most of their input signals from the cerebral cortex and
- Also return almost all their output signals back to cortex
- **Afferent** connection to the basal ganglia from cortex
 - ❖ mainly terminate in the striatum (**caudate nucleus + putamen**)
 - ✓ considered as the *doorway into the basal ganglia*
- **Efferent** connections mainly originate in **Globus pallidus**



Relation of the basal ganglia circuitry to the corticospinal-cerebellar system for movement control.

Basal Ganglia in Executing Patterns of Learned Motor Activity: PUTAMEN CIRCUIT

- ❖ Function in association with corticospinal system to control *complex patterns of learned activity*.
 - Writing letters, hammering nails, passing a football, shooting a basket ball,
 - most aspects of vocalization,
- ❖ Basal ganglia is important virtually in any of our **skilled movements**,
 - ✓ most of them are performed subconsciously.
 - ✓ **Basal ganglia controls the timing and intensity of movement**

↓↓↓↓↓
- ❖ Two important capabilities of the brain in controlling movements are:
 1. To determine how rapidly the movement is to be performed and
 2. To control how large the movement will be.
 - Eg. Speed of writing , proportional characteristics of the letter.
- ❖ If there is serious damage to basal ganglia,
 - the cortical system of motor control can no longer provide these patterns.
 - Instead, writing becomes crude, as if one were learning for the 1st time

Neural Pathways of Putamen circuit

- ✓ **principal pathway for executing learned patterns of mov't**
- ✓ begins mainly in the premotor and SA of the motor cortex and in the somatosensory areas of the sensory cortex.



pass to the putamen (bypassing the Caudate Nucleus),



Then to the internal portion of the Globus Pallidus,



Ventroanterior and ventrolateral relay nuclei of the thalamus



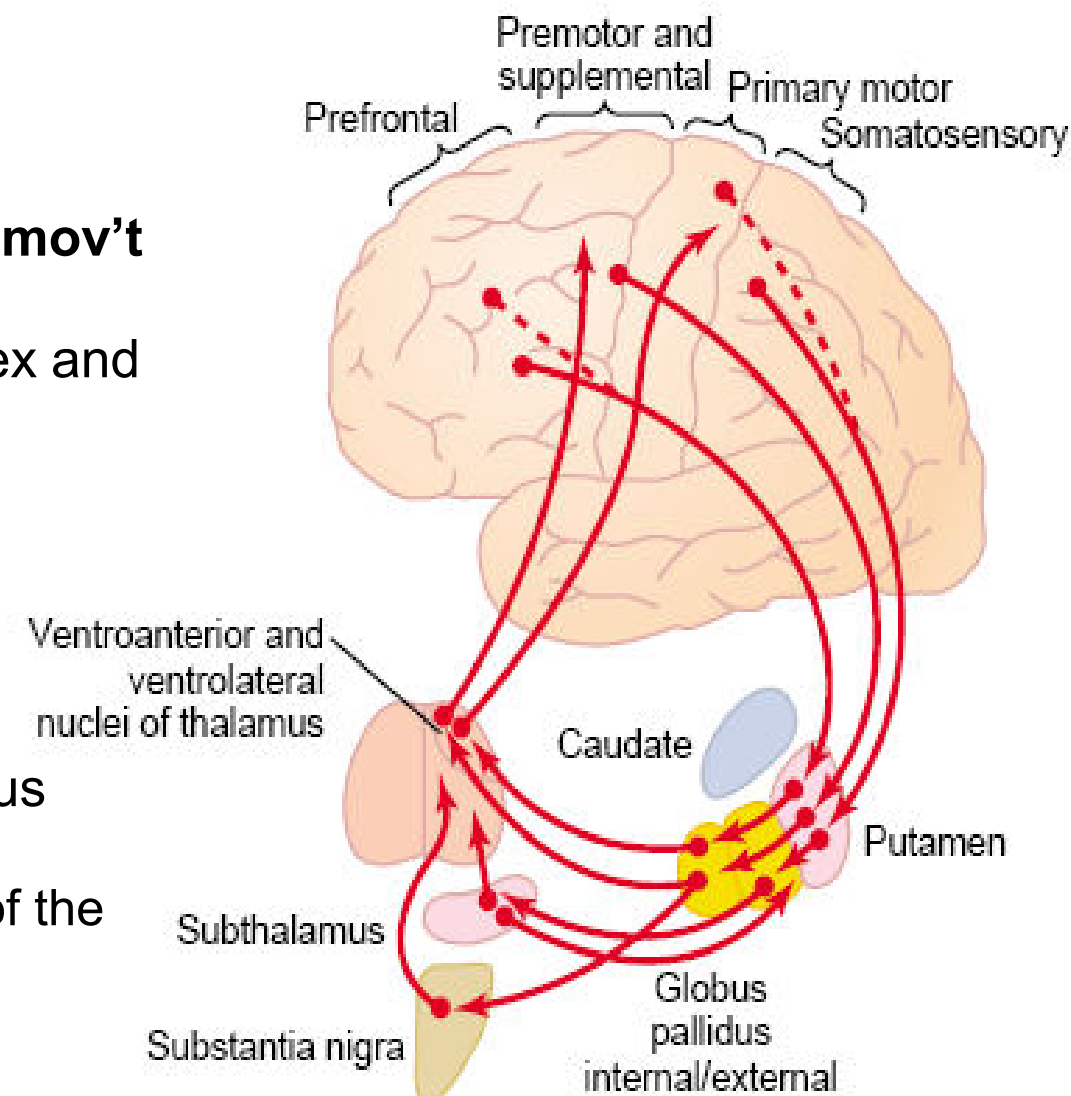
Finally return to cerebral primary motor cortex & to portions of the premotor and supplementary cerebral areas

Role of Putamen circuit

- ✓ for subconsciously executing learned patterns of movt
 - **eg. Signature, writing, lightening candle**

✓ **Damage to this circuit**

- **Motor apraxia-** Inability or difficulty to carry out familial movements in the absence of motor paralysis



Putamen circuit through the basal ganglia for subconscious execution of learned patterns of movement.

The Caudate Circuit

- ❖ Responsible for the role of Basal Ganglia in **cognitive control of motor activity**.
- ❖ what do we mean by cognition?
 - ❖ it is the thinking processes of brain, using both sensory input to the brain + information already stored in memory
- ✓ Unlike putamen circuit, it begins from all lobes of cerebrum including premotor & SMA and from somatosensory
 - ↓
- ✓ Caudate nucleus
 - ↓
- ✓ Putamen, Globus pallidus(internal)
 - ↓
- ✓ to the relay nuclei of the ventroanterior & ventrolateral thalamus
 - ↓
- ✓ Back to prefrontal, premotor & Supplementary motor area
 - ❖ **are concerned with building sequential patterns of movement within few seconds**
 - ❖ Eg. Subconsciously and abruptly determine how to scape from approaching lion
 - ❖ Subconsciously protecting the face using hands from the offense on the face

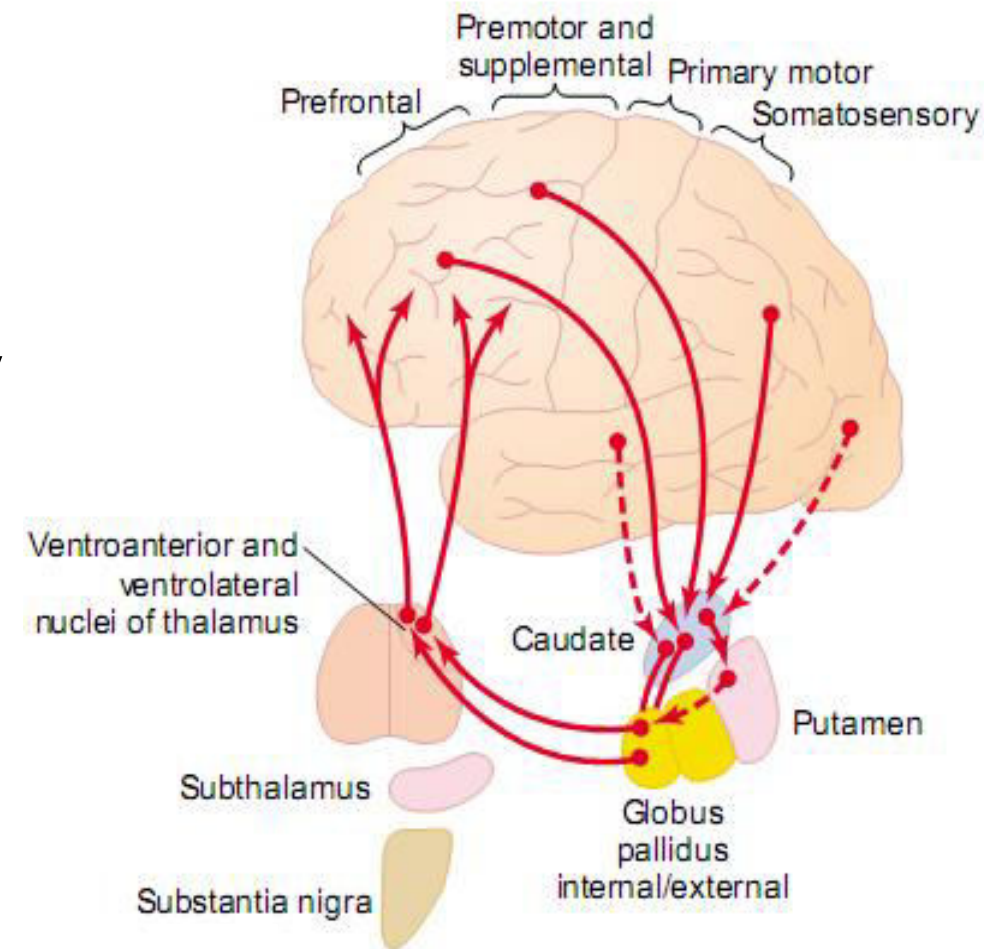
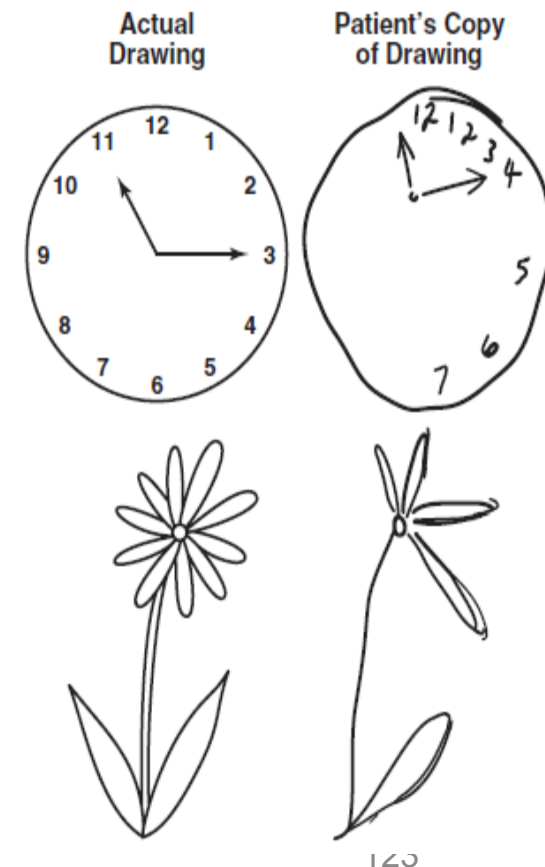


Fig Caudate circuit through the basal ganglia for cognitive planning of sequential and parallel motor patterns to achieve specific conscious goals.

Role of the caudate circuit

- **Converts thoughts and ideas in a motor plan into motor action**
 - What pattern of movement will be used
 - In what sequence to achieve a complex goal
- **Determines the time and scale movement**
 - To what extent the movement will be fast
 - For how long the movement will last
- ❖ ***Effects of Damage to caudate circuit(particularly the posterior parietal lobe)***
 - **Wearing neck tie before a shirt (failing to organize)**
 - **Failure to scale a contralateral side (when drawing)**



Functions of Specific NT in the Basal Ganglial System

- 1. **DA** pathways from Substantia Nigra(SN) to Caudate Neucleus(CN) and putamen,
- 2. **GABA** pathways from CN and putamen to GP and SN,
- 3. **Ach** pathways from cortex to CN and putamen, and also from CN to Putamen
- 4. multiple general pathways from brain stem
 - secrete **NE, serotonin, enkephalin etc ..**
- Multiple Glutamate pathways**
 - provide most of the excitatory signals
 - balance out the large numbers of inhibitory signals by the DA, GABA, and serotonin

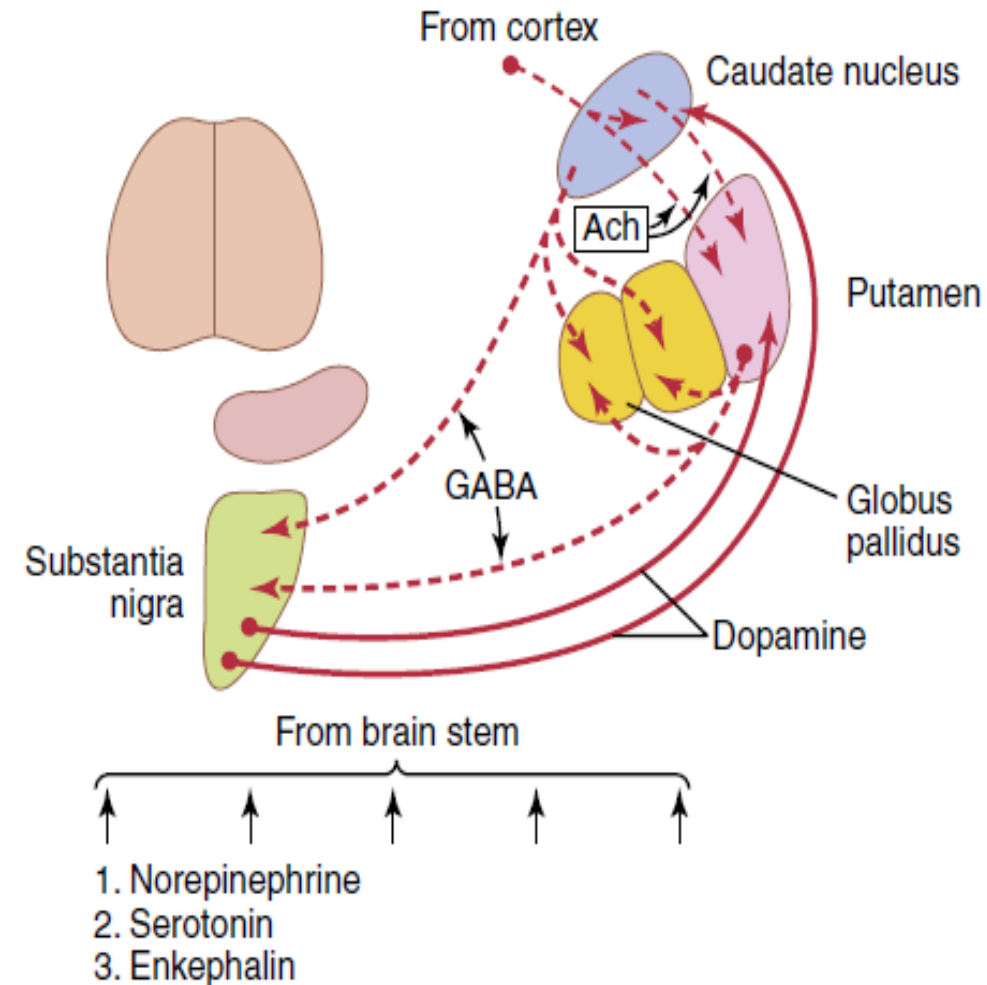


Figure 57-14 Neuronal pathways that secrete different types of neurotransmitter substances in the basal ganglia. Ach, Acetylcholine; GABA, gamma-aminobutyric acid.

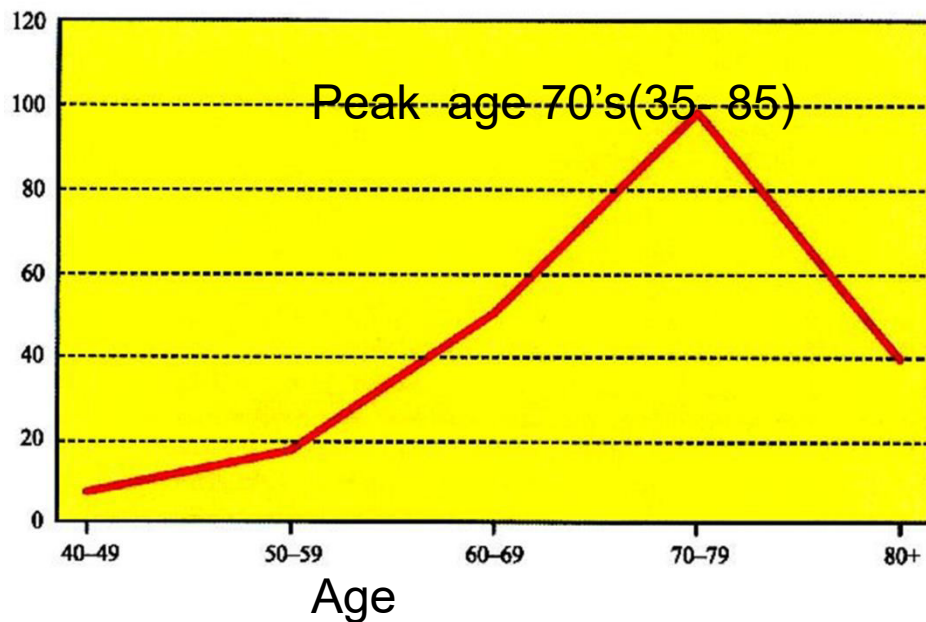
Parkinson's Disease (Paralysis Agitans) - hypokinetic movement disorder

- ❖ Degenerative disorder of CNS that impairs motor skills, cognitive processes, and other functions
- ❖ Due to widespread destruction of portion of the SN that sends DA-secreting nerve fibers to the CN and putamen



✓ **DEFICIENCY OF DOPAMIN (DA)**

- ❖ **Accelerated by Aging, Atherosclerosis, Phenothiazine tranquilizers**
- ❖ usually appears at age >60 and peak at 70's, although there are young-onset cases.



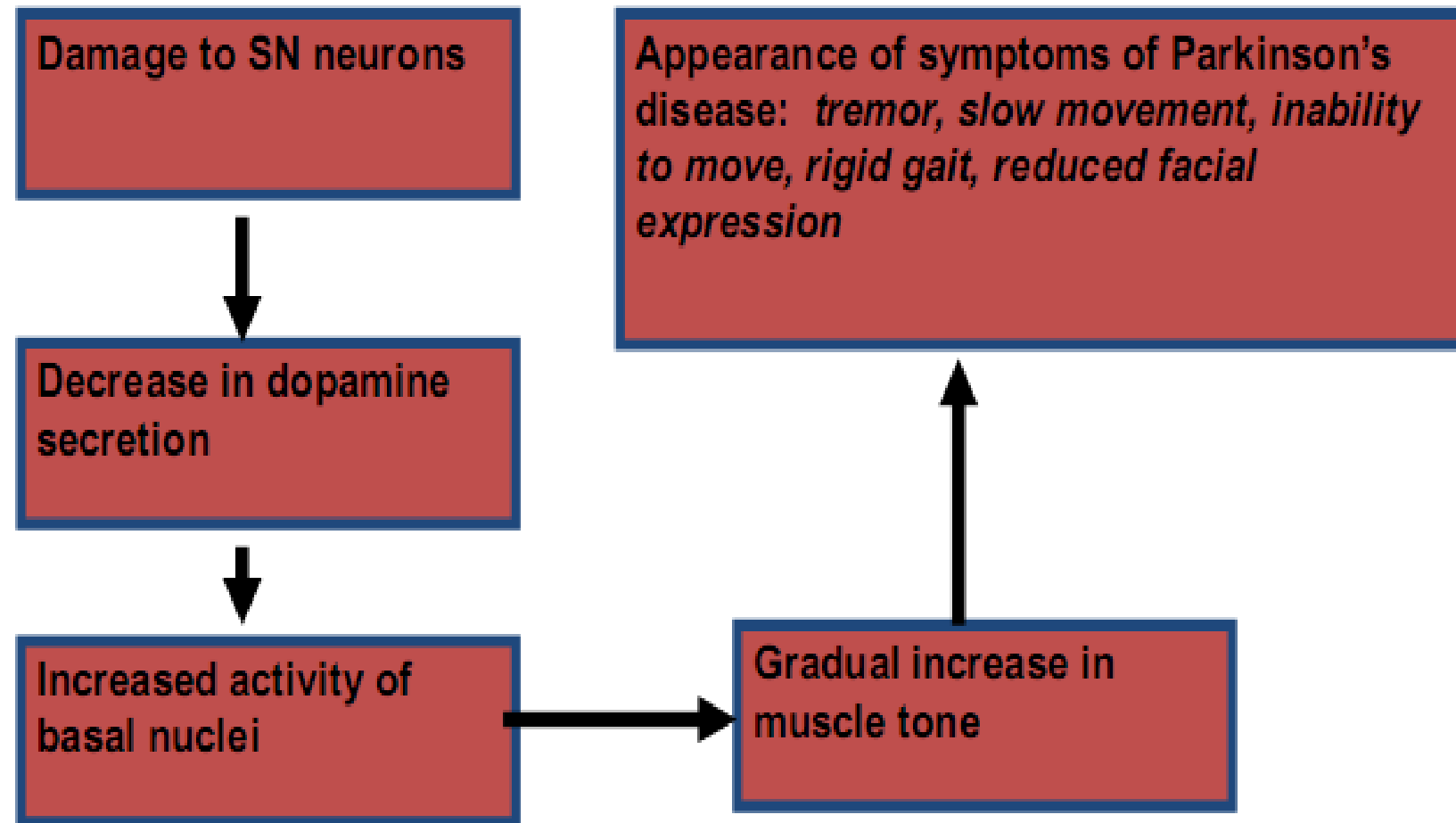
John Paul II



Muhammad Ali

- **Parkinson's Disease**

- Each side of the midbrain contains a nucleus called the *substantia nigra*.
- Neurons in the substantia nigra inhibit the activity of basal nuclei by releasing dopamine.



DA secreted in the CN and putamen is an inhibitory NT

Destruction of the DAergic neurons in SN ⇔ overly active CN & PTN

continuous output of excitatory signals to the corticospinal motor control system.

excite many or all of the muscles of the body, thus leading to *rigidity*

PD is characterized by

- **Resting Tremor (4-8Hz)- static tremor but not during sleep**
 - It is a resting tremor unlike cerebellar intention tremor
 - Due to abnormal feedback circuit between BG and cortex
- **Muscle Rigidity- prominent in flexor muscles ⇒ Parkinson's posture**
 - Muscle rigidity resist bending (lead pipe rigidity); or gives rise to serial clicks during bending (cog wheel rigidity)
 - Due to facilitation of gamma motor neurons when inhibitory lentiform nucleus is lost
 - Due to facilitation of alpha motor neurons following lack of DA.
 - ✓ The caudate nuclei becomes hyperactive, sends facilitatory discharge via extrapyramidal tract
- **Bradykinesia or Akinesia** —slowness of movement or lack of movement
 - Due to lack of dopamine⇒ caudate nucleus is locked in state of excitation.
 - ↓
 - Hence can not activate a motor program or plan, and looks paralyzed (paralysis agitans)

Table 372-1 Clinical Features of Parkinson's Disease

Cardinal Features	Other Motor Features	Nonmotor Features
Bradykinesia Rest tremor Rigidity Gait disturbance/postural instability NB: clinical dx of PD requires at least two cardinal features	Micrographia -Words/letters small and crowded Masked facies (hypomimia) Reduced eye blink Soft voice (hypophonia) Dysphagia Freezing	Anosmia Sensory disturbances (e.g., pain) Mood disorders (e.g., depression) Sleep disturbances Autonomic disturbances Orthostatic hypotension Gastrointestinal disturbances Genitourinal disturbances Sexual dysfunction Cognitive impairment/Dementia

❖ **The clinical diagnosis of PD** requires the presence of at least two cardinal features:
Bradykinesia Plus rest tremor or rigidity

❖ **Supportive criteria** — include features that increase confidence in the diagnosis of PD

Treatment of Parkinson's Disease

1. Treatment with L-Dopa

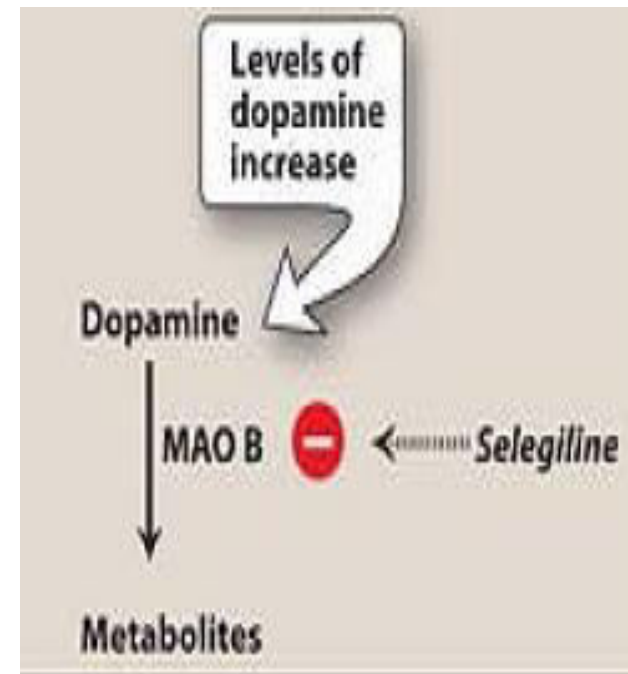
- ❖ Ameliorates many of the symptoms
- ❖ L-Dopa is converted in CNS into DA ⇨ restores normal balance b/n inhibition & excitation in the CN & PTN
 - ✓ Levodopa remains the most effective symptomatic treatment for PD and the gold standard
- ❖ Administration of DA itself does not help, because it can not pass through the BBB
- ❖ Levodopa is routinely administered in combination with a peripheral decarboxylase inhibitor
 - ✓ decarboxylase inhibitor (Eg. Carbidopa) prevents the peripheral metabolism to dopamine
 - ✓ Also prevents the Nausea-Vomiting side effect of L-Dopa

2. DOPAMINE AGONISTS

- are a diverse group of drugs that act directly on dopamine receptors
- Unlike levodopa, they do not require metabolism to an active product
- do not have comparable efficacy to levodopa
- But are relatively long-acting, are lesser side effects than L-dopa
- e.g., pramipexole, ropinirole, rotigotine

3. MAO-B INHIBITORS

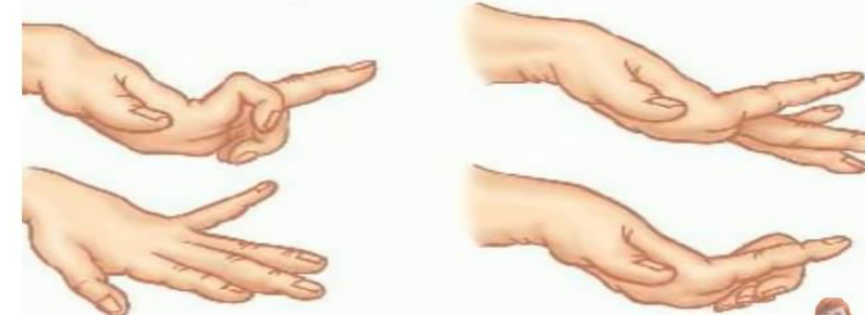
- **block central dopamine metabolism by MAO-B** and increase synaptic concentrations of the neurotransmitter.
- provide modest antiparkinsonian benefits when used as monotherapy in early disease
- are generally safe and well tolerated.
- **Selegiline and rasagiline** are relatively selective suicide inhibitors of the MAO-B enzyme.



Huntington's Disease (Huntington's Chorea)



- ❖ is described as a **hyperkinetic, semi-purposeful, nonrhythmic movement disorder**
- ❖ patients present with abnormal, rapid, dance-like movements, typically of the hands and neck
- ❖ It is a hereditary disorder usually begins causing symptoms at age 30-40 yrs
- ❖ Due to loss of most of the cell bodies of the **GABA-secreting neurons in the CN & putamen** and of **Ach-secreting neurons** in many parts of the brain
- ❖ **severe dementia** develops along with the motor dysfunctions
 - Due to loss of Ach-secreting neurons in the thinking areas of the cerebral cortex
- ❖ There is no adequate treatment for the cognitive or motor decline
- ❖ Rx: Dopamine-blocking agents may control the chorea.



Athetosis

- ❖ Is spontaneous and often continuous *writhing movements* of a hand, an arm, the neck, or the face.
 - ✓ Slow, distal, writhing, involuntary movements
 - ✓ with a propensity to affect the arms and hands
 - ✓ *Due to lesions in the globus pallidus (Damage to putamen circuit)*

Hemiballismus

- ❖ Is a sudden *flailing or waving movements* of an entire limb
- ❖ During waking it results in stumbling
- ❖ Due to a lesion in the **subthalamic Nucleus of putamen circuit**(Eg. 2^o to stroke)
- ❖ usually self-limiting and tends to resolve spontaneously after weeks or months